Welcome to the Parasitology Department



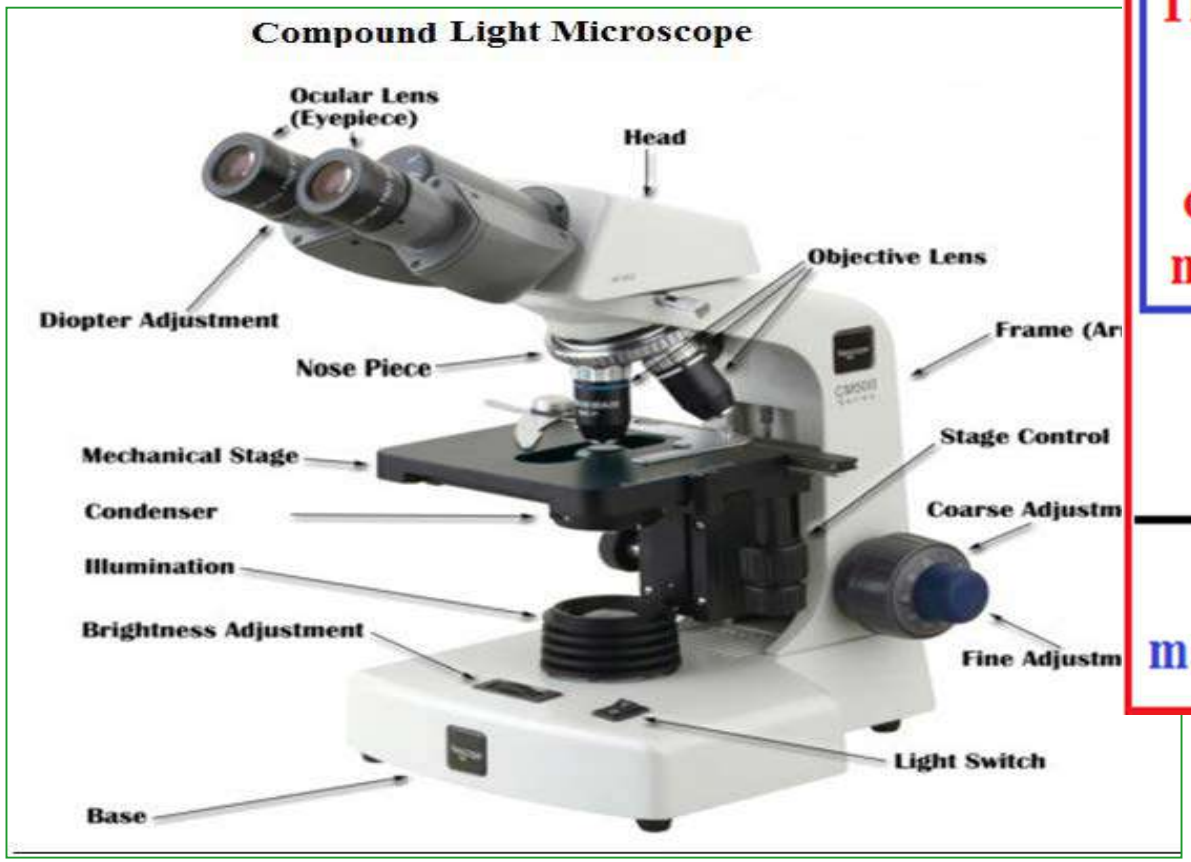
Lab 1

CamScanner

Light Microscopes used in the laboratory



1- Compound light microscope

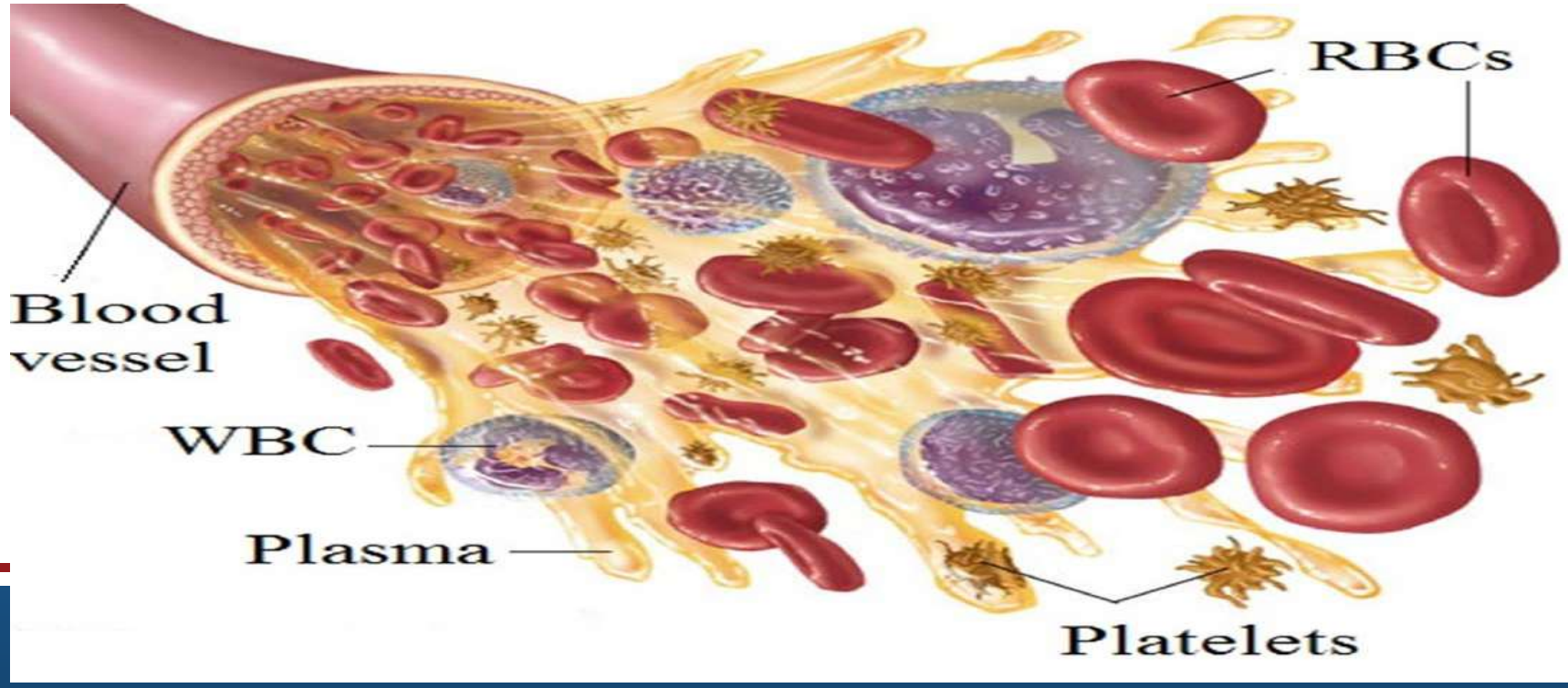


The objective lenses of the compound microscope	Low power	High power	Oil-immersion lens
Working distance ⇨	10x	40x	100x oil
Total magnification ⇨	10 x 10 x 100	10 x 40 x 400	10 x 100 x 1000

(x10 or x40 or x100)

CamScanner

Blood, lymphatics and mononuclear phagocytic system parasites



Lab 1

CamScanner

Blood, lymphatics and mononuclear phagocytic system parasites

I- Helminths

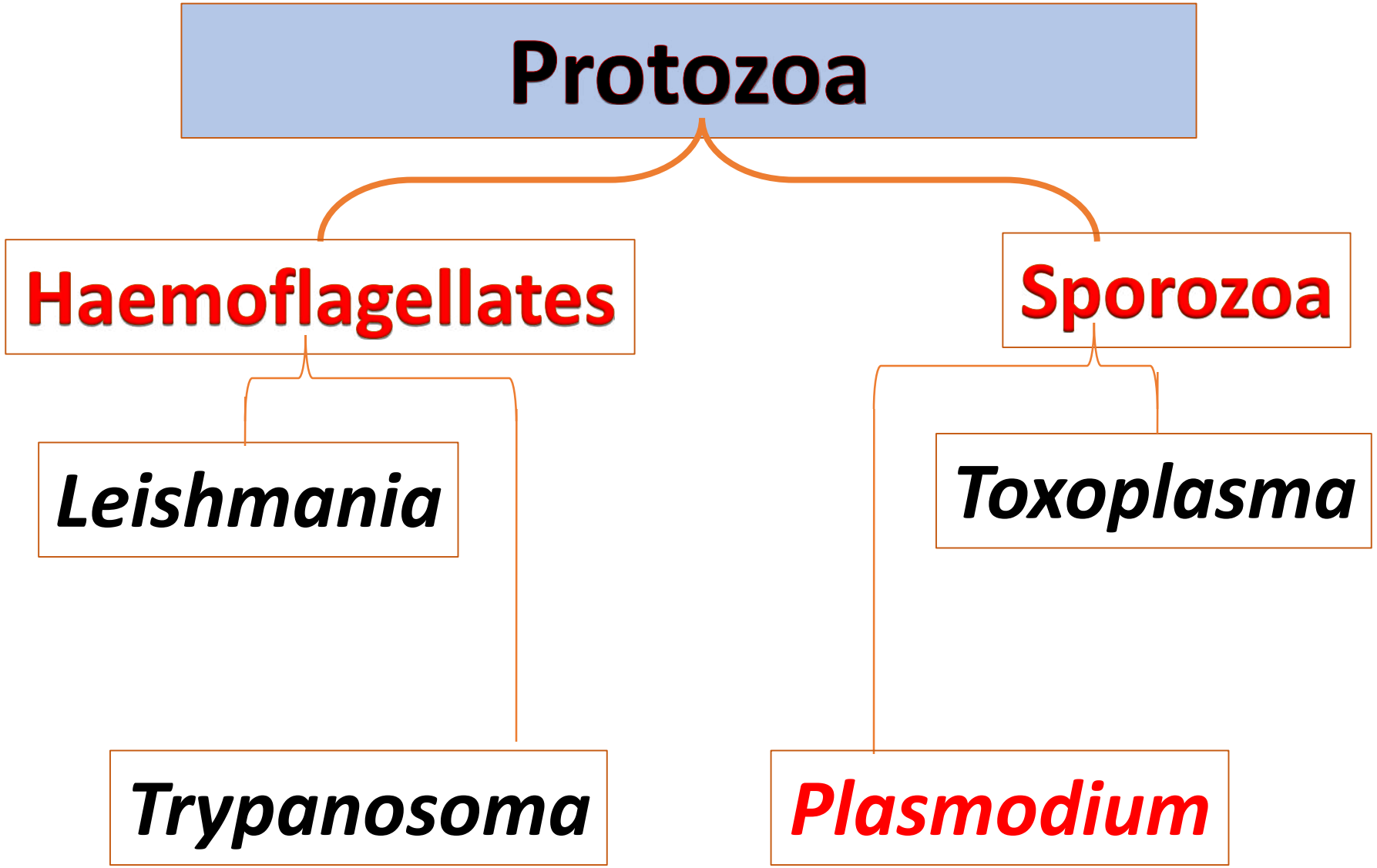
- Nematodes in lymphatic system:

Wuchereria bancrofti

II- Protozoa

- Sporozoa
- Haemo- flagellates

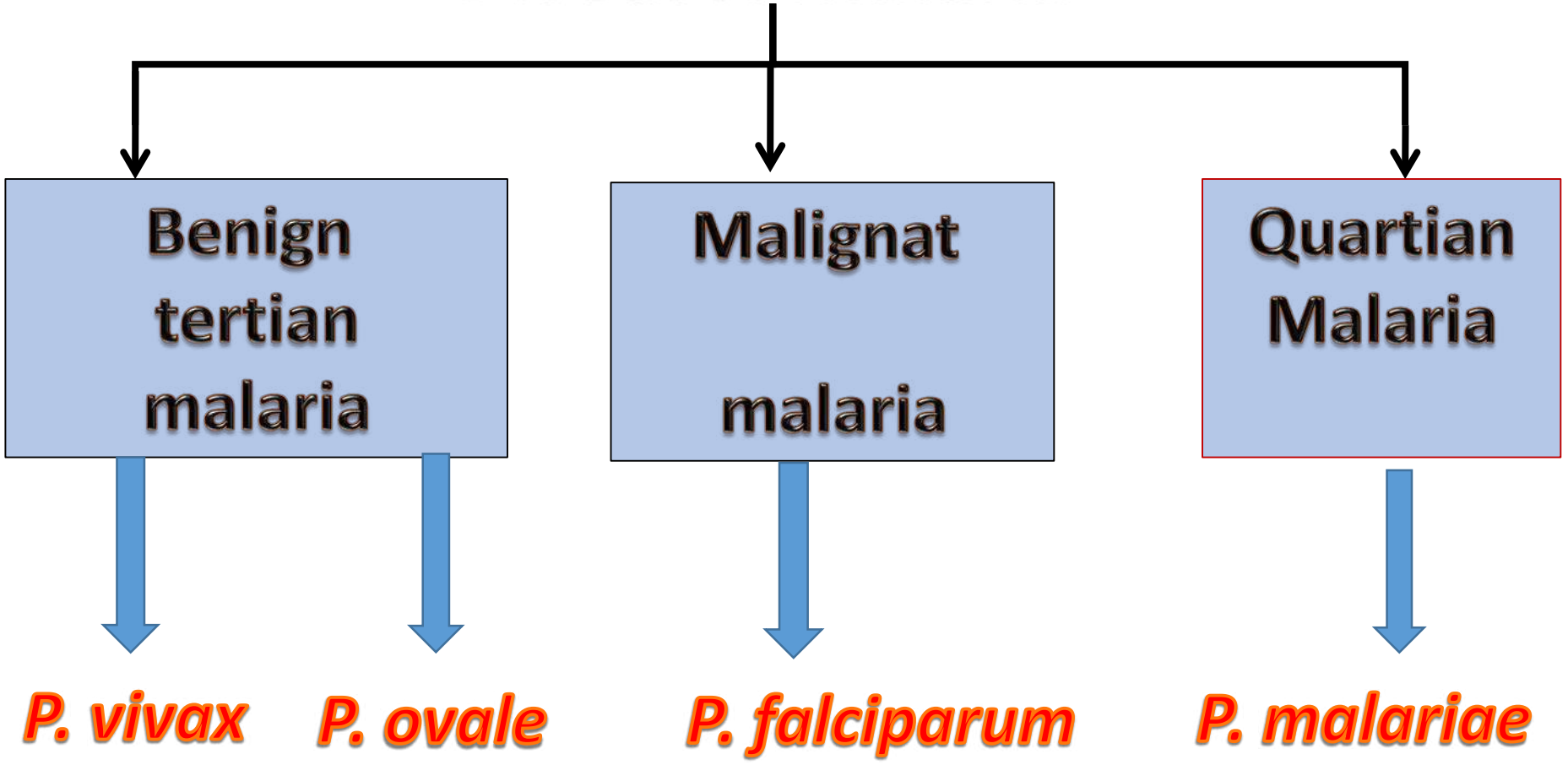
CamScanner


Protozoa
Haemoflagellates
Sporozoa
Leishmania
Toxoplasma
Trypanosoma
Plasmodium

CamScanner

Plasmodium spp.

Disease: Malaria



CamScanner

Only *Plasmodium vivax*

and

Plasmodium falciparum

will be studied in practical sessions

CamScanner

Plasmodium spp.

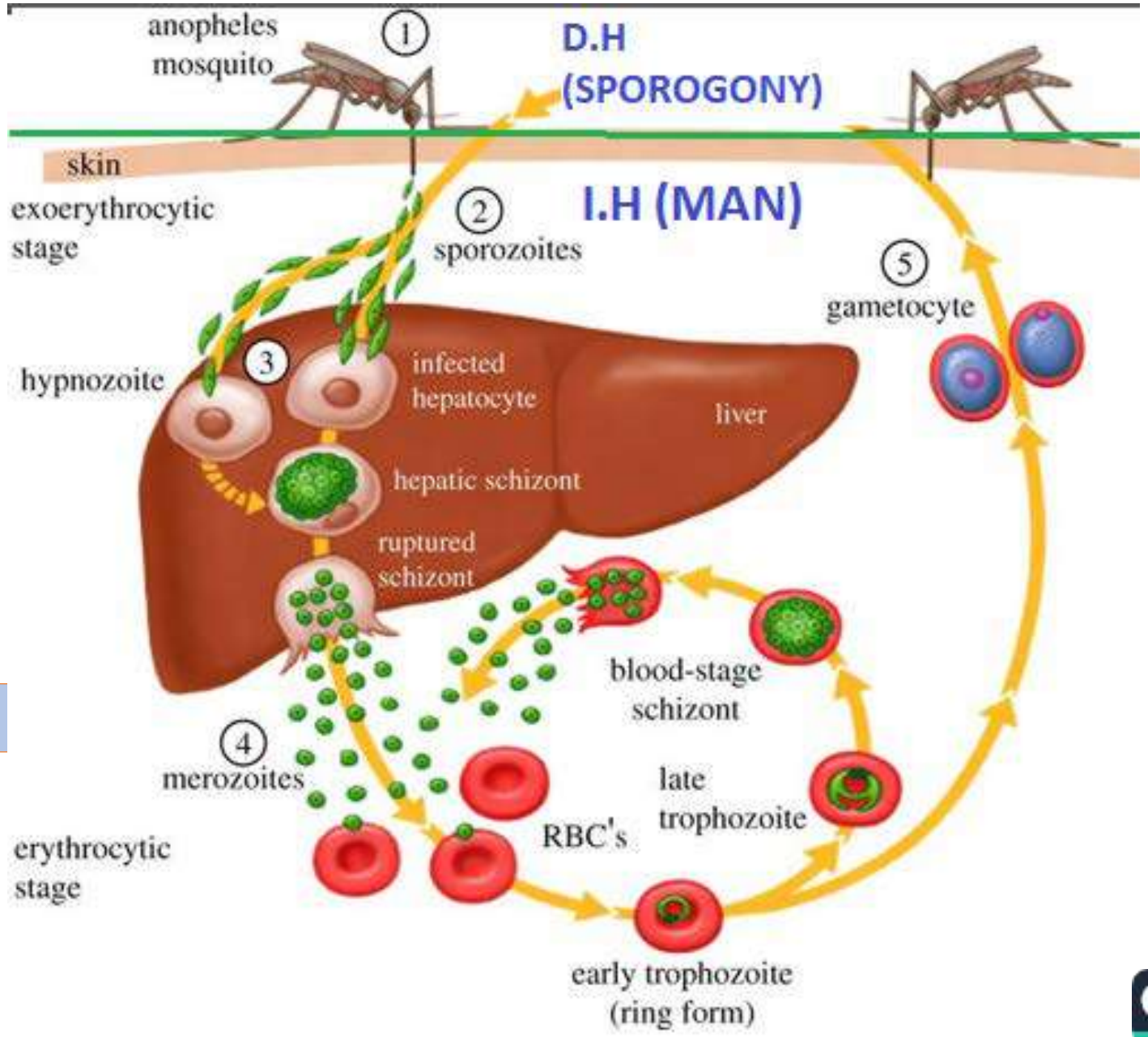
***Plasmodium* spp.** are protozoan parasites that need two hosts, one of them is a vector, to complete the life cycle.

The vector of *Plasmodium spp.* is the female of many species of ***Anopheles*** mosquito.

CamScanner

Exoerythrocytic schizogny

Erythrocytic schizogny



Gametogony

CamScanner

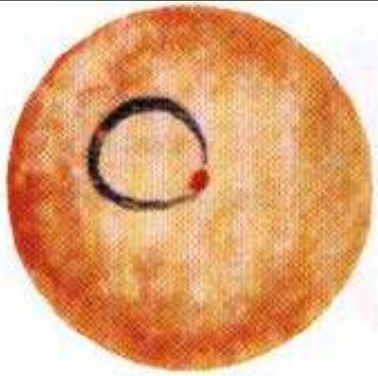
The Wellcome Trust

UNDP / World Bank / WHO
Special Programme for Research and Training in Tropical Diseases
(TDR)

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Morphology of *Plasmodium* Stages in human blood

P. vivax

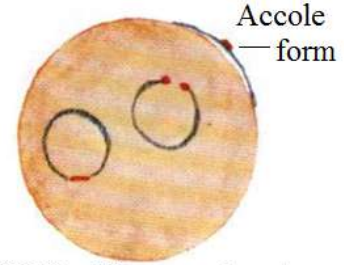


Enlarged RBC with ring stage of *P. vivax*

Ring stage= early trophozoite

-Thin rim of cytoplasm.
- Central vacuole
- Nucleus (or chromatin dot)

P. falciparum



RBC with many ring stages of *P. falciparum*

Infected RBC not ENLARGED

- Thin rim of cytoplasm.
- Single chromatin dot.



- Very thin rim of cytoplasm.
- Usually two chromati dots.



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Virtual microscopy

https://www.pathozoom.com/imageaccess/583225/bab2c4e8edb04e9e8414856d0ba08b61/12029

https://www.pathozoom.com/imageaccess/583220/bab2c4e8edb04e9e8414856d0ba08b61/12029

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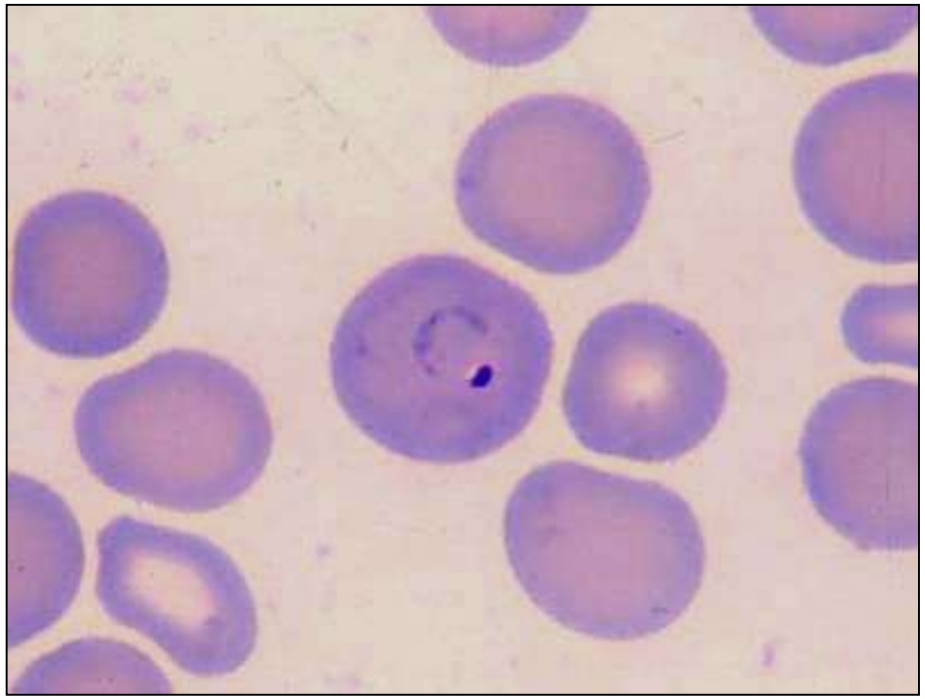
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CamScanner

Morphology of *Plasmodium* Stages in human blood

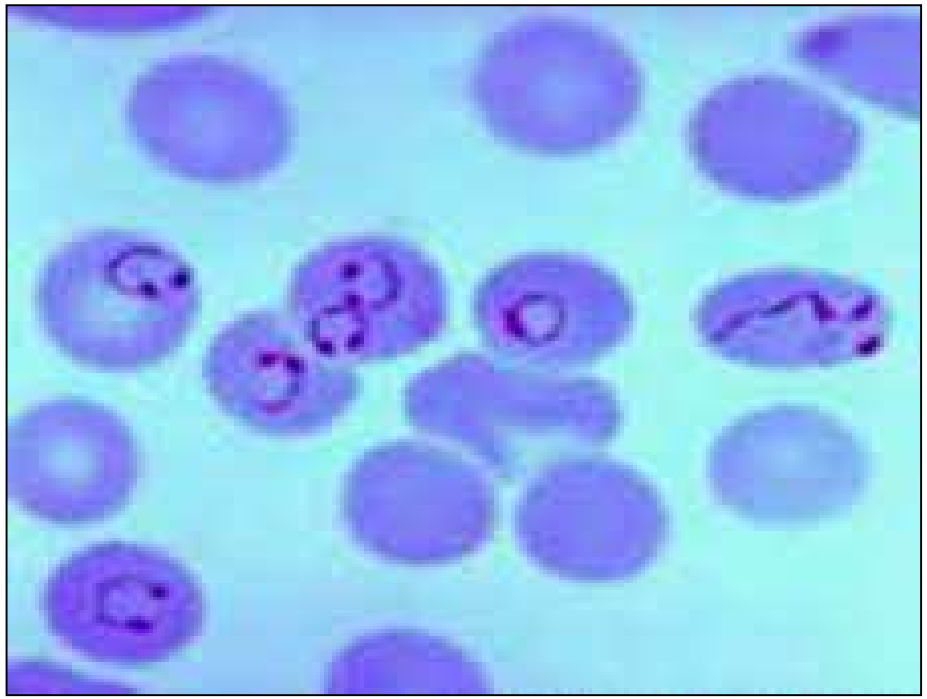
Ring stage

P. vivax



- Enlarged infected RBC
- Single infection per RBC

P. falciparum



-Infected RBC same size as others.
-Multiple infection per RBC

CamScanner

Morphology of *Plasmodium* Stages in human blood

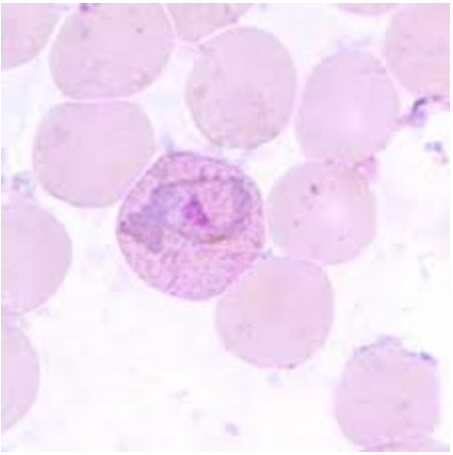
Mature (late) Trophozoite stage of *P. vivax*

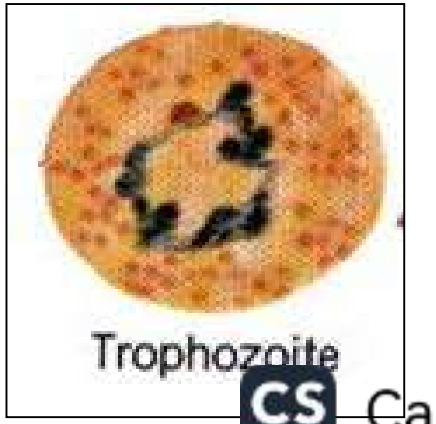
Shape: irregular.

Chromatin: single dot.

Stippling: yellowish brown scattered very fine particles (Schüffner's dots).

Infected RBC: enlarged.

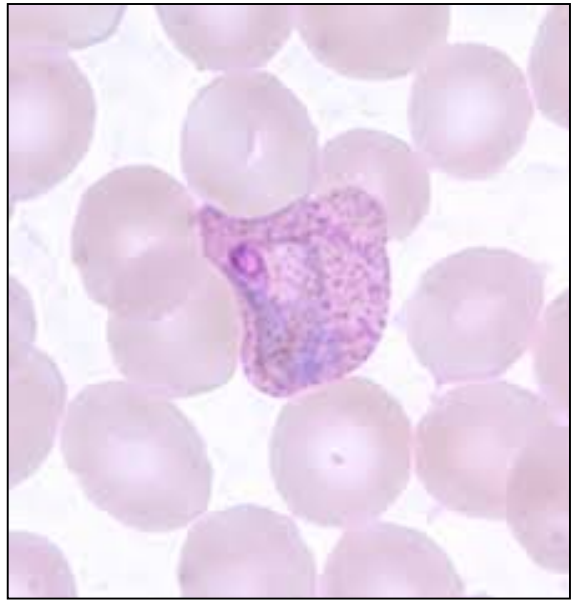


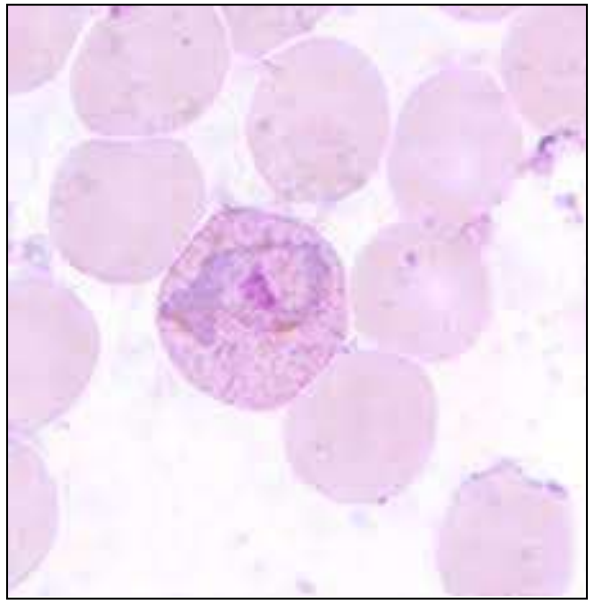


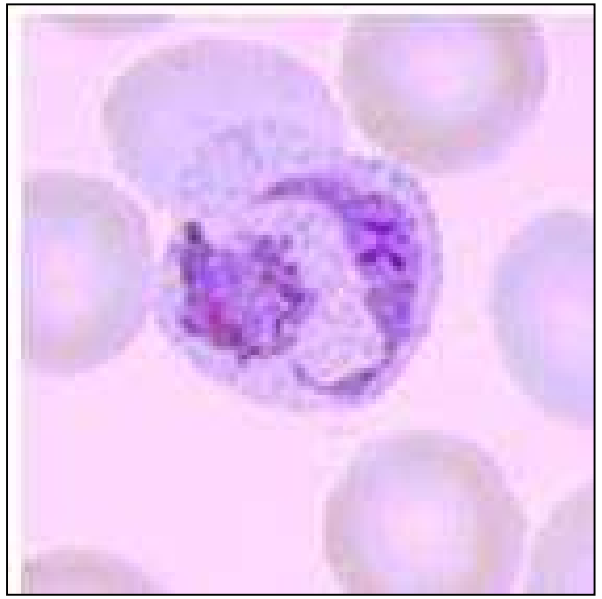
CamScanner

Morphology of *Plasmodium* Stages in human blood

Trophozoite stage of *P. vivax*





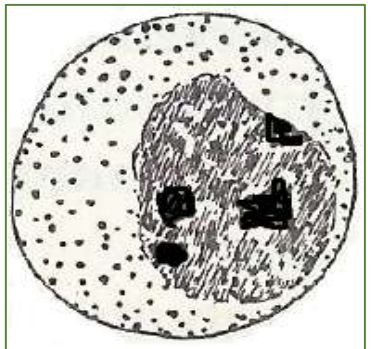


CamScanner

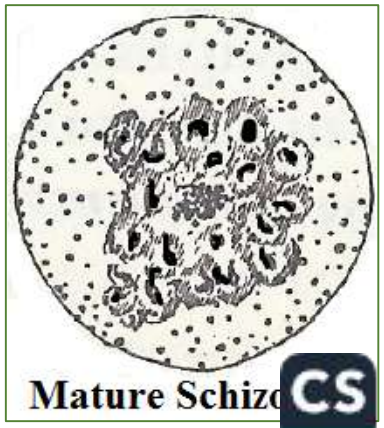
Morphology of *Plasmodium* Stages in human blood

Schizont of *P. vivax*

- Irregular outline.
- Almost fills the RBC.
- Chromatin: numerous irregular masses.
- Scattered pigment dots.



Growing Schizont

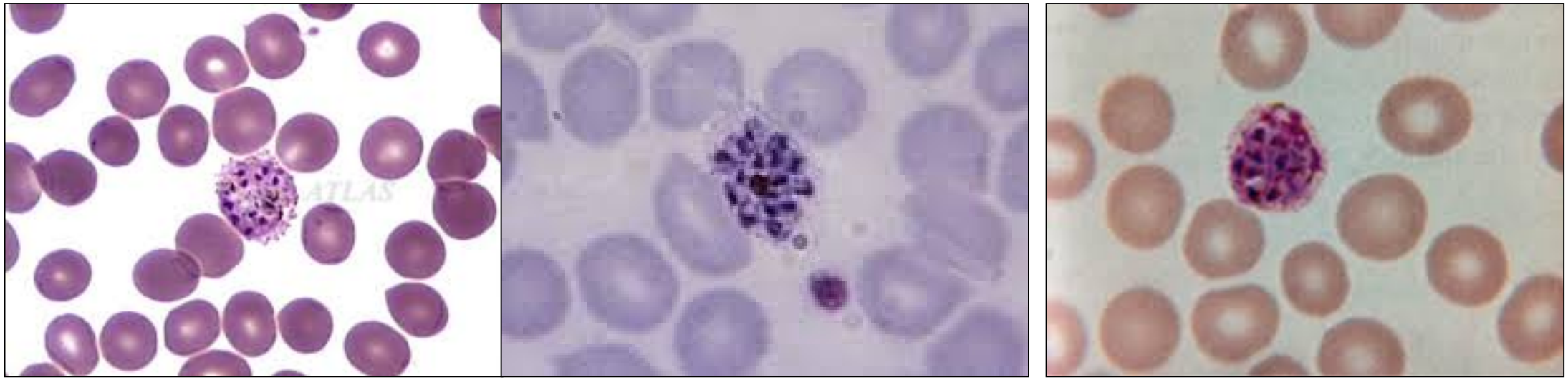


Mature Schizo

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Morphology of *Plasmodium* Stages in human blood

Schizont of *P. vivax*



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Virtual microscopy

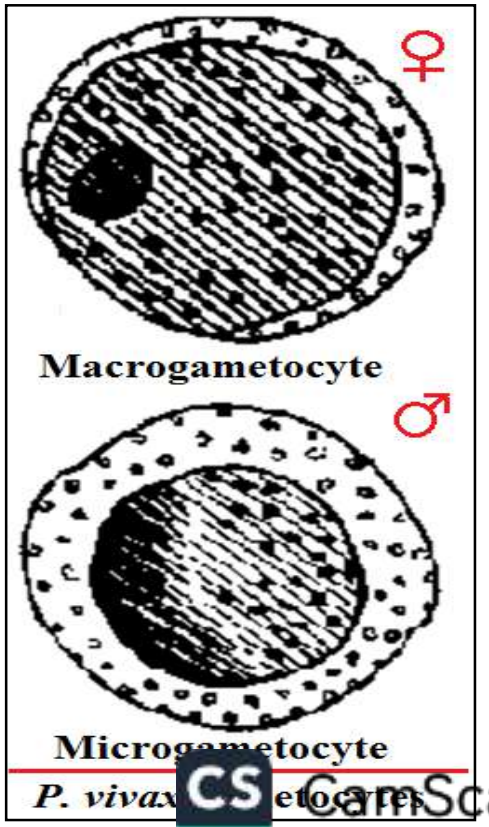
https://www.pathozoom.com/imageaccess/583224/bab2c4e8edb04e9e8414856d0ba08b61/12029

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Morphology of *Plasmodium* Stages in human blood

Gametocytes of *P. vivax*

- Round or oval compact.
- Fill about ¾ the RBC.
- Dark blue cytoplasm.
- Chromatin single peripheral mass.
- Scattered pigment dots at periphery.



P. vivax gametocytes

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Virtual microscopy

https://www.pathozoom.com/imageaccess/583213/bab2c4e8edb04e9e8414856d0ba08b61/12029

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Morphology of *Plasmodium* Stages in human blood

Gametocytes of *P. vivax*

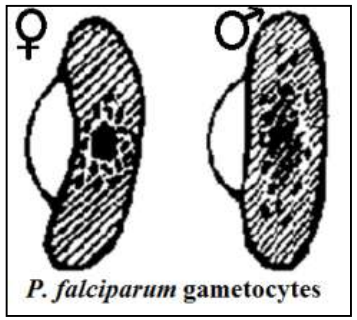


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Morphology of *Plasmodium* Stages in human blood

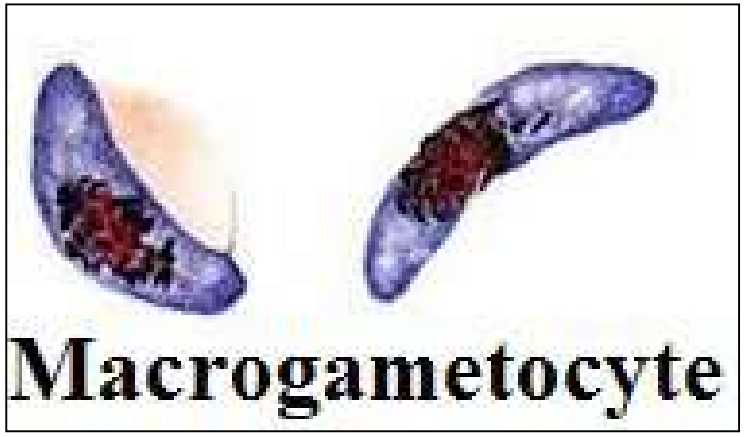
Gametocytes of *P. falciparum*

Size: Larger than RBC



P. falciparum gametocytes

Shape: crescent- or kidney-shaped


Macrogametocyte


Microgametocyte

CamScanner

Virtual Microscopy

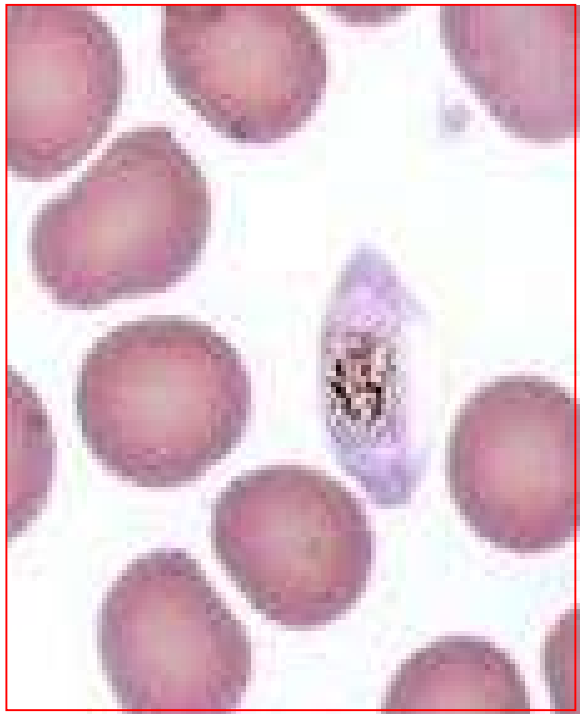
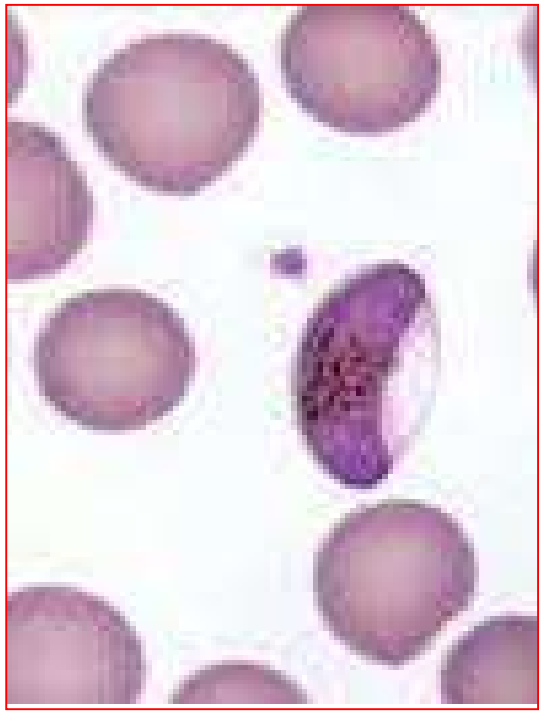
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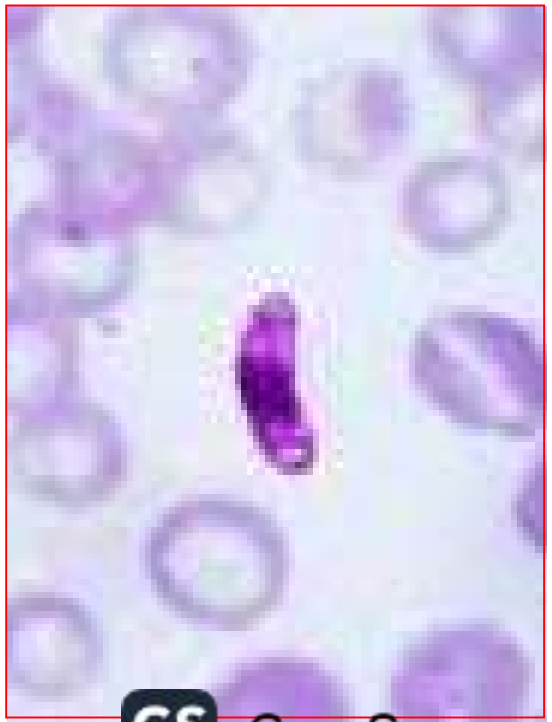
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Morphology of *Plasmodium* Stages in human blood

Gametocytes of *P. falciparum*



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To wrap up

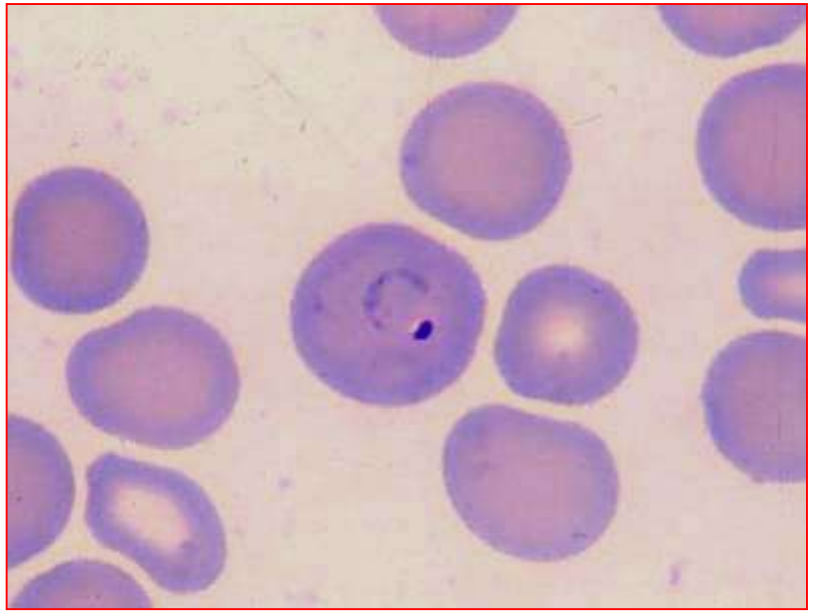
CamScanner

6 slides

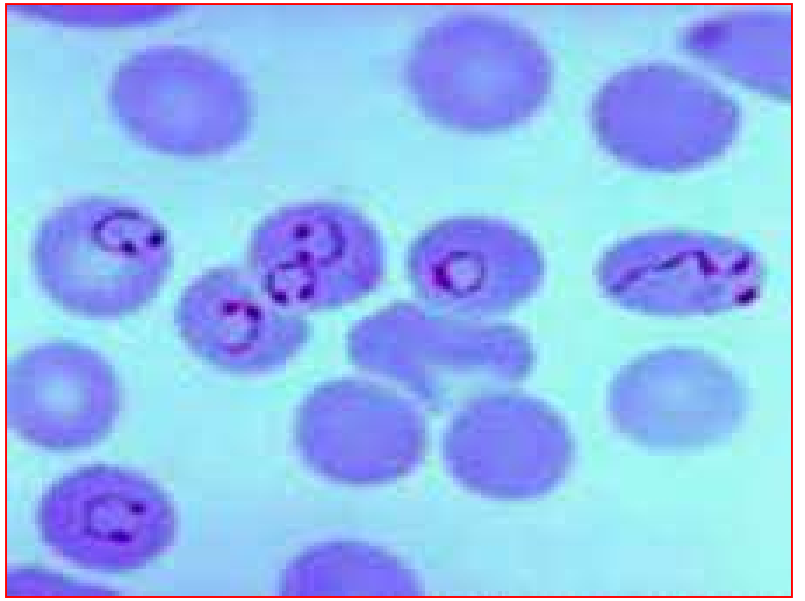
Erythrocytic stages of *P.vivax* and *P. falciparum* in blood films

CamScanner

Ring stage



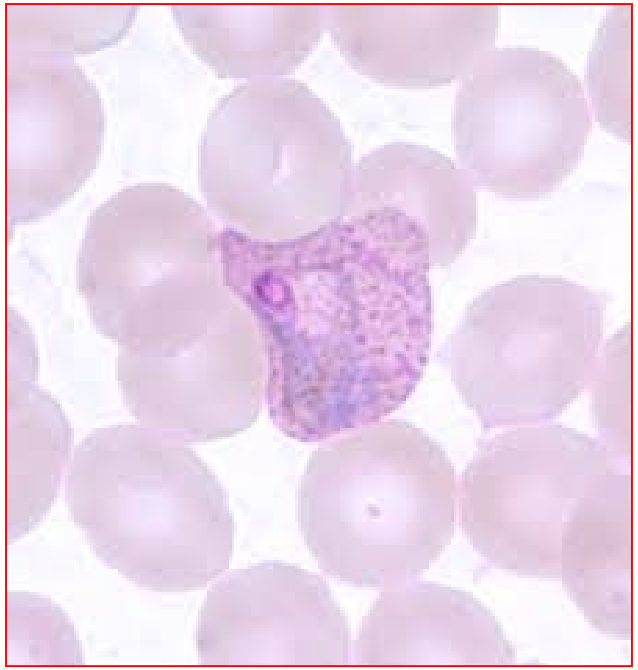
Ring of *P. vivax* in blood film



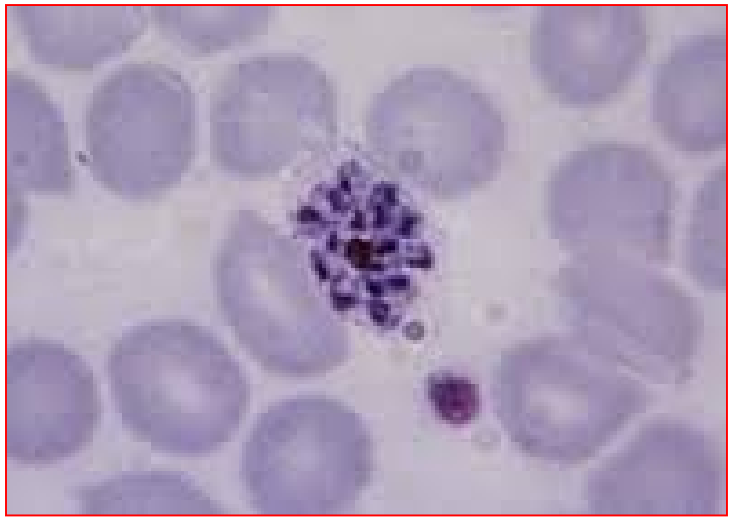
Ring of *P. falciparum* in blood film

CamScanner

Plasmodium vivax



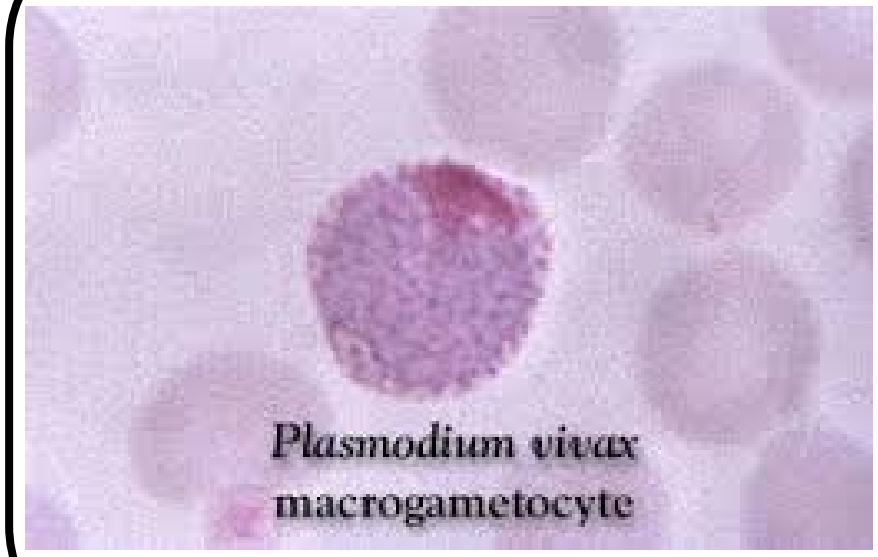
Trophozoite of *P. vivax* in blood film



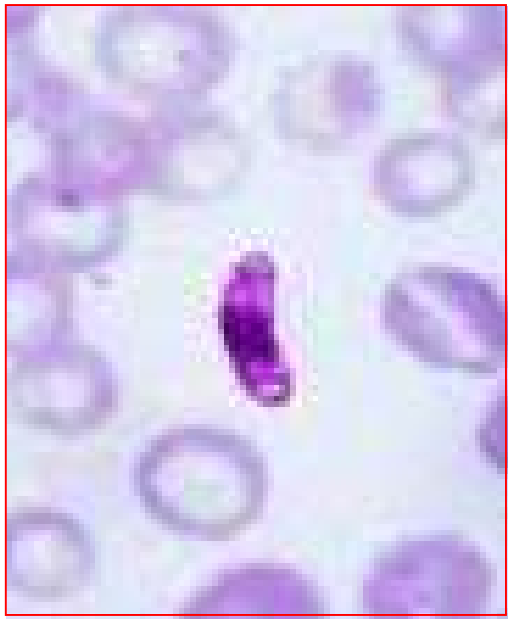
Schizont of *P. vivax* in blood film

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Gametocytes of *Plasmodium*



Gametocyte of *P. vivax* in blood film



Gametocyte of *P. falciparum* in blood film

CamScanner

Self assessment

- Write your answer on a paper
  - Keep it on your log book

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Interpret the following statements:

True relapse of malaria happens in *Plasmodium vivax* and *P. ovale* infections and not in other species.

The phenomena of true relapse that occurs in *P.vivax* and *P.ovale* is explained by the concept of **hypnozoites** .

Hypnozoites are not found in the development of *P.falciparum* and *P. malarie*, so there is no true relapses.

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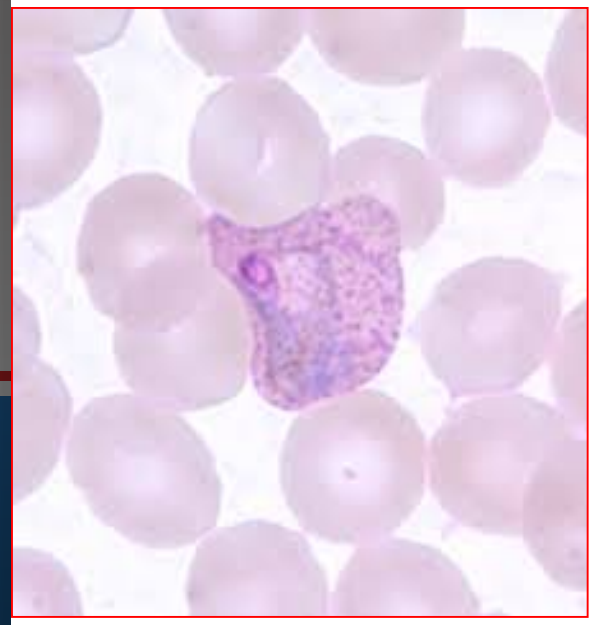
Interpret the following statements:

The occurrence of paroxysm (fever every 3rd or 4th days) in malaria.

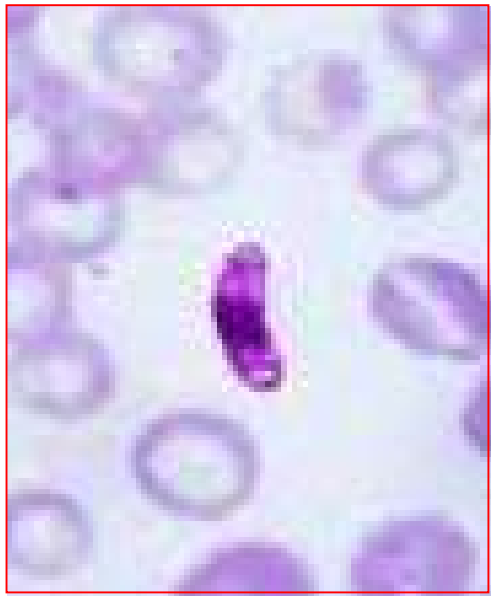
The destruction of RBCs (**schizont**) and release of the merozoites, pigments at the end of erythrocytic schizogony cycle every 3rd or 4th days produce the malarial paroxysms that characterized the malaria.

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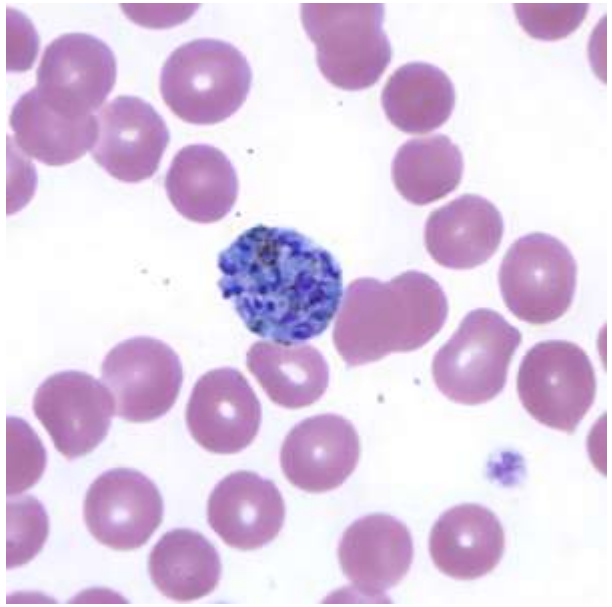
The stage responsible for the malaria paroxysm is:



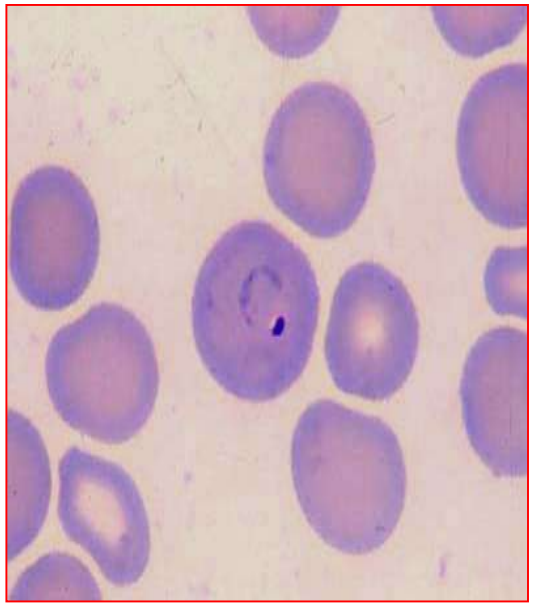
a



b

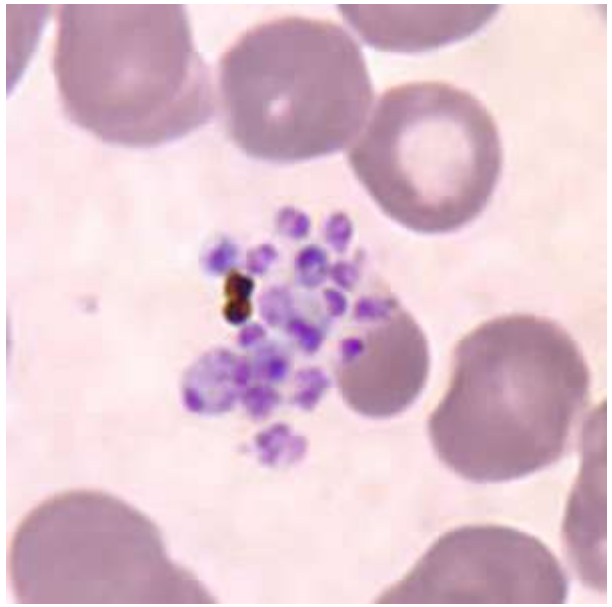


C



d

CamScanner



Ruptured schizont of *P. vivax* in a thin blood smear, showing free merozoites and pigment.

https://www.cdc.gov/dpdx/malaria/index.html

CamScanner

- https://forms.office.com/Pages/ResponsePage.aspx?id=qoXLTRRxHUmdHrlgE0RO8onJd56ApDtBrUPoiPpkaF5UQ0sxUTNHSjIzUERFNUMxTlNMR0JSWEEwQy4u

- https://forms.office.com/r/0ysT6wTWFn



THANK YOU

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Parasitology Department

Blood Module

2024/2025

Lab 2

CamScanner

Lab 2

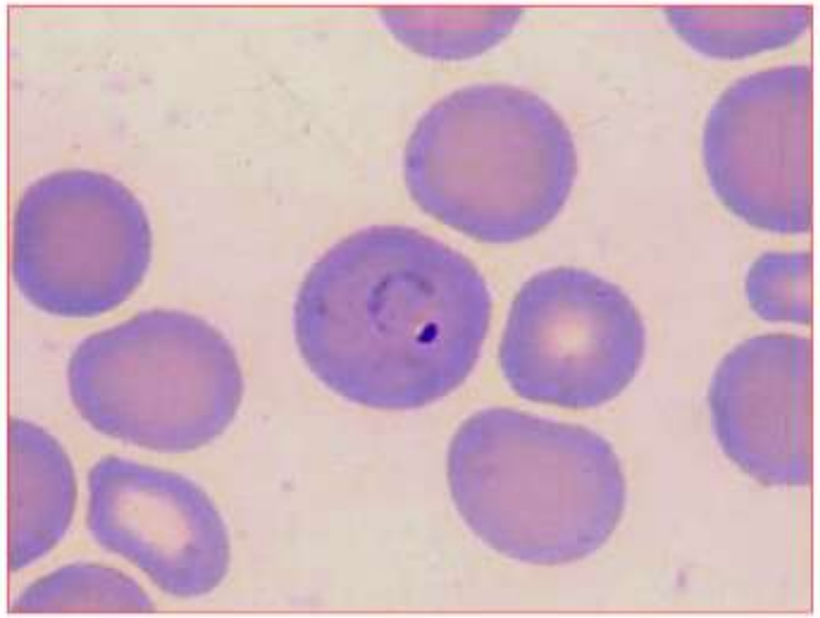
Section I

Prelab Assessment

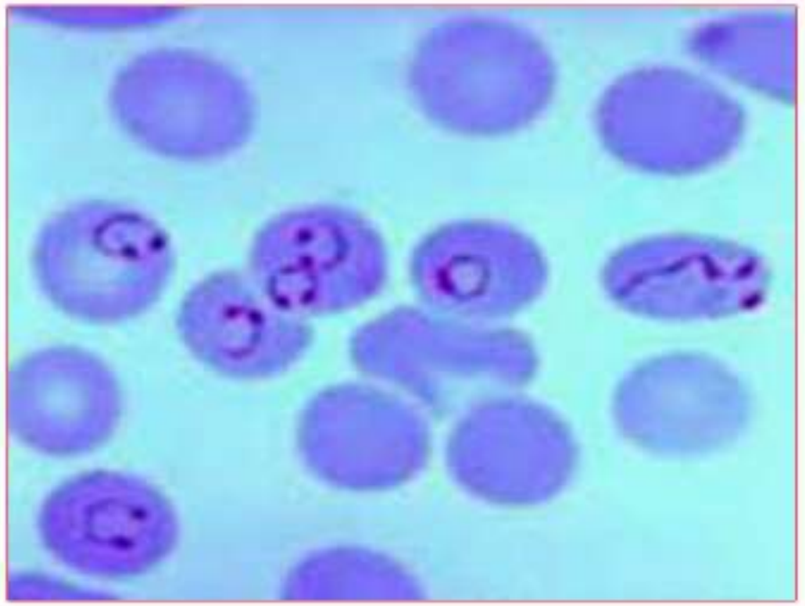
CamScanner

Lab 2

Identify



Ring of *P. vivax* in blood film

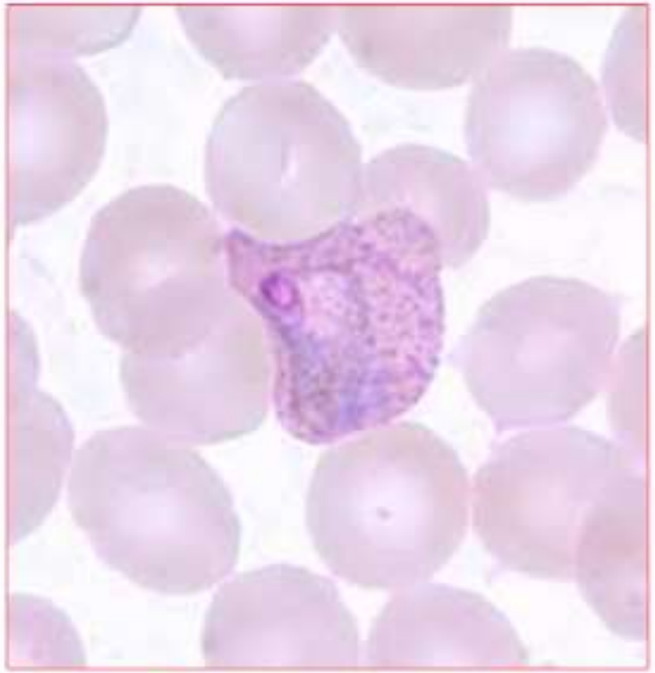


Ring of *P. falciparum* in blood film

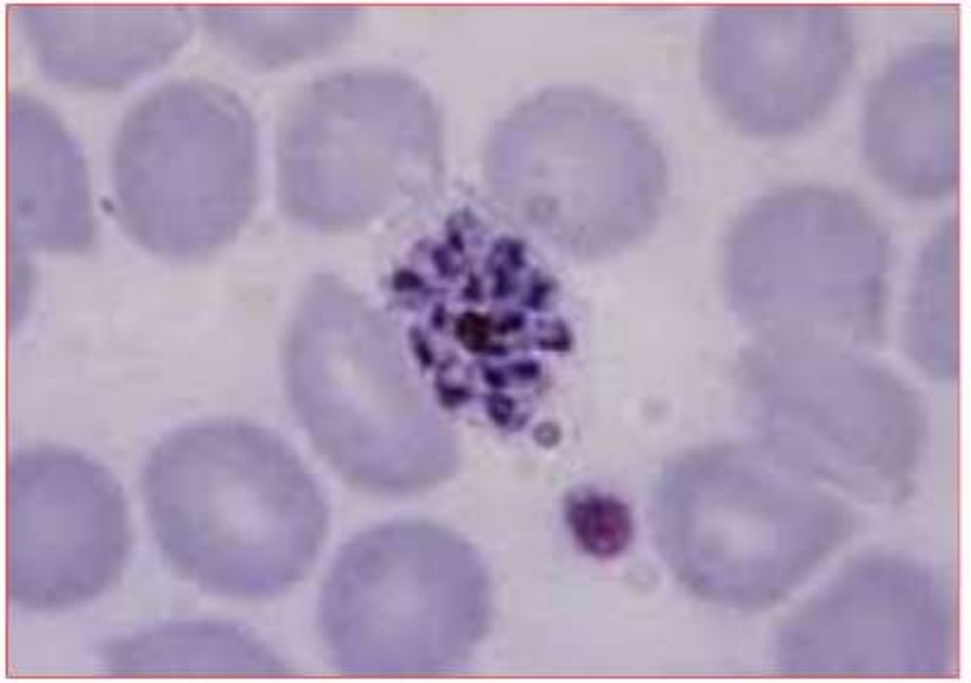
CamScanner

Lab 2

Identify



Trophozoite of *P. vivax* in blood film

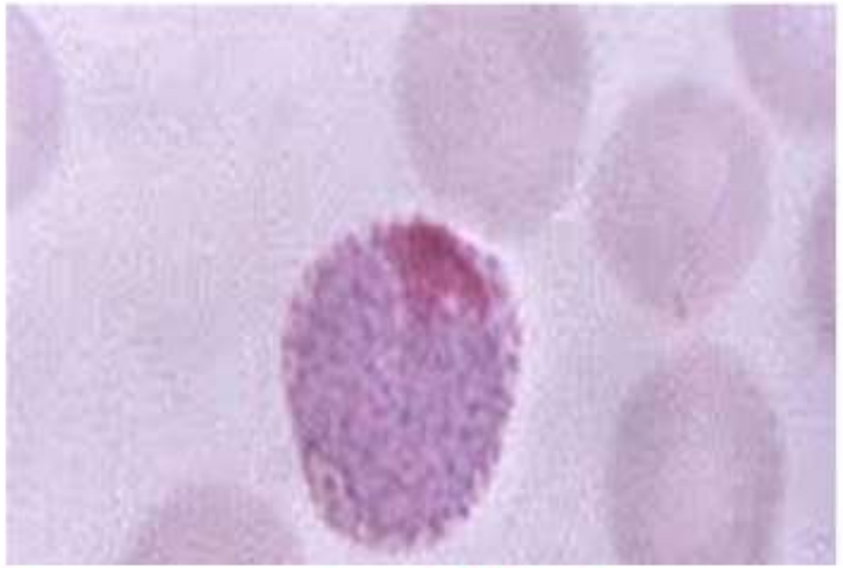


Schizont of *P. vivax* in blood film

CamScanner

Lab 2

Identify



Gametocyte of *P. vivax* in blood film



Gametocyte of *P. falciparum* in blood film

CamScanner

Lab 2

Section II

Mosquitoes

CamScanner

Lab 2

Mosquitoes (12 slides)

- *Culex* spp.: 6 slides

Mouthparts of adult male and female, egg raft, larva, pupa and complete adult.

- *Aedes* spp.: 2 slides

Egg and larva.

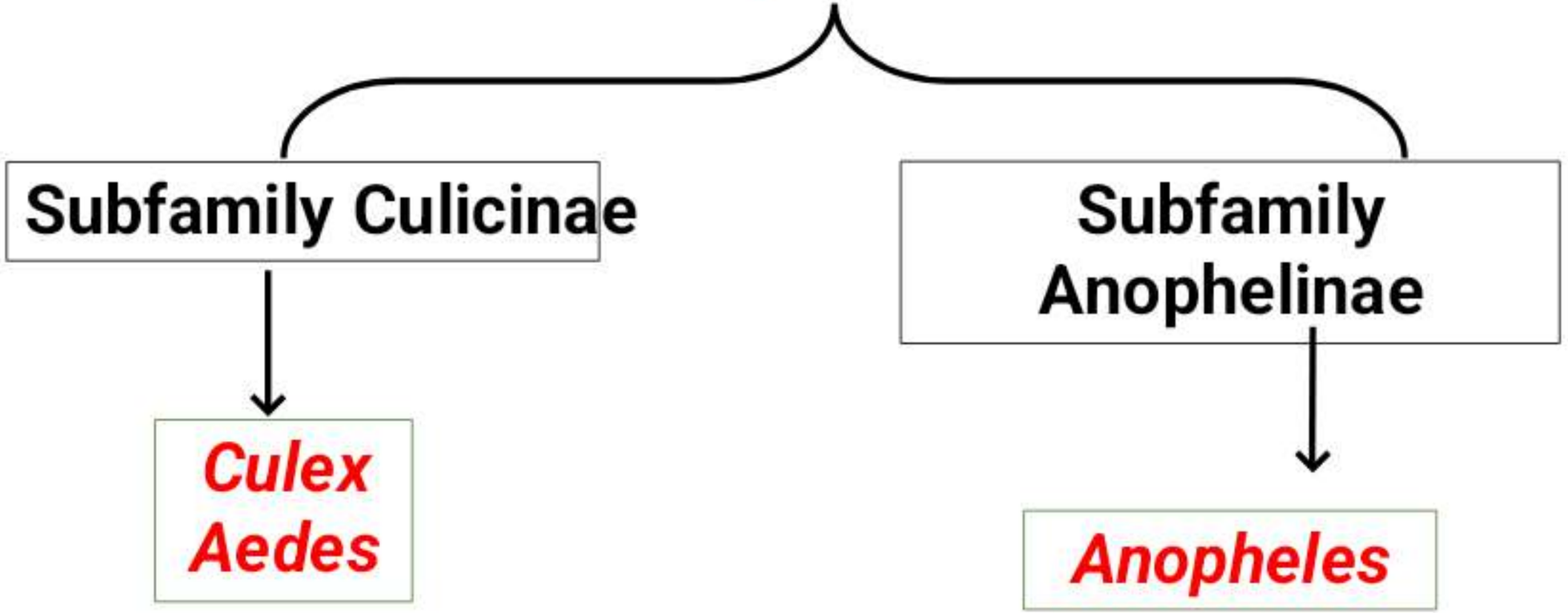
- *Anopheles* spp.: 4 slides

Mouthparts of adult male and female, egg and larva.

CamScanner

Lab 2

Mosquitoes

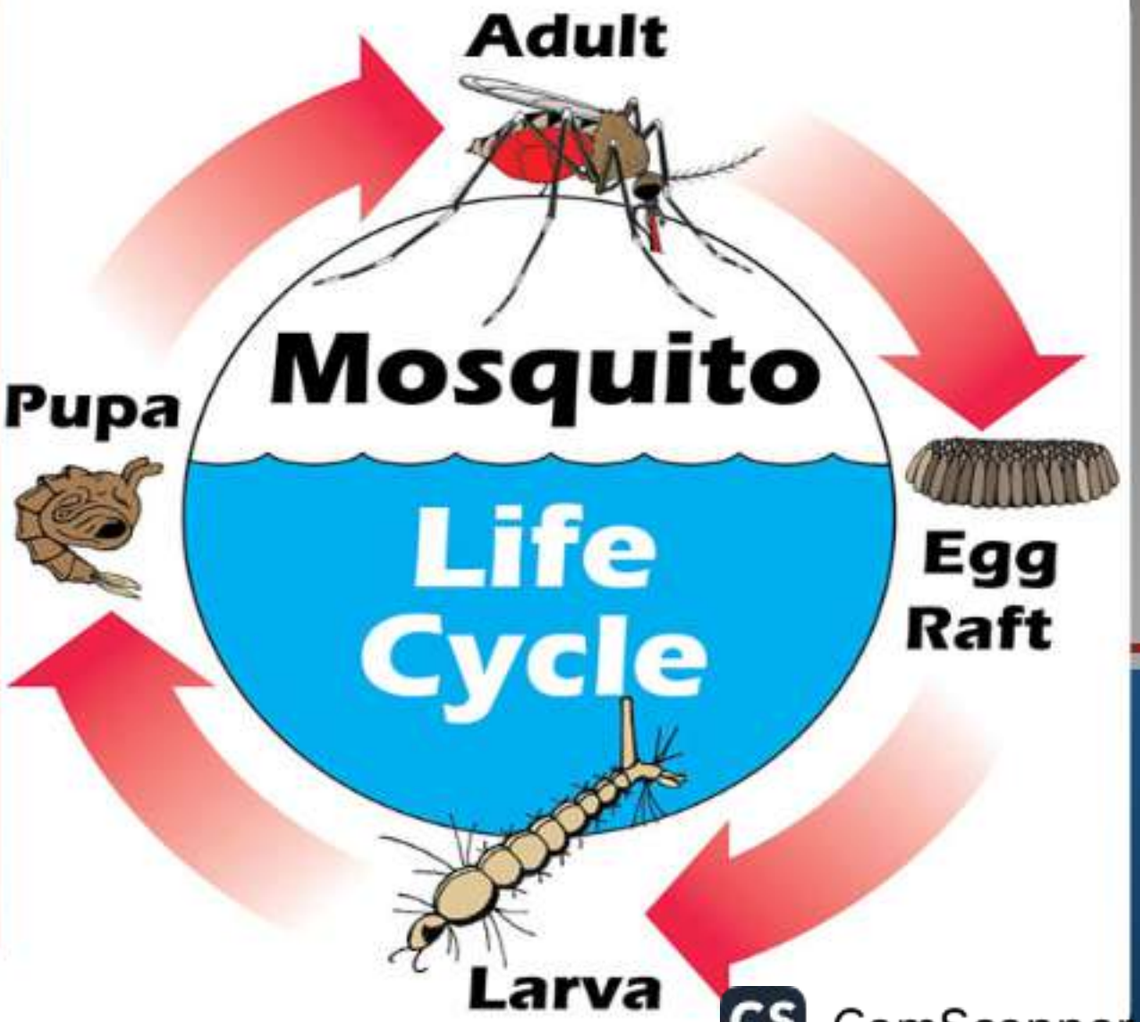


CamScanner

Lab 2

Life Cycle

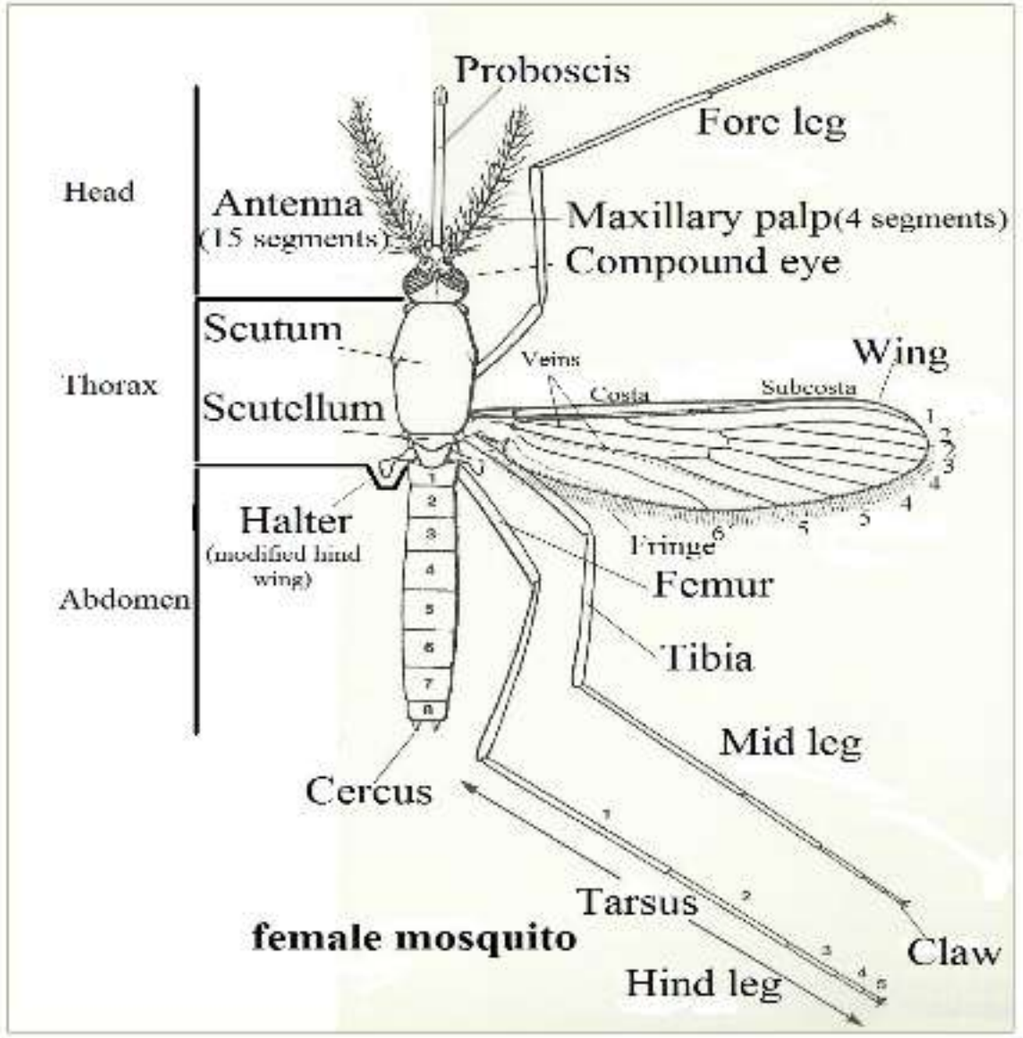


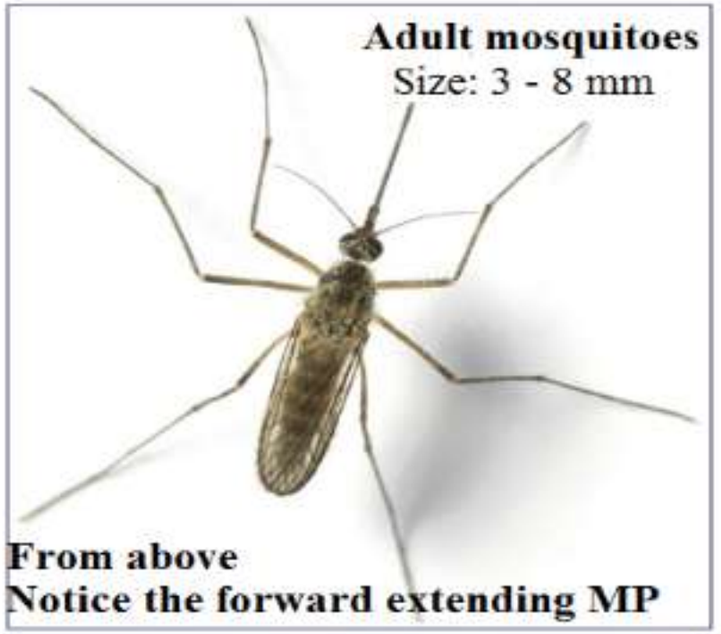


CamScanner

Lab 2

Adult







CamScanner

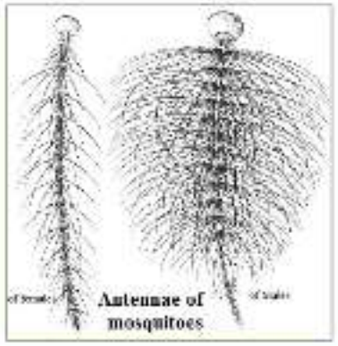
Lab 2

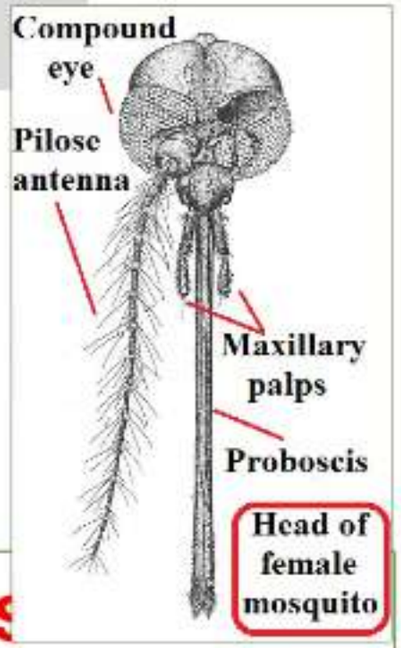
Head and mouthparts of adult mosquitoes:

Sex and genus of adult mosquito are identified by examining:

The antennae

- Multisegmented
- Plumose in males
- Pilose in females





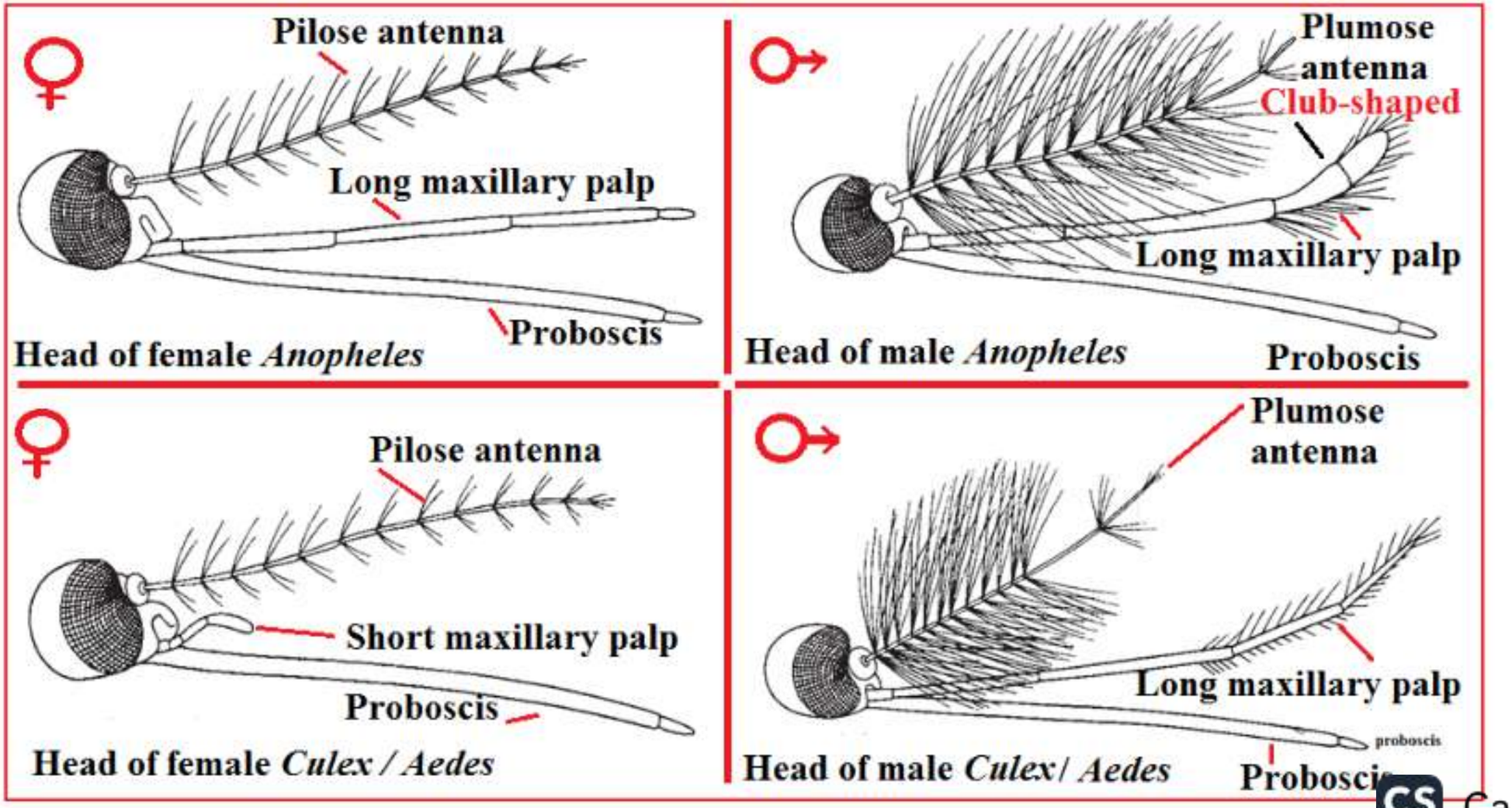
The maxillary palps

- A pair of 4-segmented structure
- Short or long
- Thin or club-shaped

CamScanner

Lab 2

Mouthparts



CamScanner

Lab 2

How to identify mouthparts of mosquitoes?

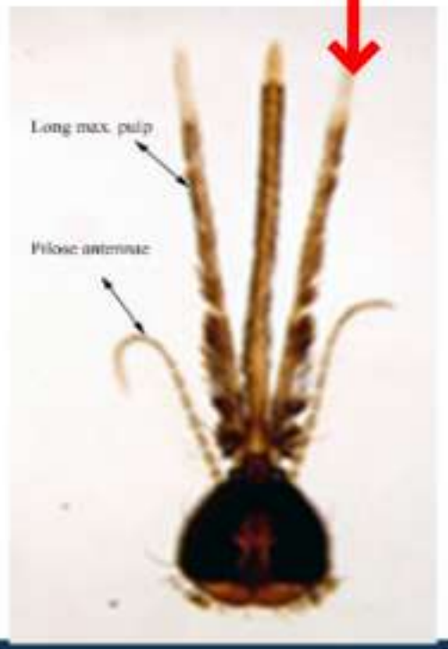
First identify the sex by identifying type of antennae

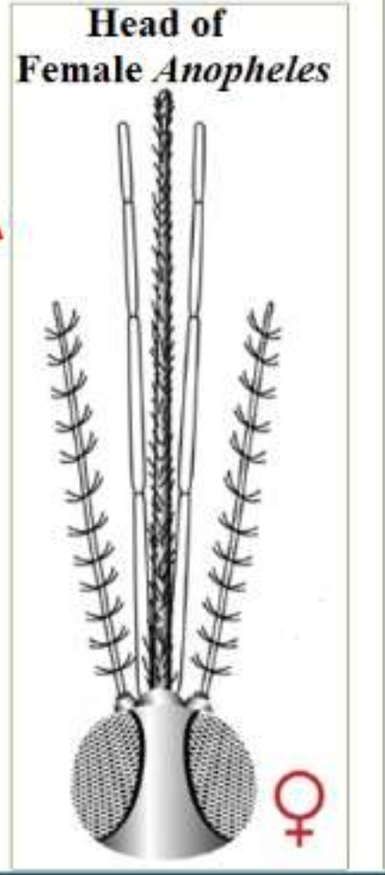
CamScanner

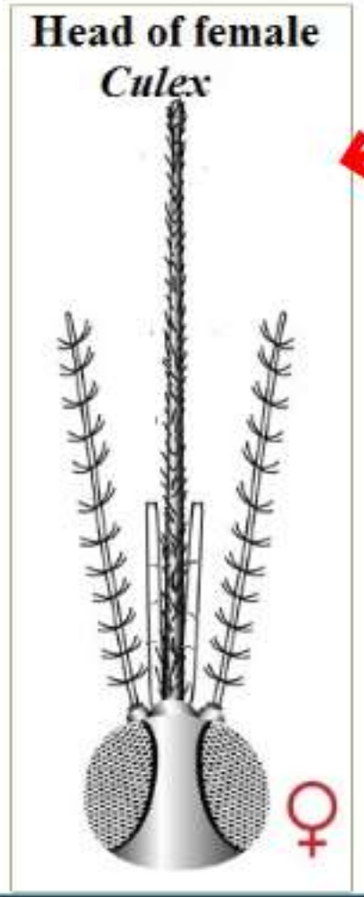
Lab 2

MP of female mosquitoes = Pilose antennae (short hairs)

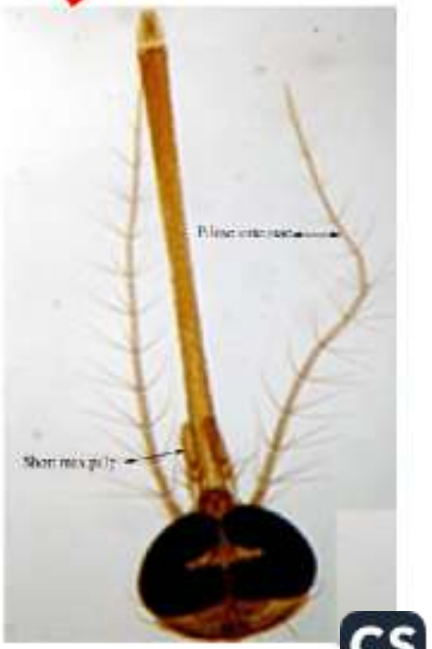
Long maxillary palp
Anopheles







Short maxillary palp
Culex



CamScanner

Lab 2

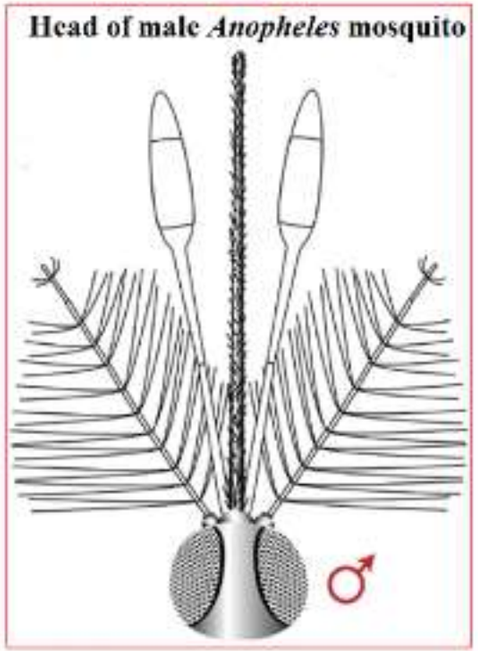
Mouthparts of male mosquitoes

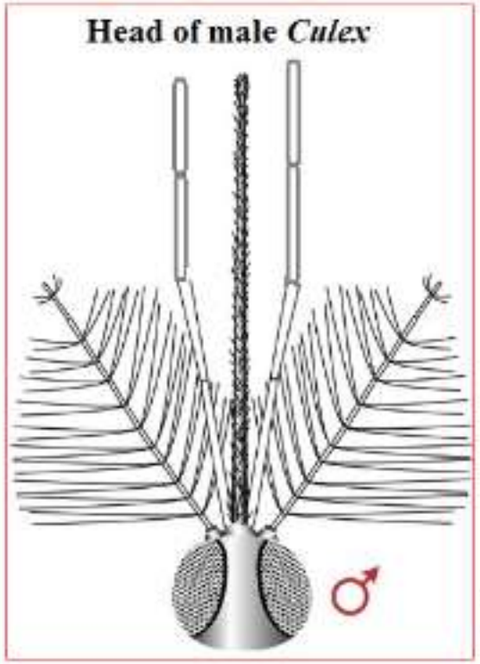
= Plumose antennae (Long hairs) + Long maxillary palps



Club end maxillary palp

Anopheles







Thin end maxillary palp

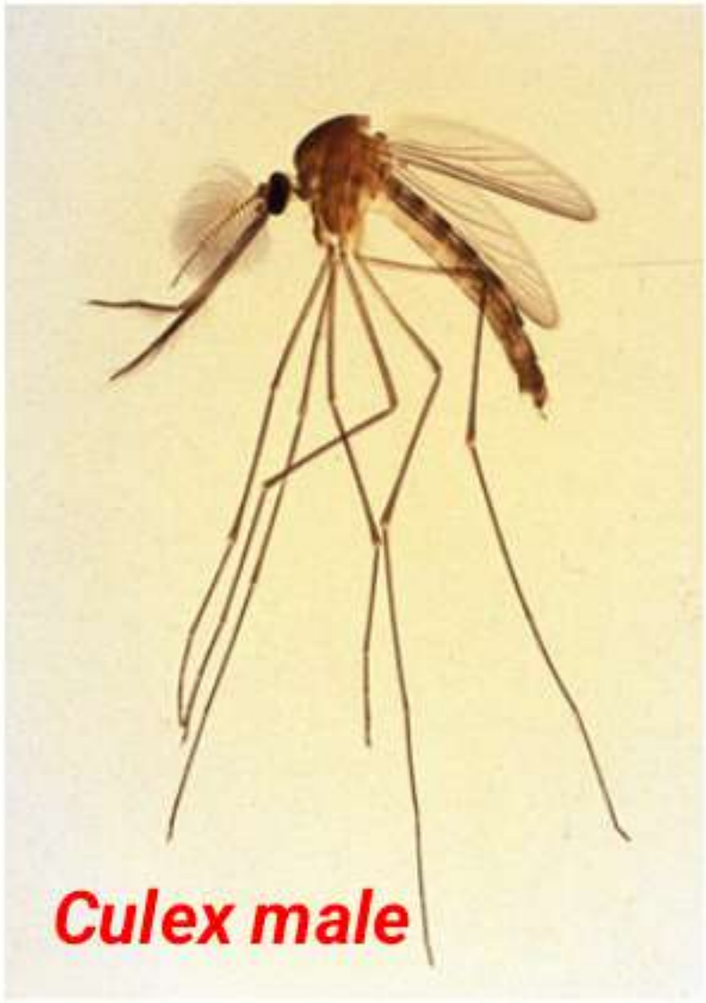
Culex / Aedes

CamScanner

Lab 2



Anopheles male

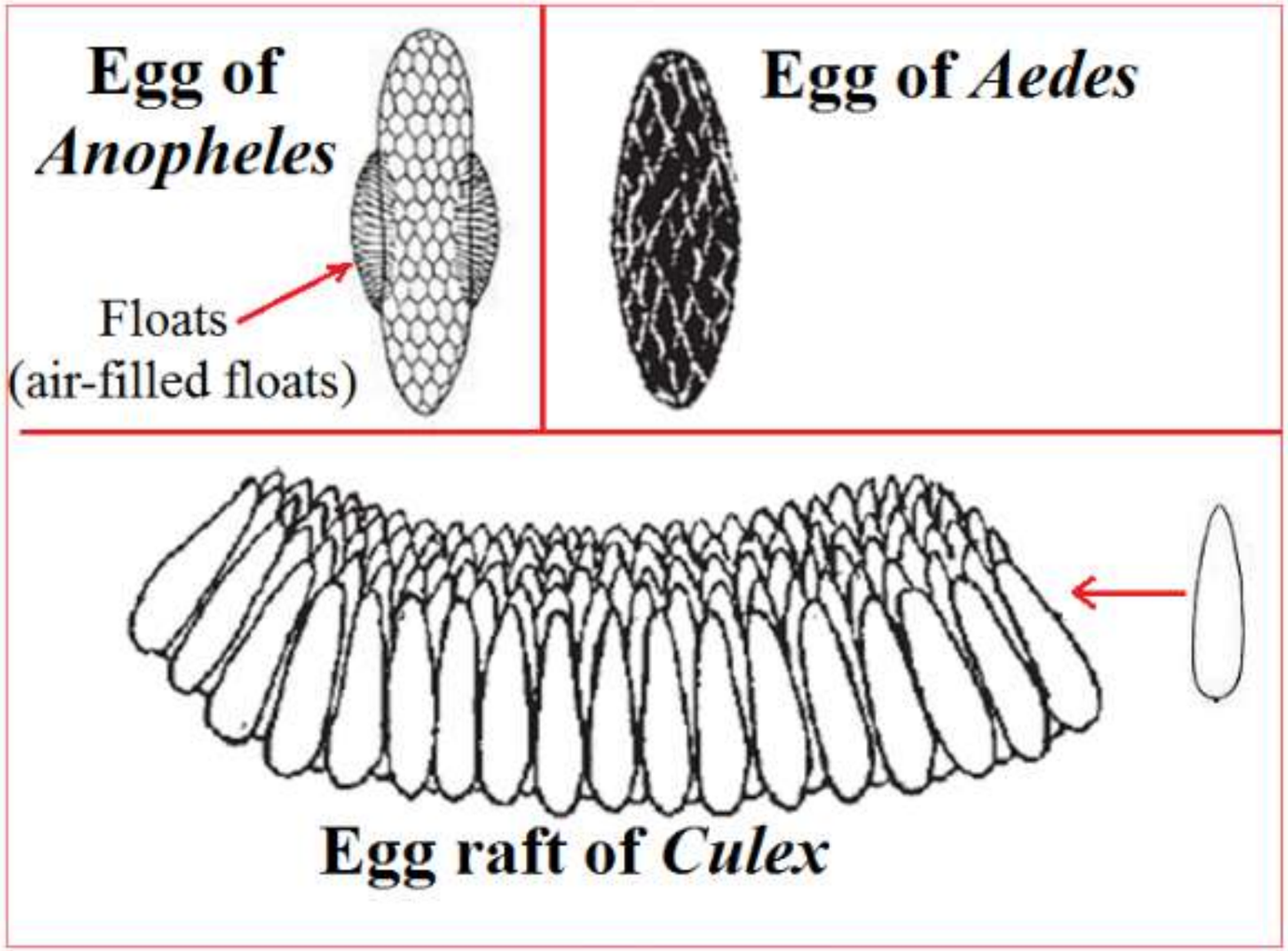


Culex male

CamScanner

Lab 2

Eggs

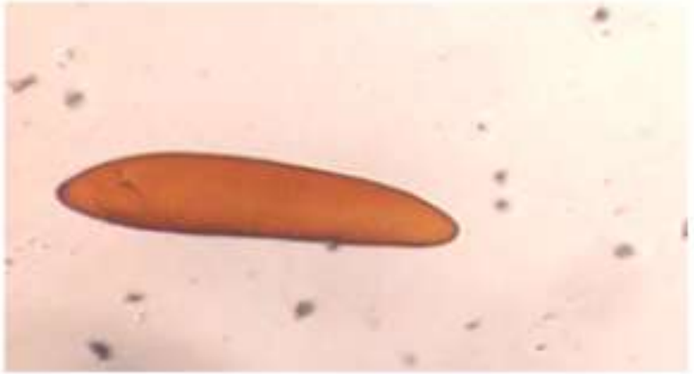


Size: 1 mm

CamScanner

Lab 2

Aedes



Aedes egg

Single, spindle in shape

Anopheles



Anopheles egg

Single, slipper in shape, brownish in color with air sac

Culex

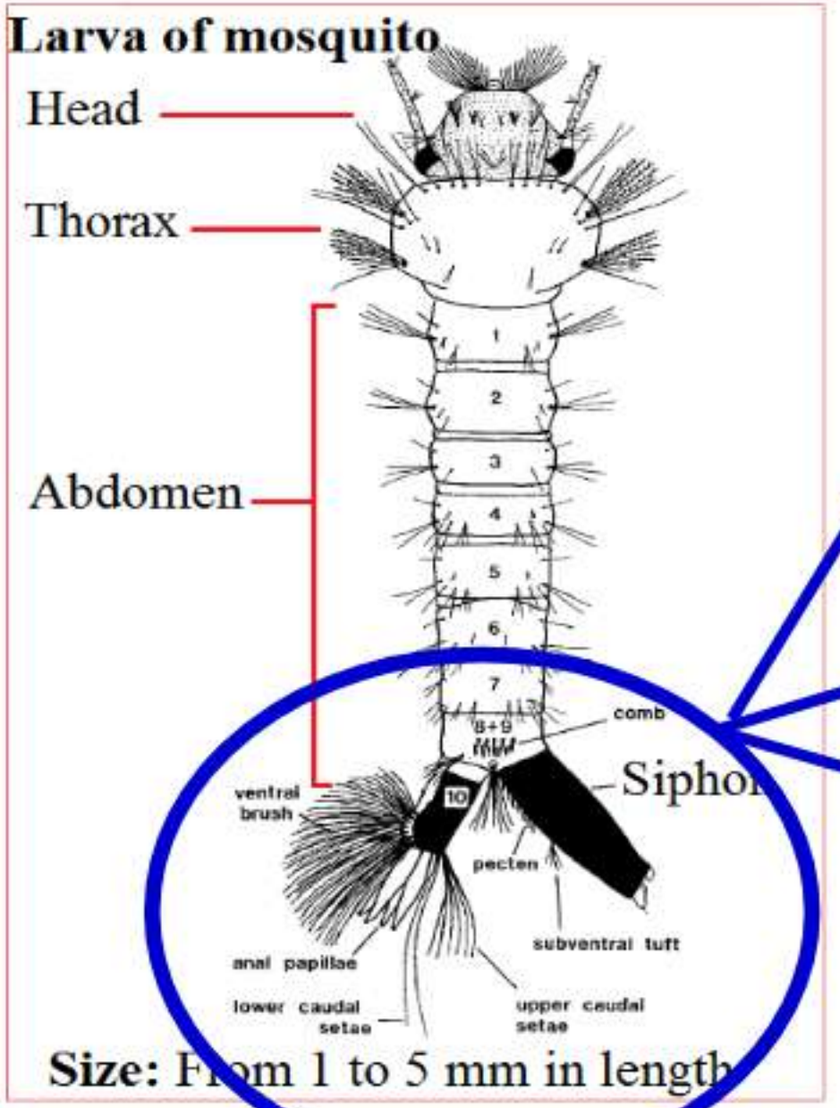


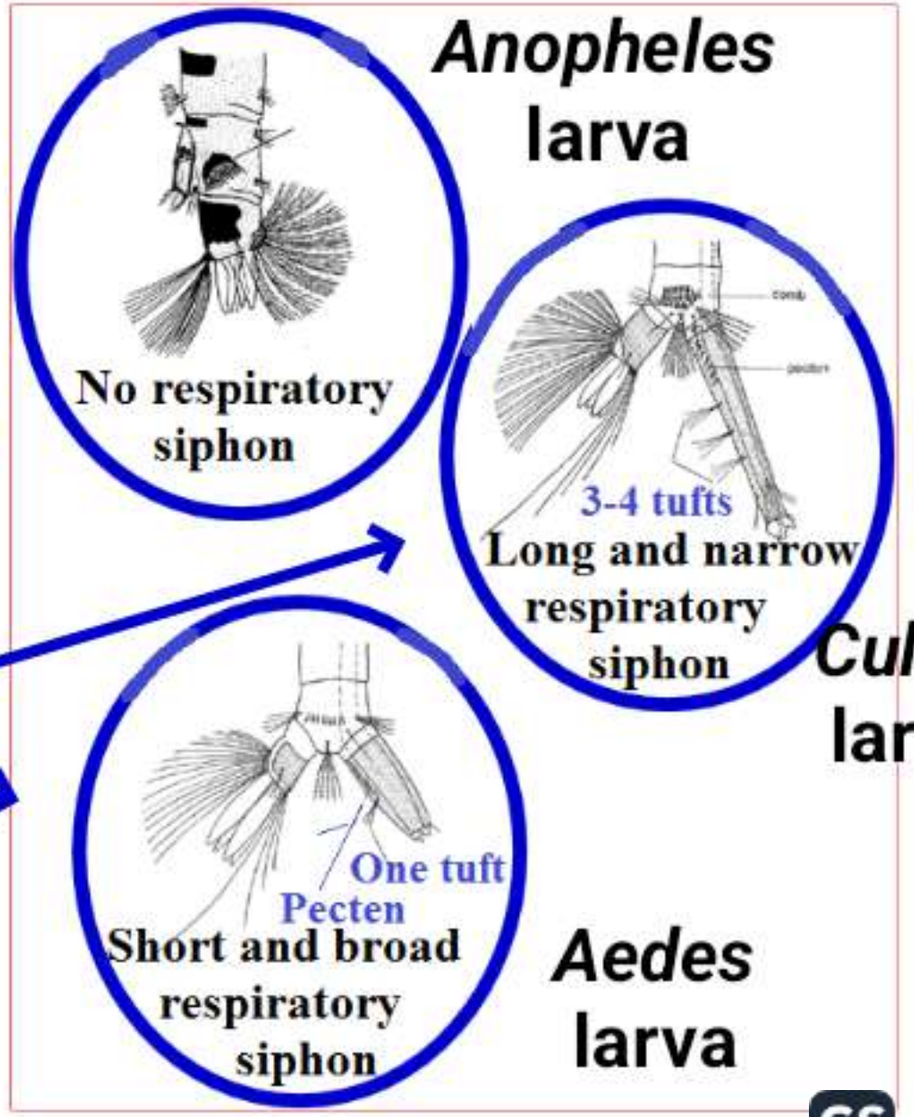
https://youtube.com/shorts/YURYsEMLkd4?si=hzLIuPgxXybz9bxq

CamScanner

Lab 2

Larvae





CamScanner

Lab 2

LARVAE of MOSQUITOES



AEDES



ANOPHELES

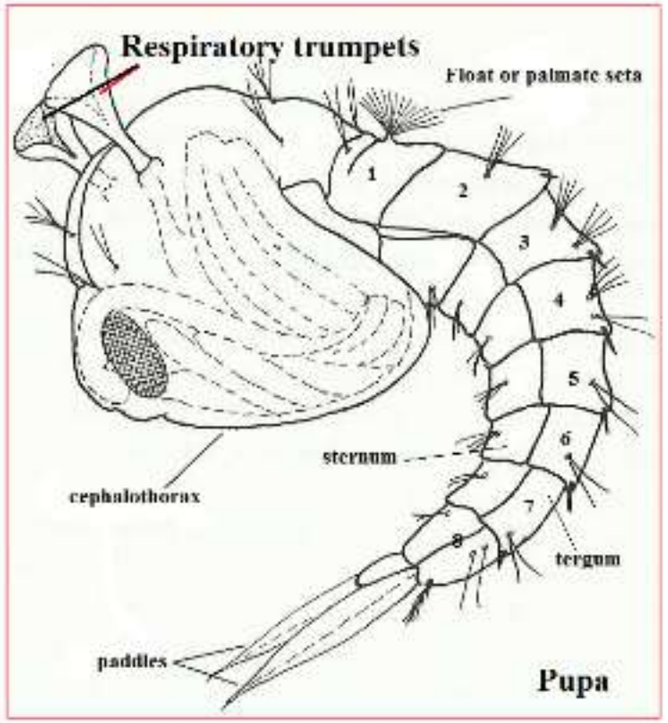


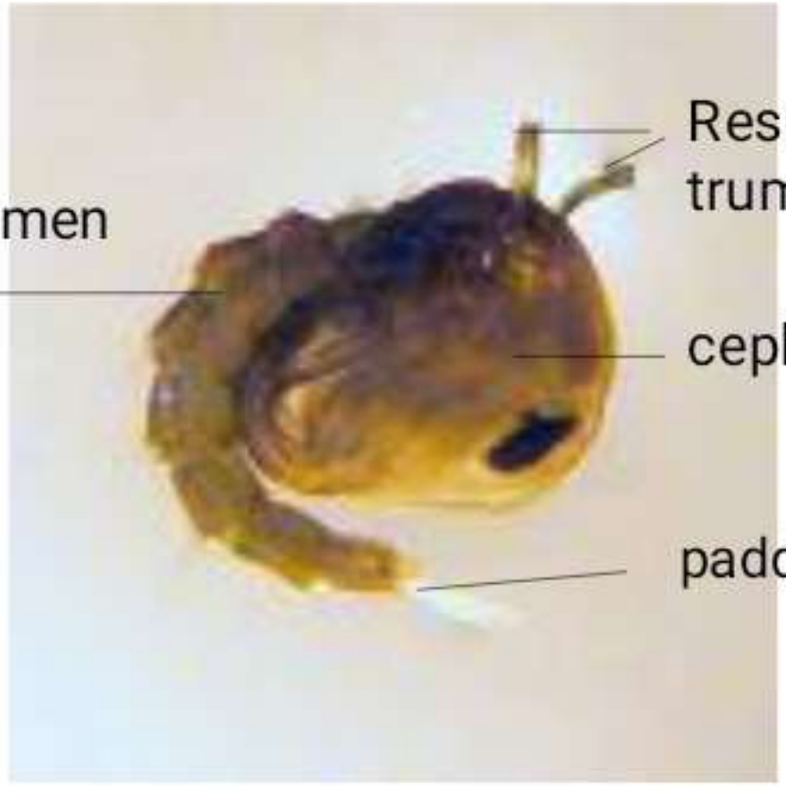
CULEX

CamScanner

Lab 2

Pupa of Mosquitoes





- Comma shaped
- Formed of cephalothorax
- Has 2 respiratory trumpets
- Ends in 1 pair of paddles

CamScanner

Virtual Microscope

- https://ain-shams.smartzoom.com/share/296b8be63d192b9b19a1ea6fabdbdd624a12d5eb5e861205b612bb3f48be1c3e
- https://ain-shams.smartzoom.com/share/e9cf00417ed7eda9a25ad117239f2bc62e8cbf9b5b18eb8c10d5c198c7ed9c69
- https://ain-shams.smartzoom.com/share/c6f25884296cc9545a48a99420c2dd0856daaf2dda496d4ea7cd213ef35fefcc
- https://ain-shams.smartzoom.com/share/854fc5750c1533cc4f72e967d8724a1211bde223c13418bb79f39bc3fdb3bc8f
- https://ain-shams.smartzoom.com/share/f3ec36f0de73f3369247d35bc8eff35a56e2dba008b7108c181ac35dba56939b
- https://ain-shams.smartzoom.com/share/13eb49cefc373431ded475f9e8e6f64ff2df016510d2eecd66b35d169f5109d5
- https://ain-shams.smartzoom.com/share/56ebbb5a9c1b53243ee7f56caeddf9d11fe01a99c582f359b4da32085b9f9bed

CamScanner

Lab 2

Section III

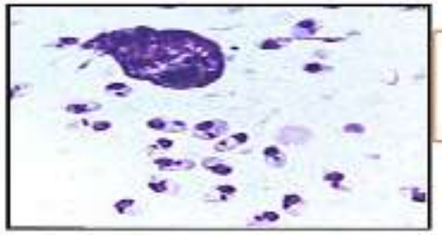
Trypanosoma spp.

CamScanner

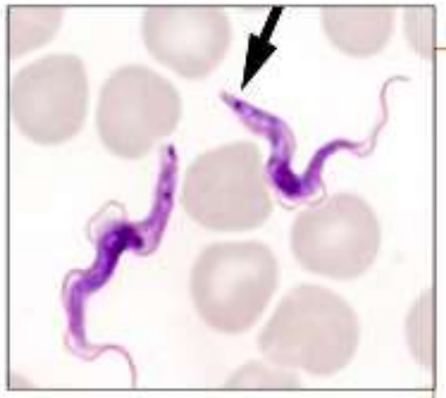
Lab 2

Protozoa

Haemoflagellates



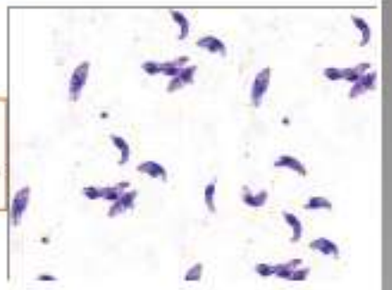
Leishmania spp.



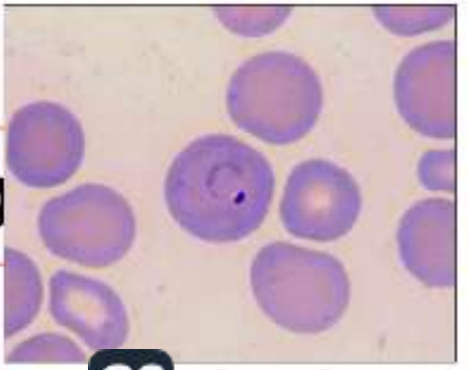
Trypanosoma spp.

Sporozoa

Toxoplasma spp.



Plasmodium sp



CamScanner

Lab 2

Trypanosoma spp.

***Trypanosoma brucei* (*T.b.gambiense* & *T.b. rhodesiense*)**

Disease: Sleeping sickness in Africa

Trypanosoma cruzi

Disease: Chagas disease in Latin America

CamScanner

Lab 2

Trypanosoma brucei
(*T.b.gambiense & T.b. rhodesiense*)
Or Polymorphic Trypanosomes
Or African trypanosomes
(1 slide)

Trypanosoma cruzi
Or Monomorphic trypanosomes
(2 slides)

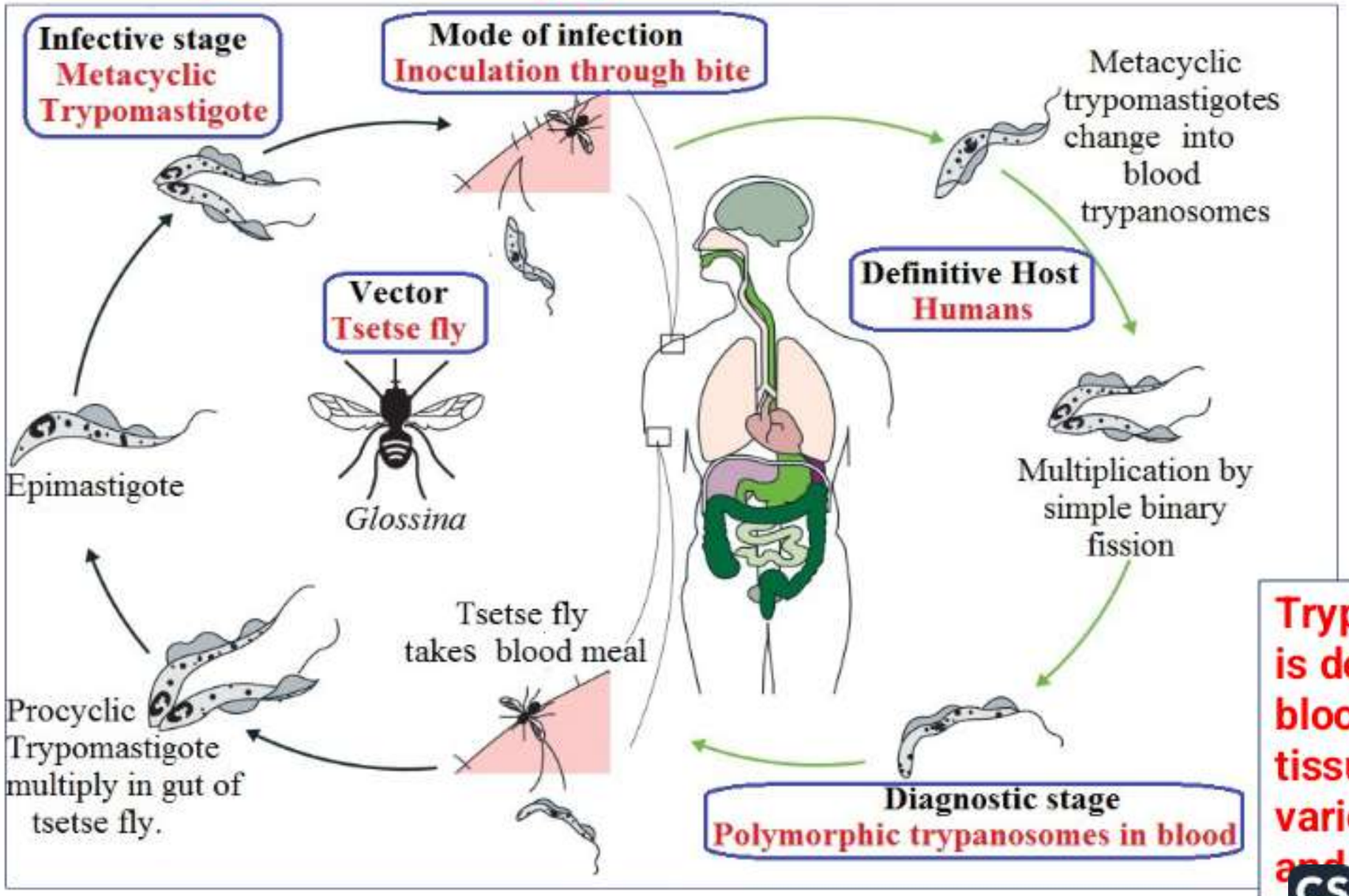
CamScanner

Lab 2

Life Cycle of *Trypanosoma brucei*



Glossina

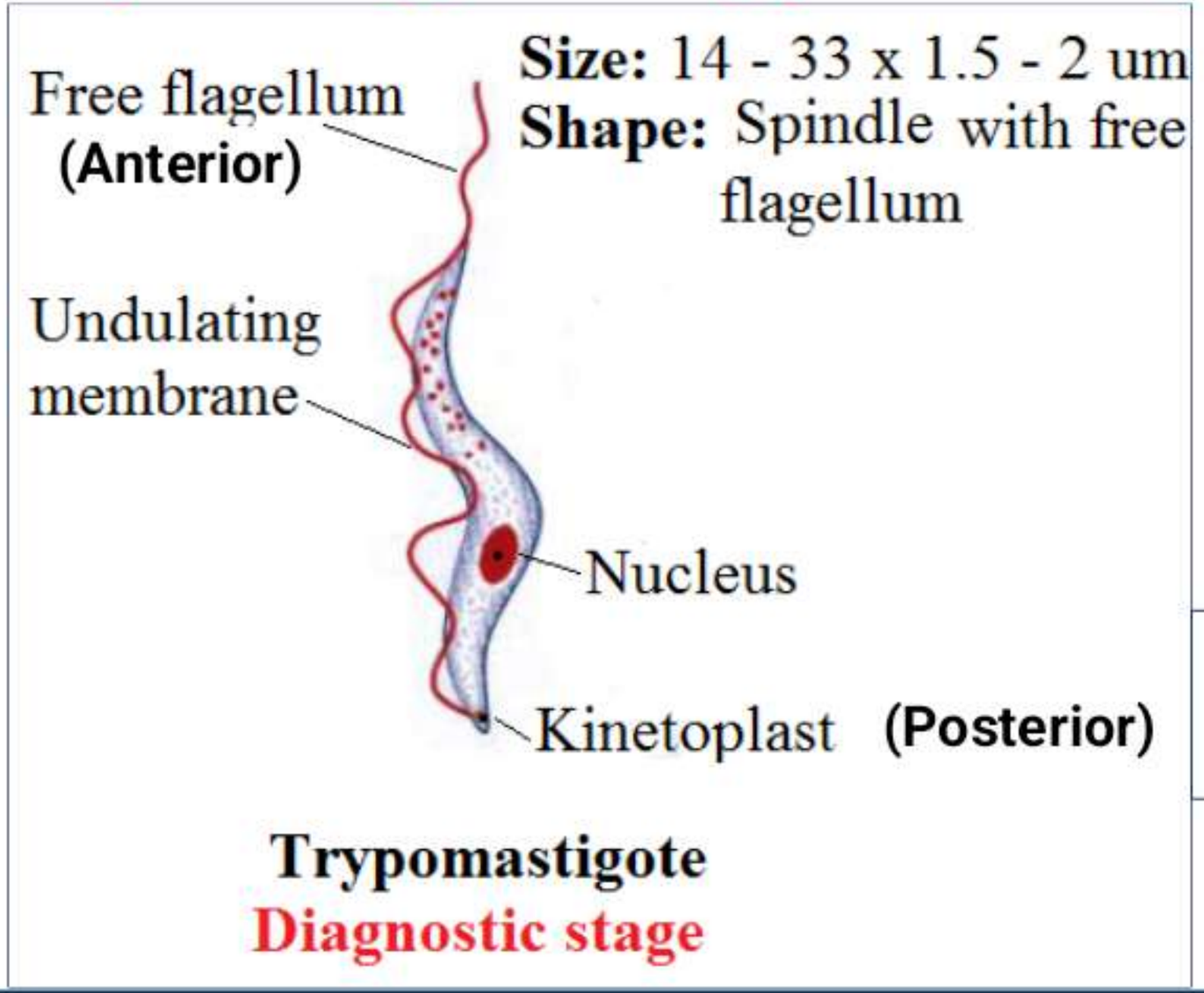


Trypomastigote is detected in blood, lymph, tissue space of various organs and CNS of vertebrate hosts

CamScanner

Lab 2

Morphology of Trypomastigote



- Extra erythrocytic stage
- Small kinetoplast

CamScanner

Lab 2

Trypanosoma brucei in blood film

The Polymorphoic Trypomastigates exist in three forms (Polymorphic).



Short stumpy form (15 um in length)
(Sluggish without a free flagellum)

Intermediate form (20 um in length)
(With short free flagellum)

Long slender form (30 um in length)
(Active with long free flagellum)

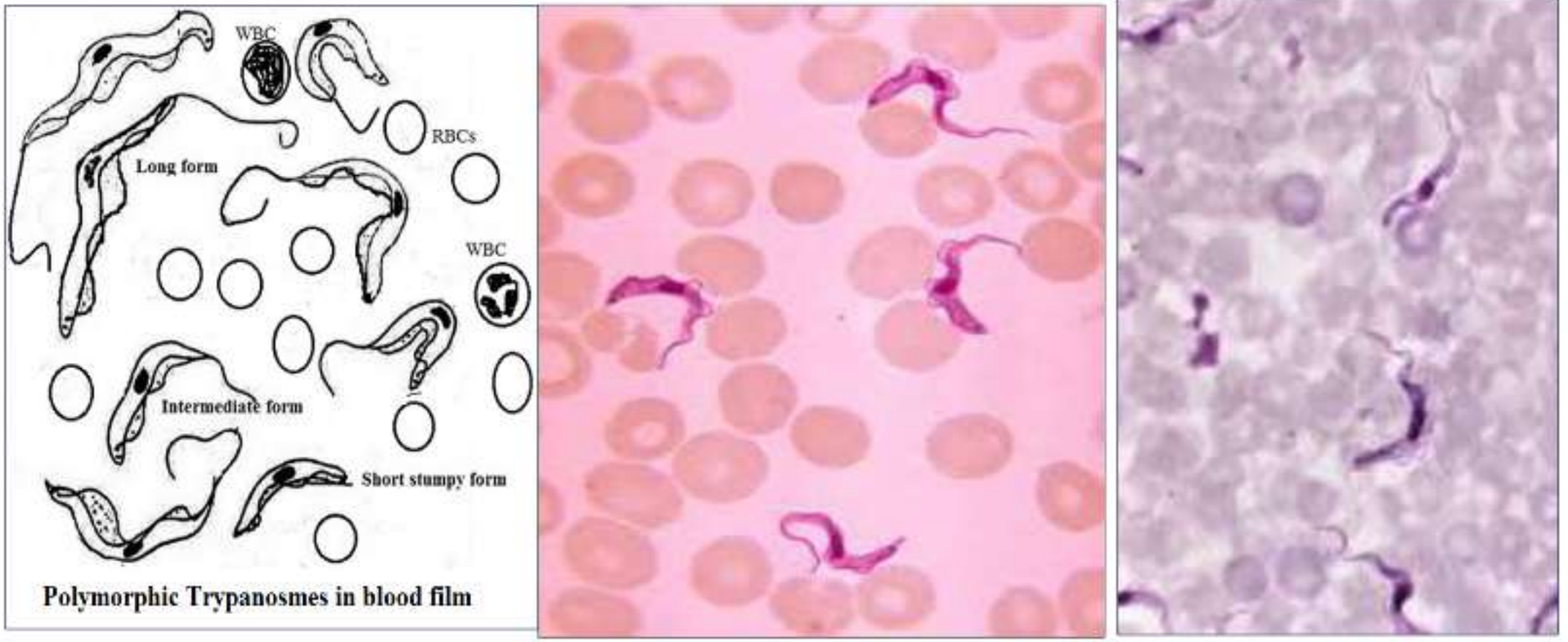
CamScanner

https://ain-shams.smartzoom.com/share/9241e0f1aea69a1bcadc94032812fbe3e1ebffbd52d67c29ef5b60ca471b4845

CamScanner

Lab 2

Trypanosoma brucei



Polymorphic Trypanosomes in blood film

CamScanner

Lab 2

Glossina (Tse tse fly)

Adult

Brown or black in colour



with anterior protruding non-retractile mouthparts

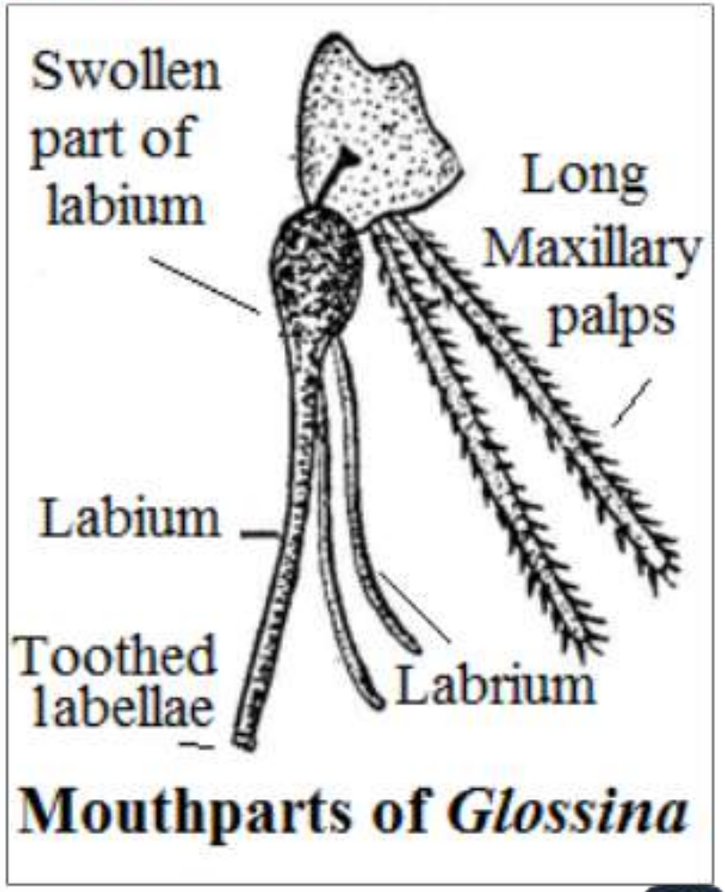
CamScanner

Lab 2

Glossina

Mouth parts





Mouthparts of *Glossina*

CamScanner

Lab
2

Trypanosoma cruzi

=

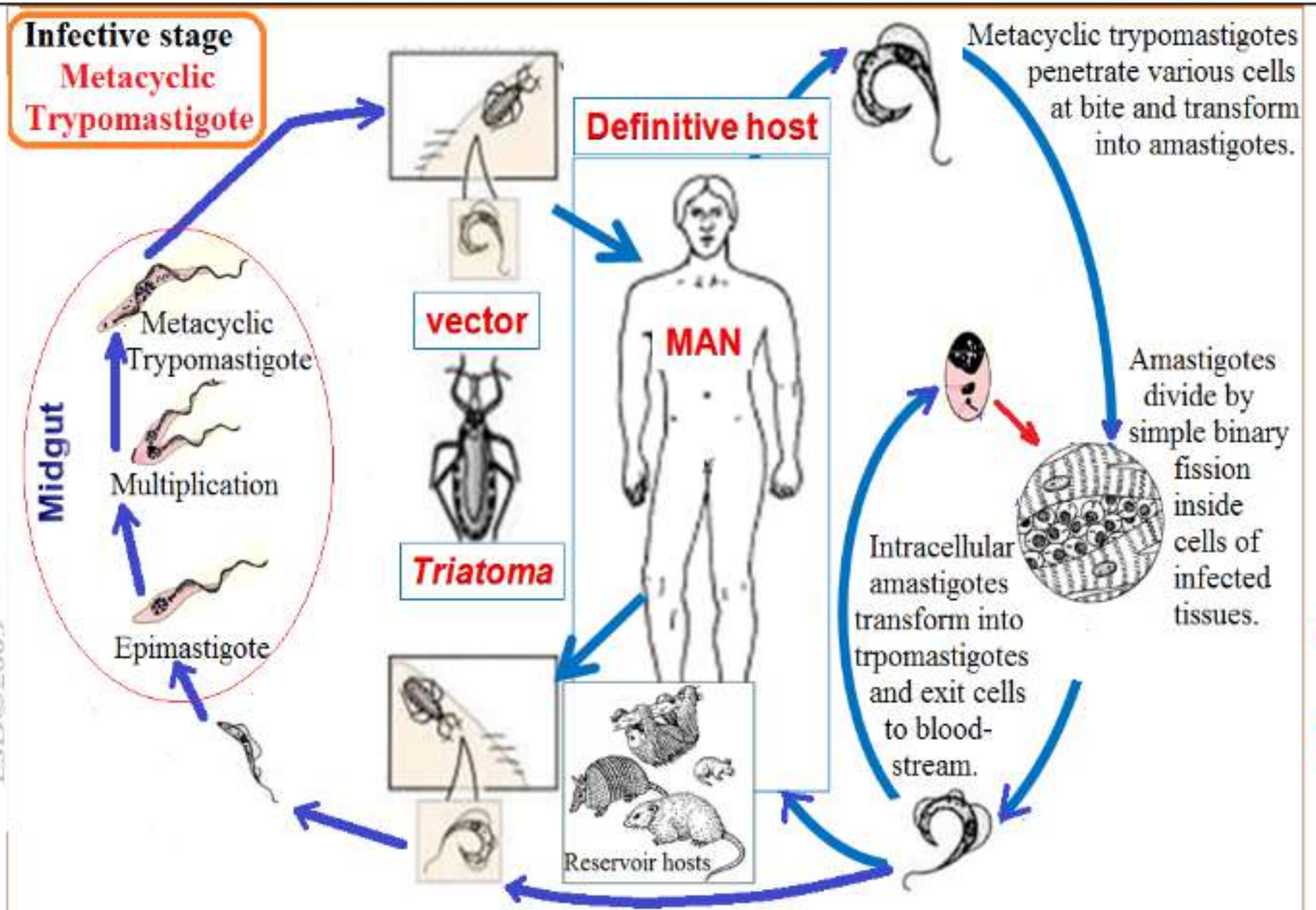
Monomorphic Trypanosomes

CamScanner

Lab 2

Life Cycle of Monomorphic Trypanosomes





The parasite exists in two forms in vertebrate hosts: Amastigote inside cells & Trypomastigote In blood

CamScanner

Lab 2

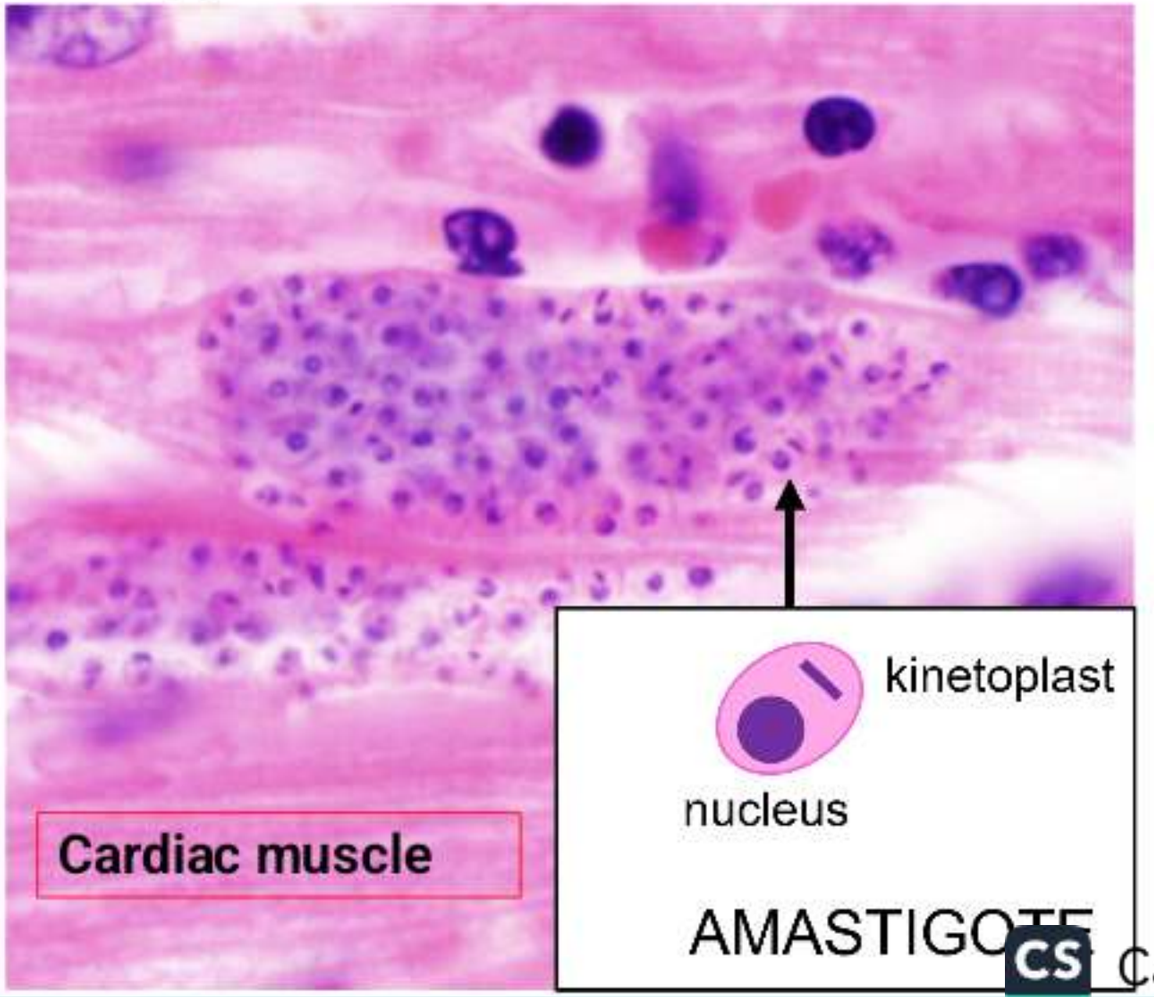
Monomorphic Trypanosomes

Amastigote forms



- Round or oval
- 2 – 6 um
- NO free flagellum

Found intracellularly in heart and skeletal muscles, brain, nerve ganglia and cells of GIT

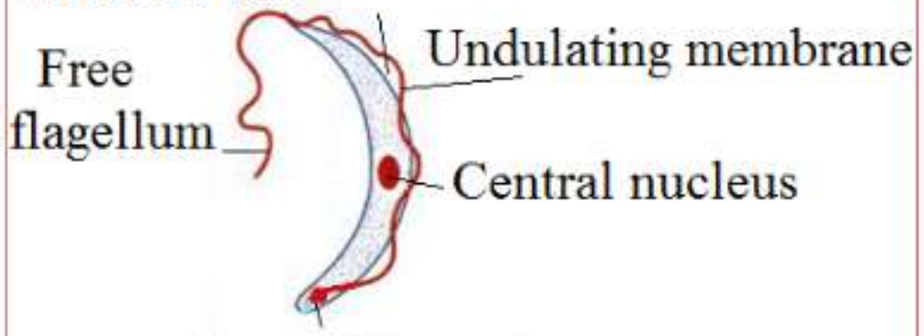


CamScanner

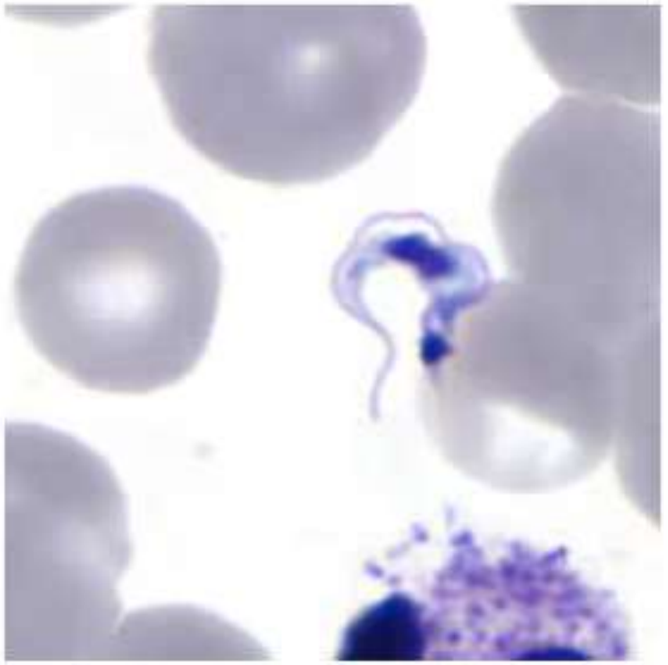
Lab 2

Monomorphic Trypanosomes

C- or U-shaped
Size: 20 um



Monomorphic Trypomastigote of *Trypanosoma cruzi*



- **Extra erythrocytic stage**
- **Prominent kinetoplast**

Monomorphic Trypanosomes in blood film (very few in number)

CamScanner

- https://ain-shams.smartzoom.com/share/63eac687a86993649d5b231c3096ccb5e594c82084d0b9406eb01b99e959cfe0
- https://ain-shams.smartzoom.com/share/folder/454675399668e18eef67744e006dfd24c053c7c8df6eaef9cc517dbdf5ce4ddf/i19455/

CamScanner

To wrap up

CamScanner

Lab 2

Mouth parts of mosquitoes



Male culex



Male Anopheles



Female culex

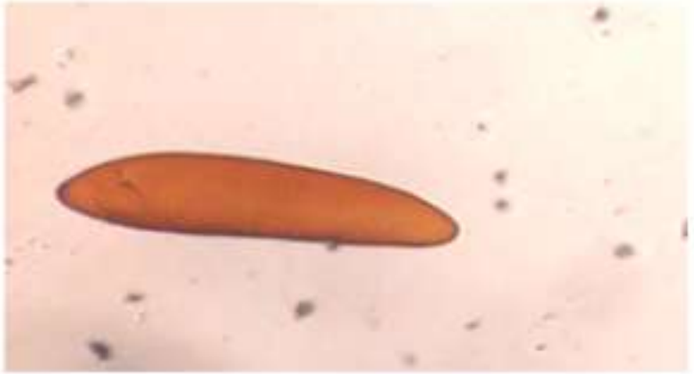


Female Anopheles

CamScanner

Lab 2

Aedes



Aedes egg

Single, spindle in shape

Anopheles



Anopheles egg

Single, slipper in shape, brownish in color with air sac

Culex



CamScanner

Lab 2

LARVAE of MOSQUITOES



AEDES

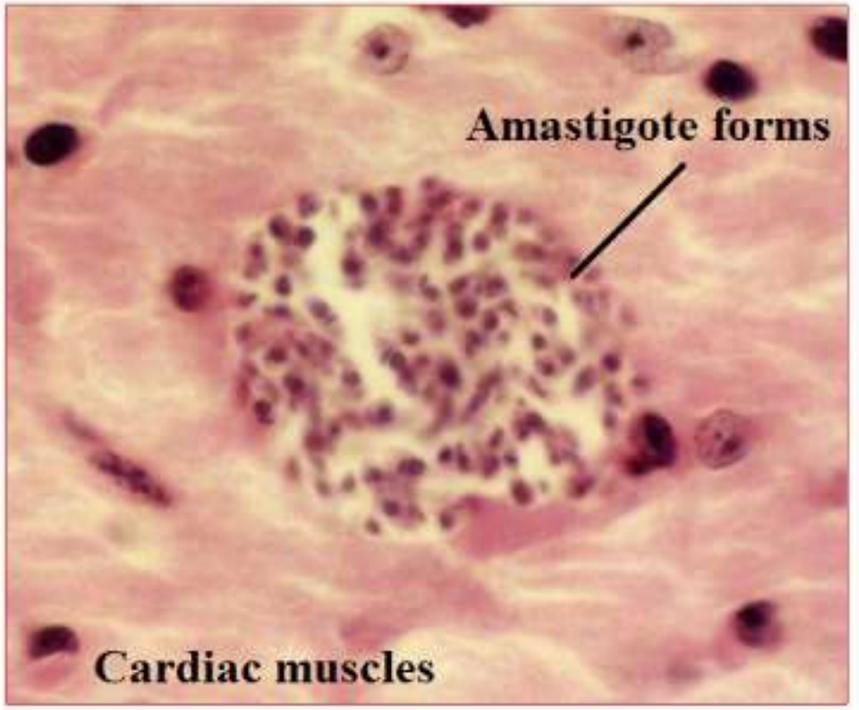


ANOPHELES



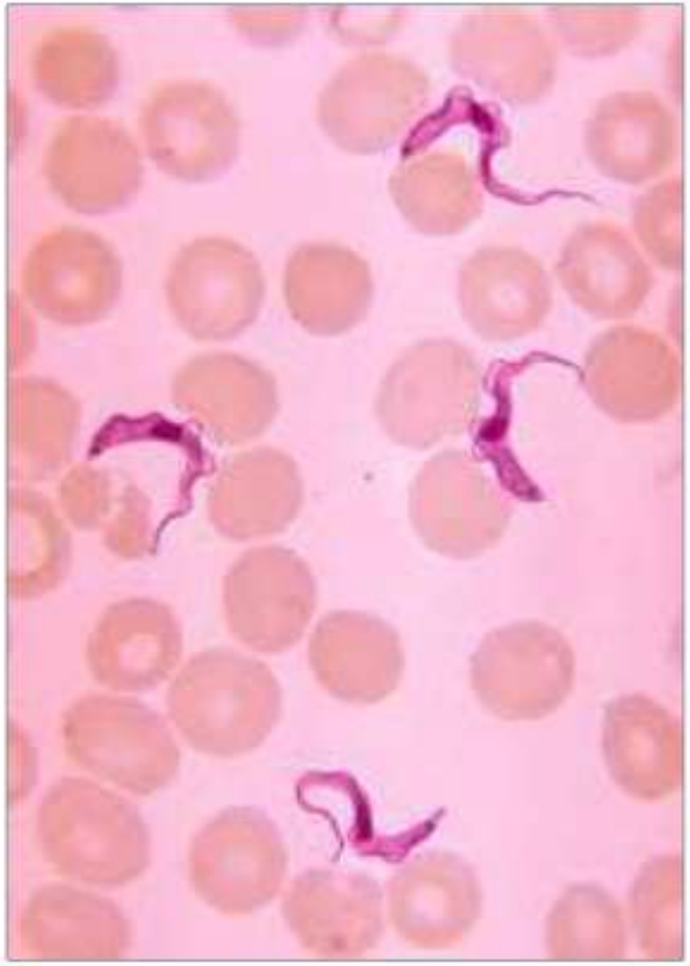
CULEX

CamScanner

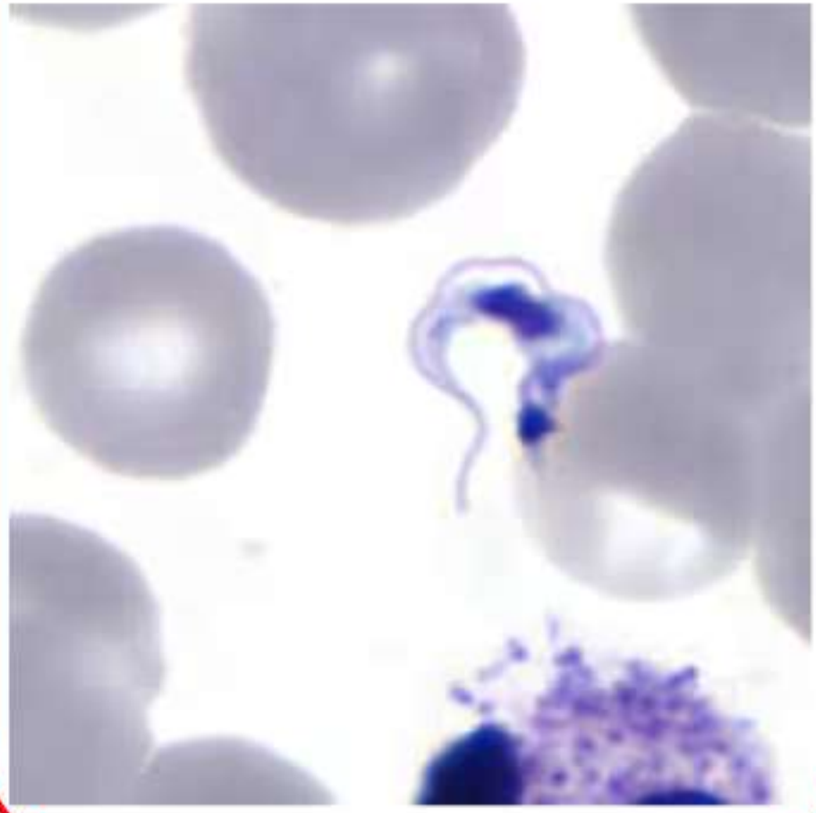


Amastigotes of *Trypanosoma cruzi* in heart muscles

CamScanner



Polymorphic Trypanosomes
In blood film



Monomorphic Trypanosomes
In blood film

CamScanner



Mouth parts of *Glossina*

CamScanner

Section IV







CamScanner

Self assessment

- Write your answer on a paper
  - Keep it on your log book

CamScanner

Lab 2

Choose the correct answer:

The arrowed structure is:

A. Siphon

B. Respiratory spiracles.

C. Respiratory trumpets.

D. Proboscis.



Answer: A

CamScanner

Lab 2

Choose the correct answer:

The vector of *Plasmodium falciparum* is:



A



B



C



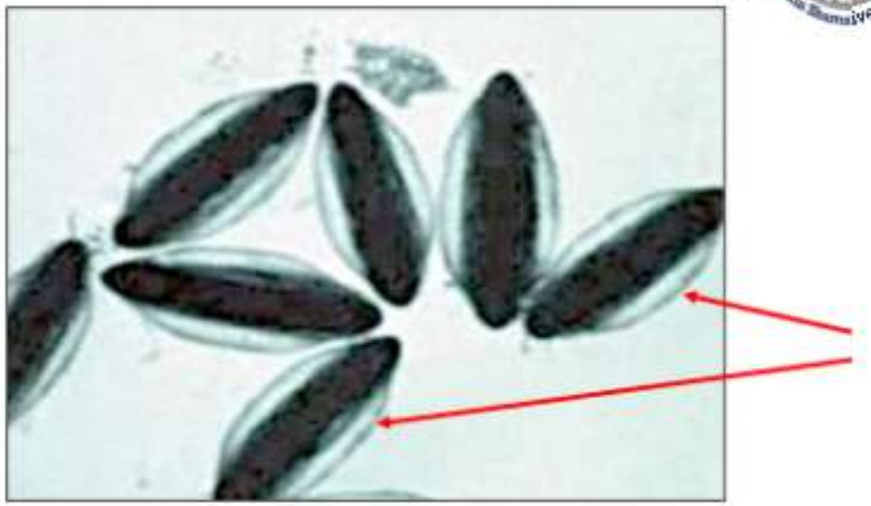
D

Answer: D

CamScanner

Lab 2

1. Identify stage
2. Label the arrowed structures



1. Eggs of *Anopheles*
2. Air cells/ Floaters

https://www.cdc.gov/dpdx/malaria/index.html

CamScanner

Lab 2

Interpret the following statement:

The use of non-volatile oils to combat aquatic stages of mosquitoes.

Because they prevent the aquatic stages of mosquitoes from respiration, as the oils form a film on the surface of the water. Also act as respiratory poison which suffocate and poison aquatic forms (eggs, larvae, pupae) by inhalation.

CamScanner

Lab 2

Interpret the following statement:

The use of *Gambusia affinis* fish as a natural enemy of mosquitoes.

As the fish predates on aquatic stages of mosquitoes. It prefers the larvae and pupae of mosquitoes as a food.

CamScanner

Lab 2

Interpret the following statement:

Culex pupa are not affected by Paris green.

The Paris green is a stomach poison, while the pupa does not feed so it is not affected by Paris green.

CamScanner

The following sign is common in this parasitic infection:

A- Sleeping sickness

B- Visceral leishmaniasis

C- Benign tertian malaria

D- Chagas' disease

E- Lymphatic filariasis



CamScanner

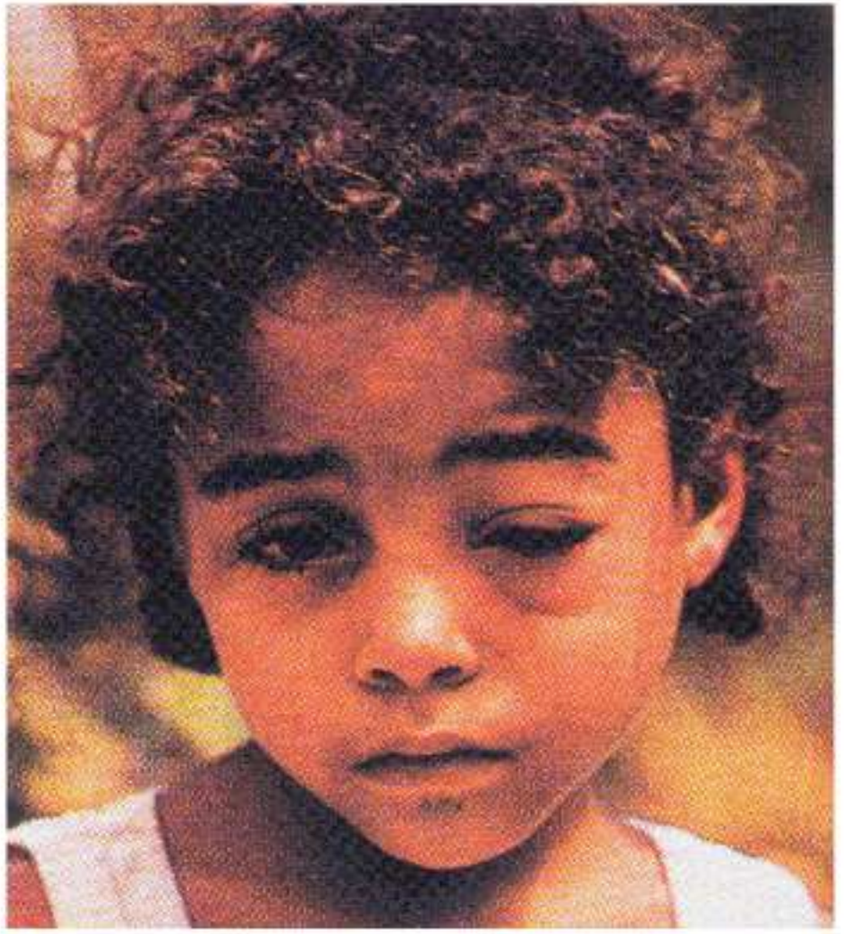
Chancre is the primary lesion found in:

A- Sleeping sickness
B- Kala azar
C- malignant malaria
D- Chagas' disease
E- Lymphatic filariasis

CamScanner

Parasitic infection causing this sign is:

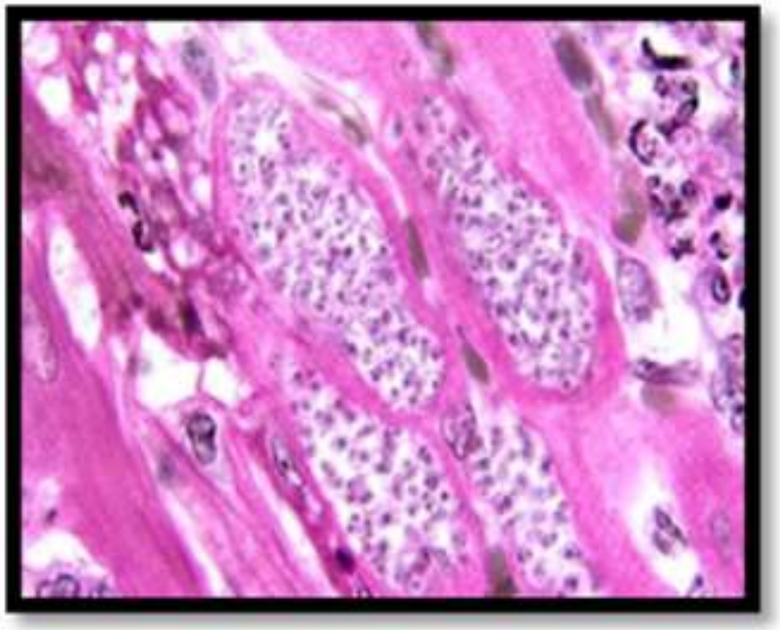
A- Visceral leishmaniasis
B- Chagas' disease
C- Lymphatic filariasis
D- Sleeping sickness
E- Babesiosis



CamScanner

Detection of this parasitic stage indicates:

a. African trypanosomiasis
b. American trypanosomiasis
c. Visceral leishmaniasis
d. Toxoplasmosis
e. Malaria



CamScanner

Lab 2

Choose the correct answer:

In chronic Chagas' disease, the main lesions are in:

A- Digestive and respiratory tracts

B- Heart and liver

C- Heart and digestive tract

D- Liver and spleen

E- CNS

Answer: C

CamScanner

Thank you for your attention!



CamScanner

Parasitology Department

Blood Module

2024/2025

Lab 3

CamScanner

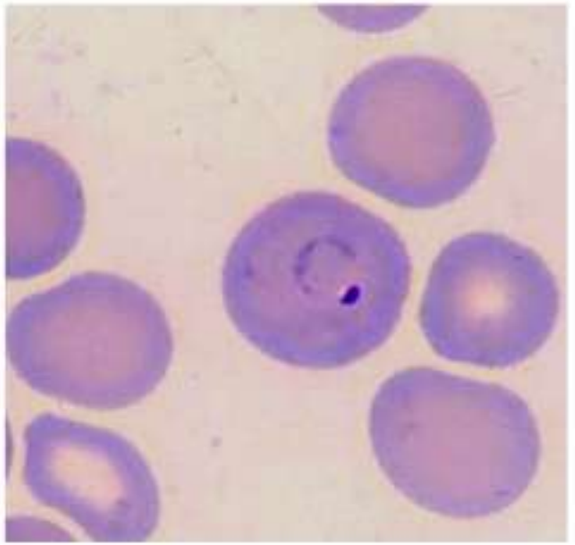
Lab
3

Prelab Assessment

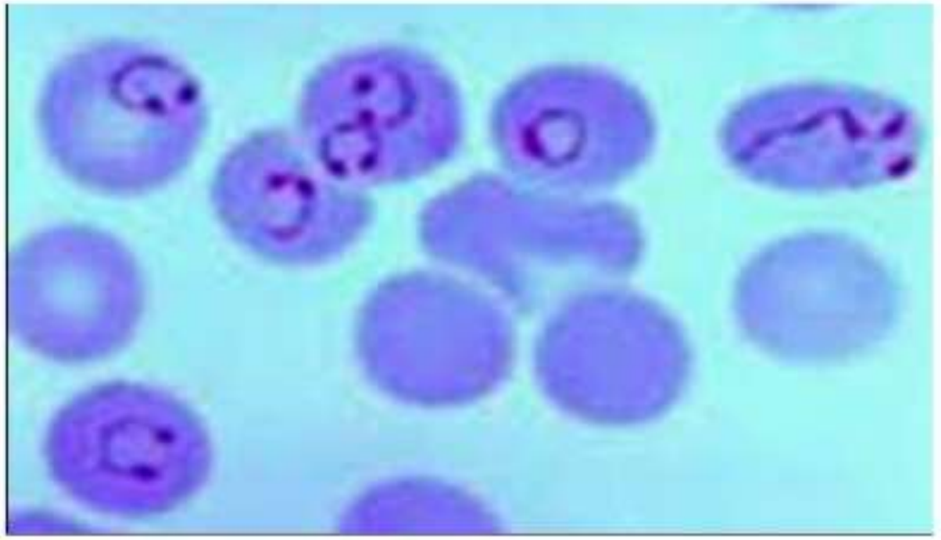
CamScanner

Lab 3

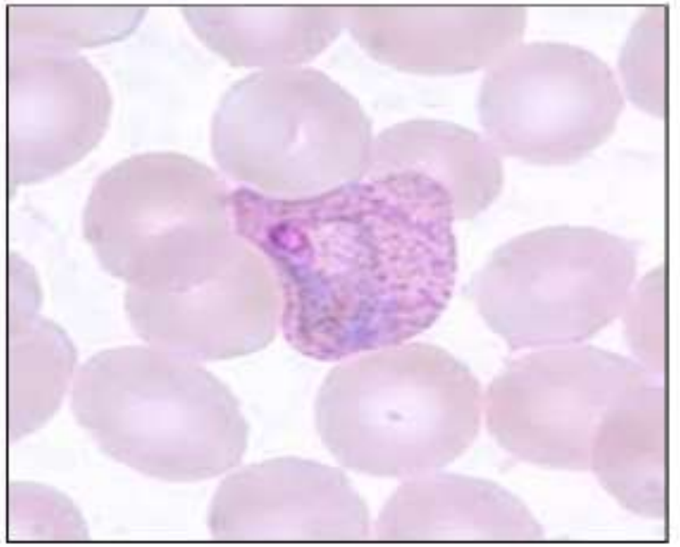
Identify



A



B

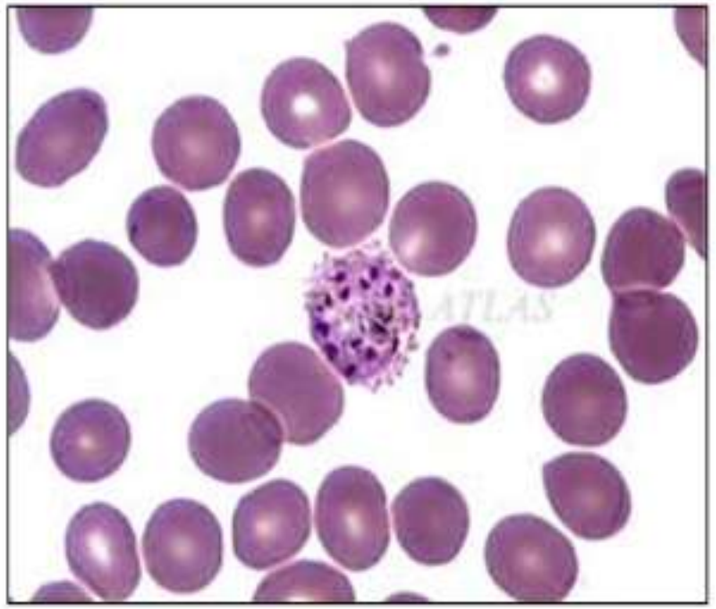


C

CamScanner

Lab 3

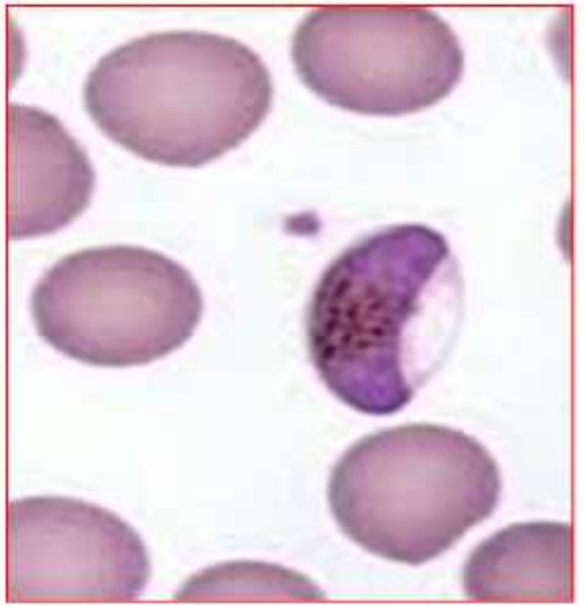
Identify



A



B



C

CamScanner

Lab 3

Identify



Female *Culex*



Female *Anopheles*



Male *Culex*



Male *Anopheles*

CamScanner

Lab 3

Identify



Egg raft of *Culex*



Eggs of *Aedes*



Eggs of *Anopheles*

CamScanner

Lab 3

Identify



Long narrow thin Siphon

Larva of *Culex*



Short broad thick Siphon

Larva of *Aedes*



NO Siphon

Larva of *Anopheles*

CamScanner

Lab 3

Identify



Pupa of mosquitoes

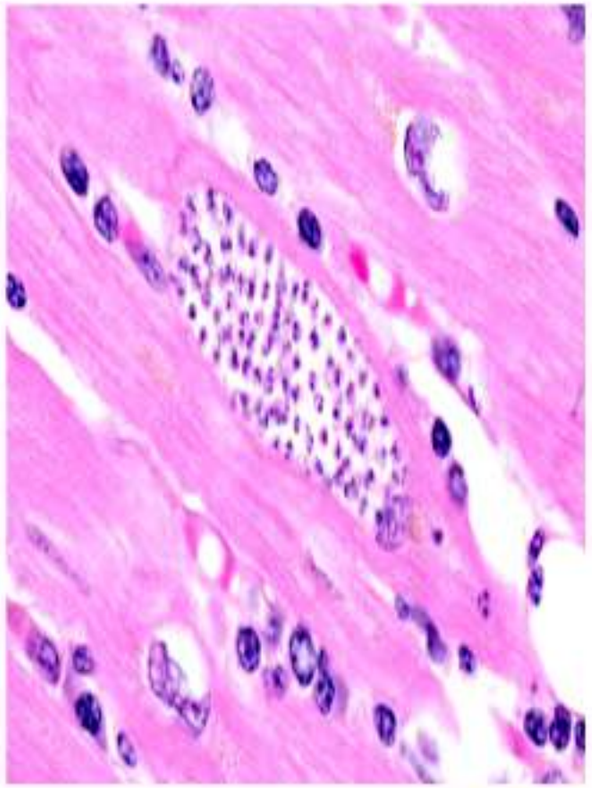
CamScanner

Identify

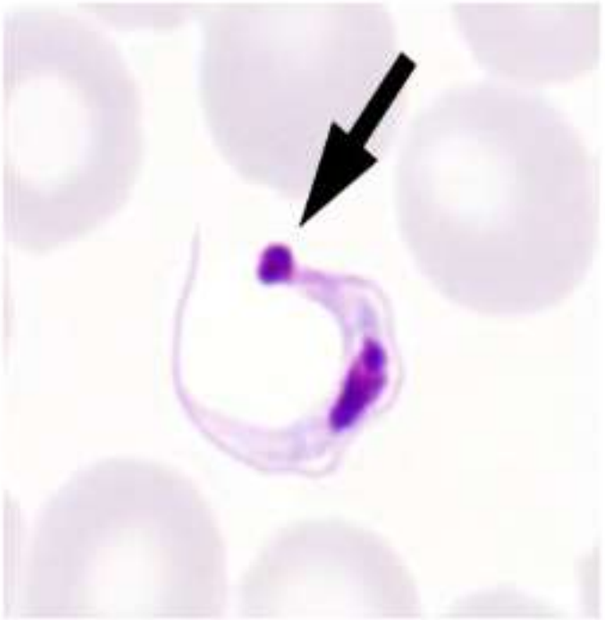


CamScanner

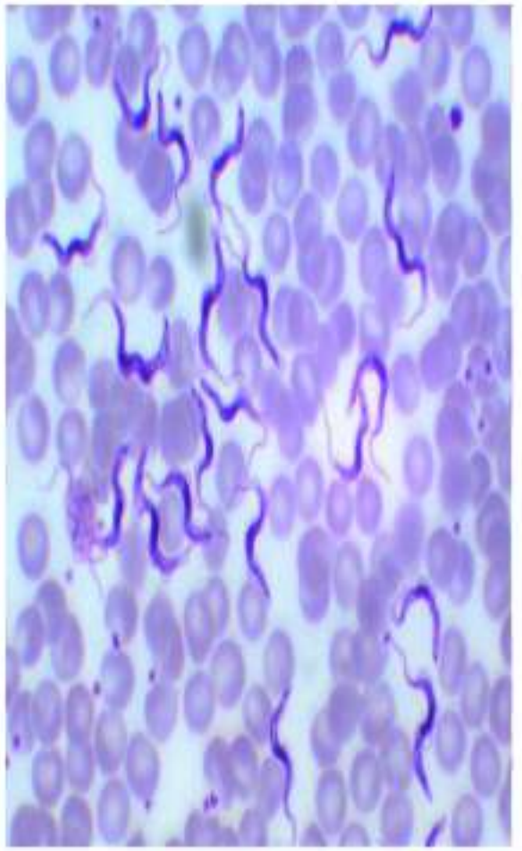
Identify



Amastigote form *of Trypanosoma cruzi* in heart muscle



Monomorphic *Trypanosoma* in Blood



Polymorphic *Trypanosoma* in Blood

CamScanner

Blood, lymphatics and mononuclear phagocytic system parasites

I- Helminths

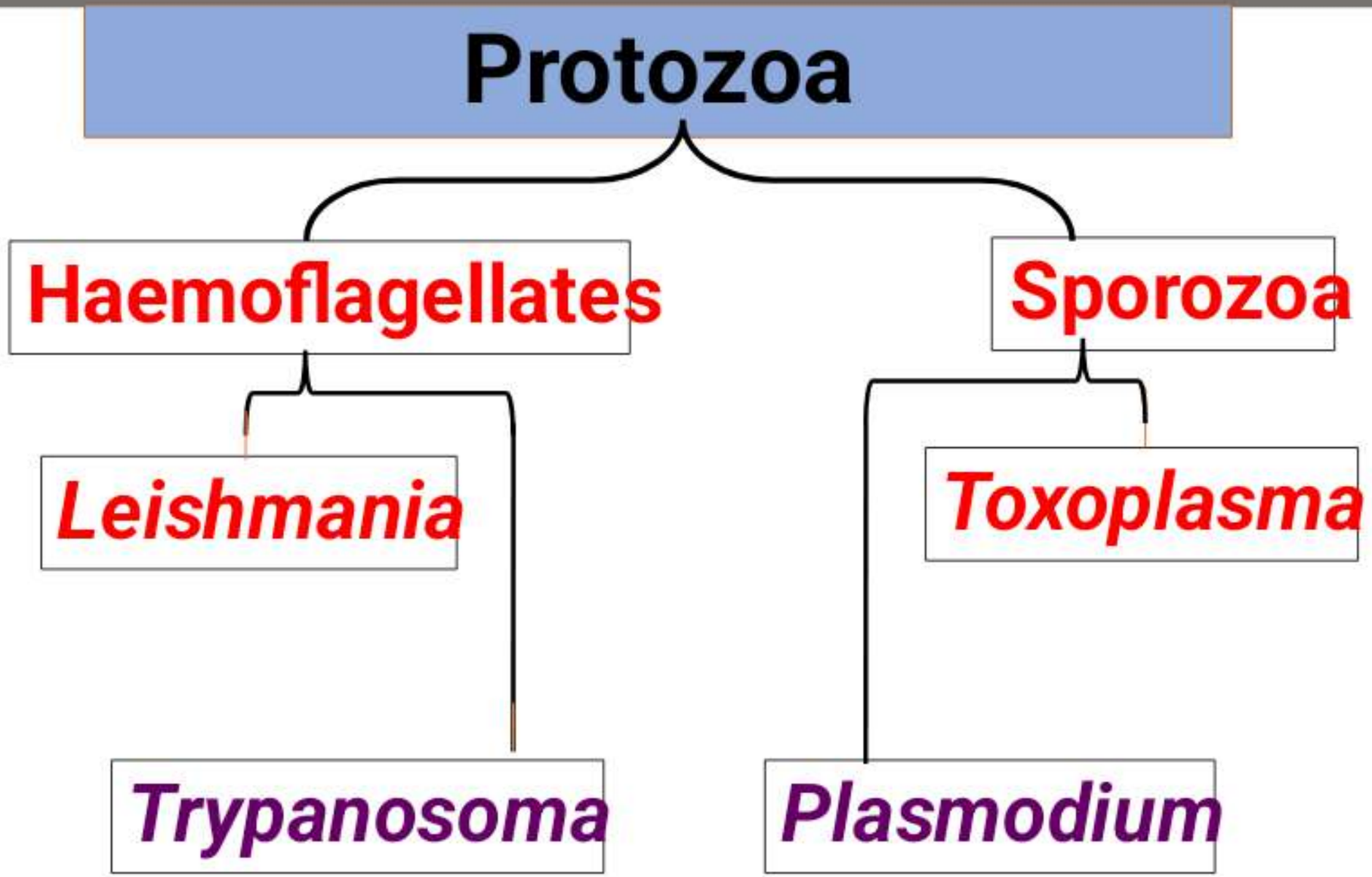
- Nematodes in lymphatic system:

Wuchereria bancrofti

II- Protozoa

- Sporozoa
- Haemo- flagellates

CamScanner


Protozoa
Haemoflagellates
Sporozoa
Leishmania
Toxoplasma
Trypanosoma
Plasmodium

CamScanner

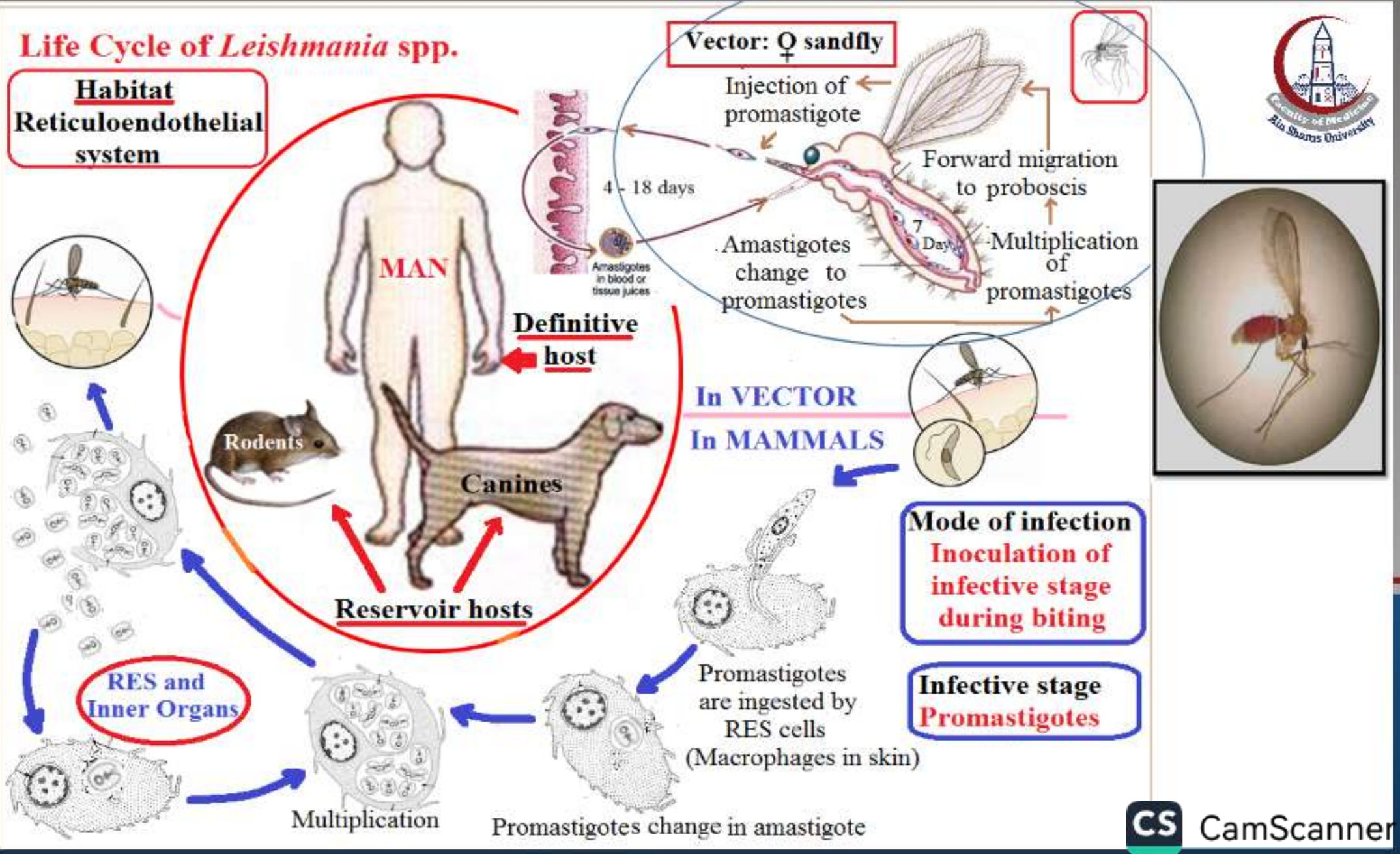
Leishmania spp.

A haemoflagellated protozoal parasite

A parasite of the reticuloendothelial (RES) system.
Causing cutaneous, muco-cutaneous and visceral Leishmaniasis.

CamScanner

Life Cycle of *Leishmania* spp.



CamScanner

Life cycle of *Leishmania sp.* in the sand fly

Images © Shortt H, Swaminath C. Indian Journal of Medical Research 1928;15:827–36; Lane R; Greenwood D; The Royal Society and Killick-Kendrick R; Killick-Kendrick R.

Animation © The Wellcome Trust.



click to play

CamScanner

Sand fly

Female



- Sandy in colour.
- Large eyes.
- Humpy thorax.
- Long, slender legs
- Long, lanceolate wings with pointed tips.

Male



CamScanner

- https://ain-shams.smartzoom.com/share/3308b3ea5fde45d9b339857a18ace5f704db4c2a290aedc09806c6962eb50c74
- https://ain-shams.smartzoom.com/share/fb504c18270c282fa98425dd6317e960c74307ff89634d1f26de094ddf07ad00

CamScanner

Morphology

The parasite exists in two morphological forms

Amastigote form

- Within RES in vertebrate hosts (intracellular)

Promastigote form

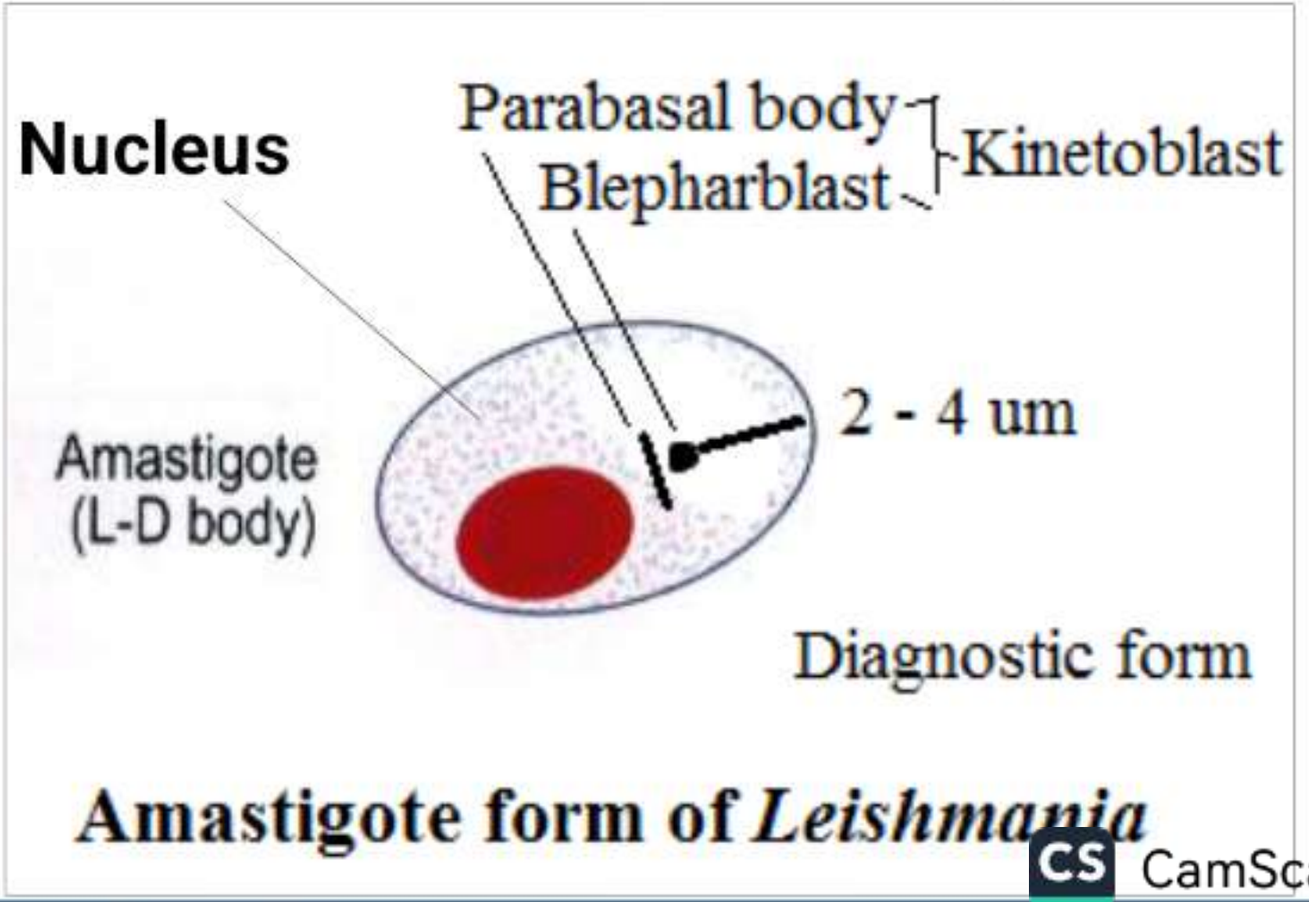
- In vectors
- In culture

CamScanner

Morphology

Amastigote form (Diagnostic stage in smear)

- Oval or rounded
- 5 µm
- Central nucleus
- Kinetoplast to one side (rod-like parabasal body and a dot-like blepharoplast)

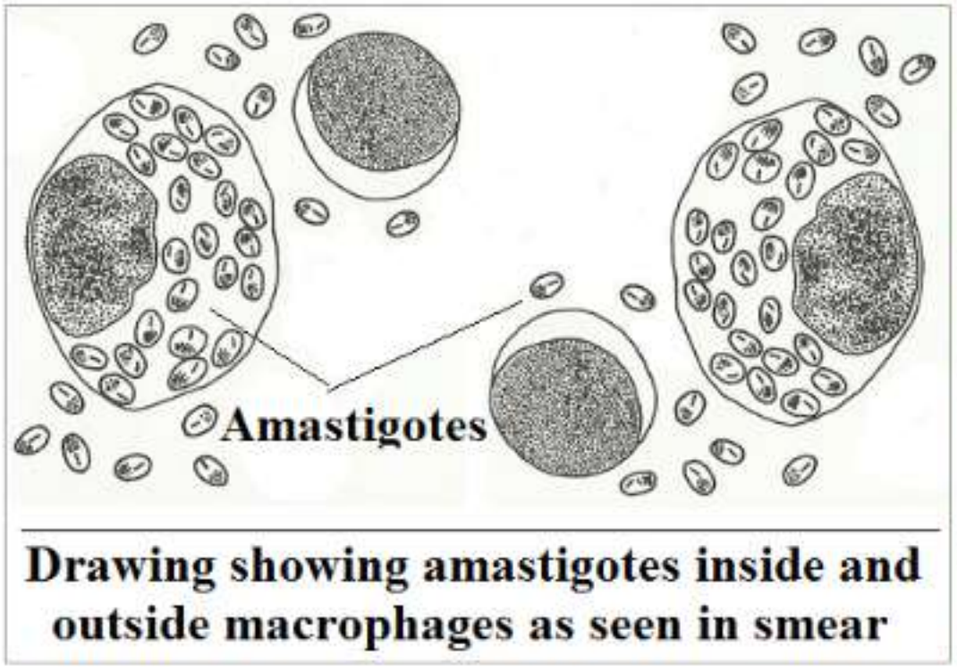


Amastigote form of *Leishmania*

CamScanner

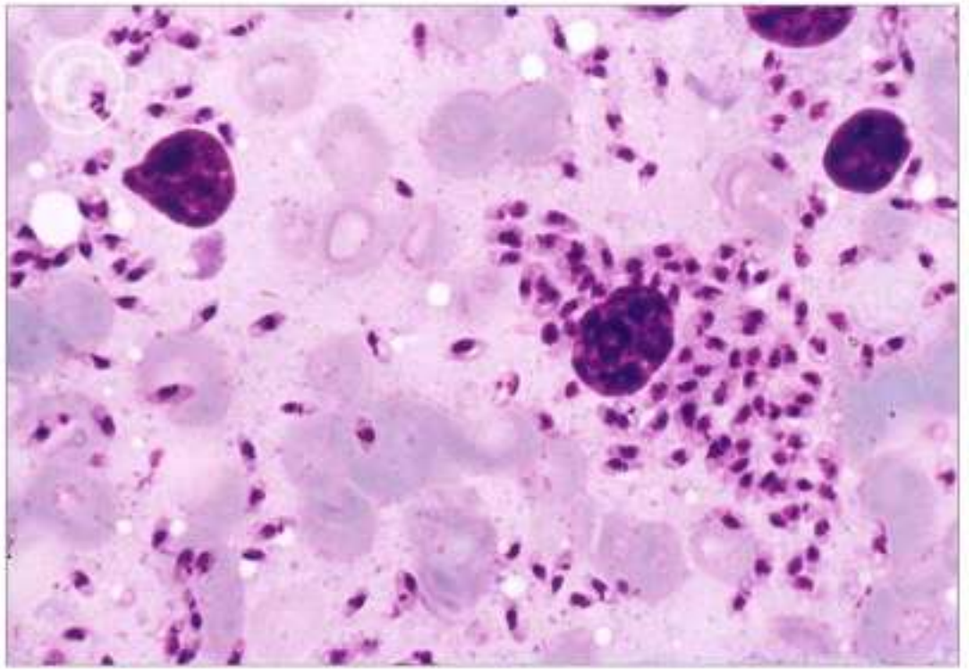
Morphology

Amastigote forms of *Leishmania* spp. in smear



Drawing showing amastigotes inside and outside macrophages as seen in smear

Diagram



Light Microscope

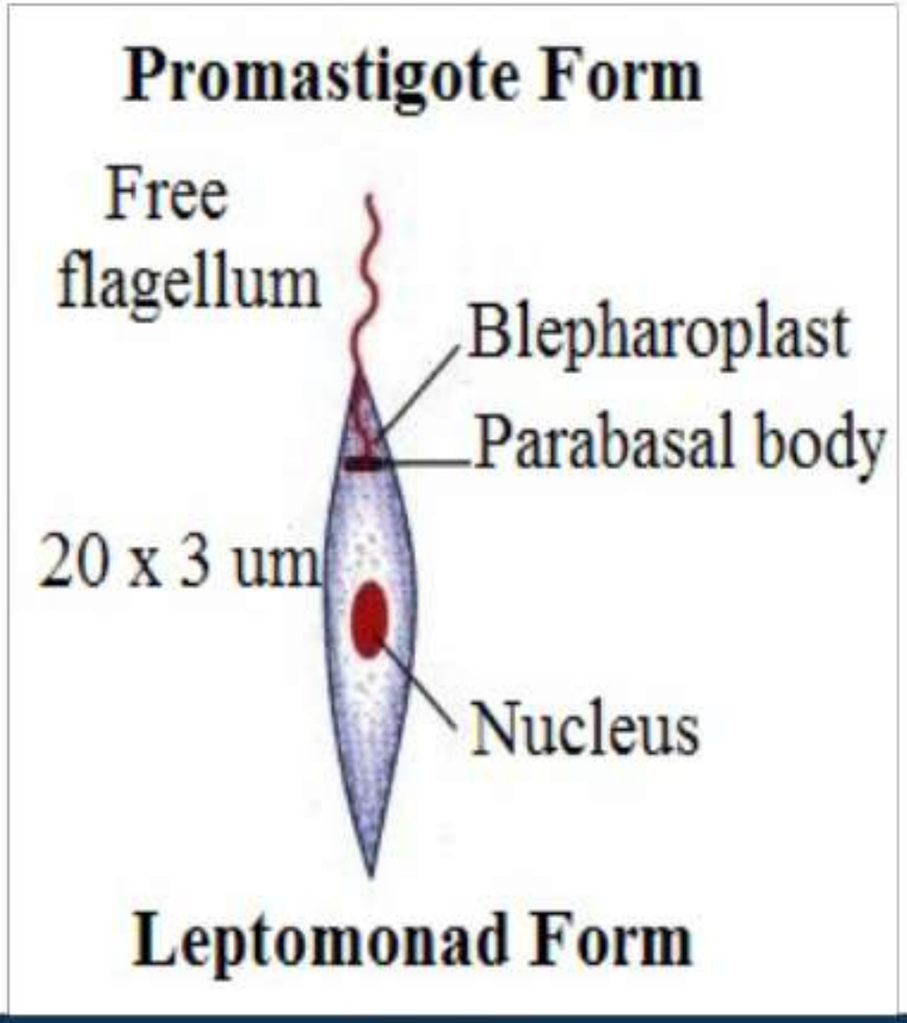
CamScanner

Morphology

Promastigote form of *Leishmania* spp.

Diagnostic stage in culture

- Spindle shaped
- 20 µ
- Central nucleus
- Anterior kinetoplast
- Anterior free flagellum
- No undulating membrane

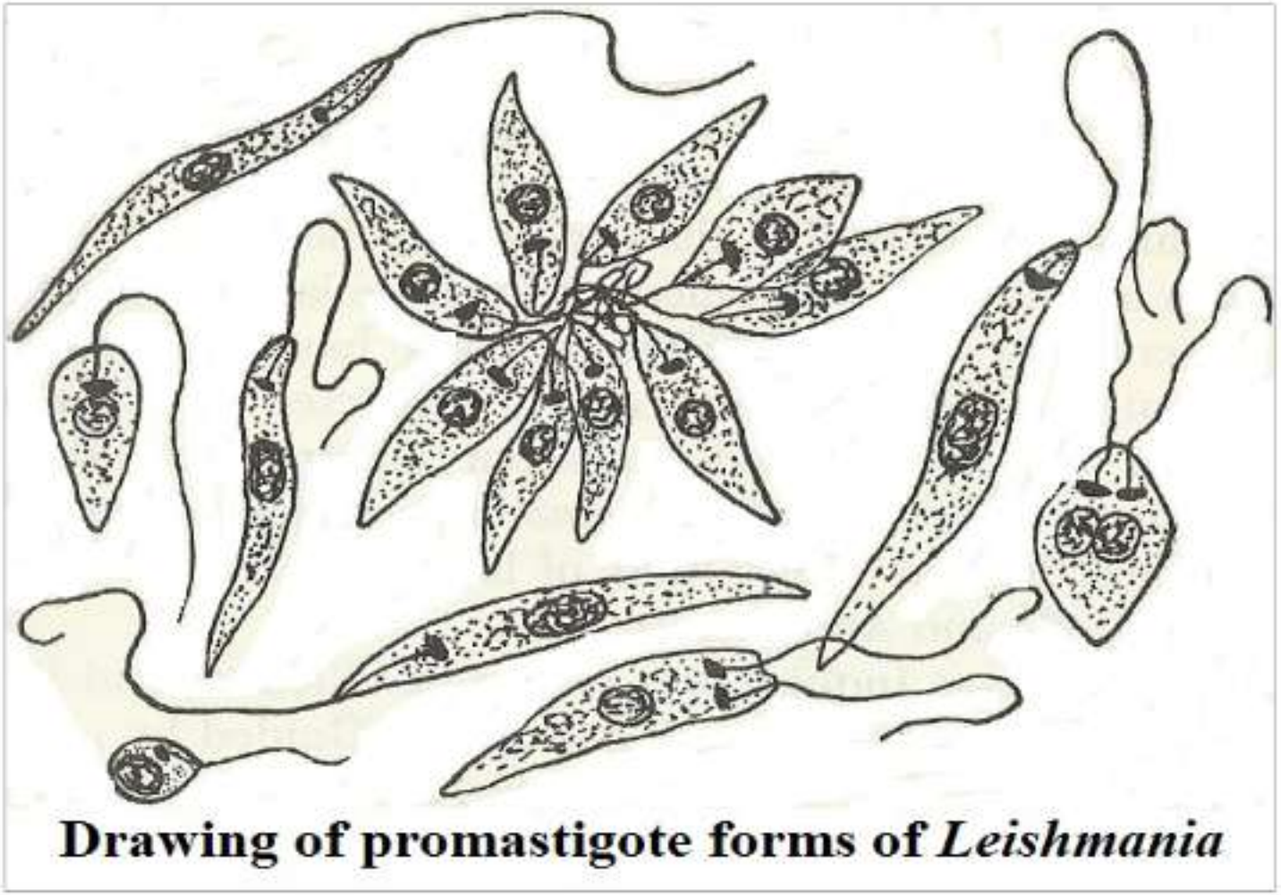


Infective stage in vector

CamScanner

Morphology

Promastigotes of *Leishmania* spp.
From culture (Diagnostic stage)



Drawing of promastigote forms of *Leishmania*

CamScanner

Morphology

Promastigotes of *Leishmania* spp. From culture (Diagnostic stage)





Promastigotes forming rosette

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- https://ain-shams.smartzoom.com/share/9b51337cad5876a61a4d368426c57c9a4013e80b8a337e8ba69cd7c34f1d22ad
- https://ain-shams.smartzoom.com/share/59e33750e4ec83b77ebd281855cb9503838bcd710ef6de69821bbeb3c3ff67d5
- https://ain-shams.smartzoom.com/share/3ac99734af84adbd695be9f4a69dcfc27b75c08c6d3c2e10889ac42699f18cf7
- https://ain-shams.smartzoom.com/share/b087dccdf3e35d270b2cfb8c56956eb194a2927059dafb0aed3a600ed5537413

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- https://ain-shams.smartzoom.com/share/9aeb5a29b13a612585afd51af376050d065810302c5c2b3574df092b0d9b0043

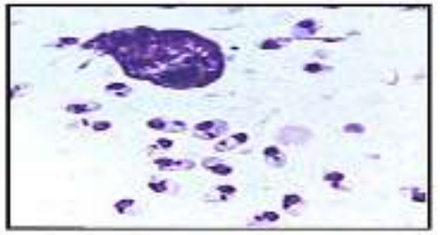
- https://ain-shams.smartzoom.com/share/0f2fd60a2efc5236f379938e6e6b8ae01857f82c3f7161c6deed59db45c537a4

CamScanner

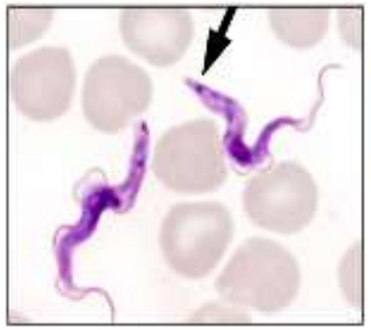
Lab 3

Protozoa

Haemoflagellates



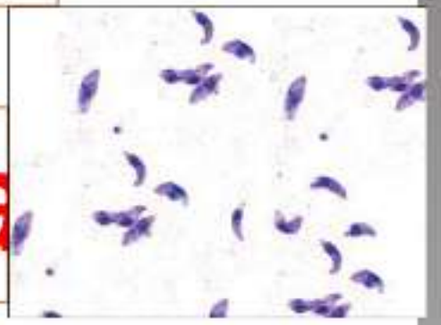
Leishmania



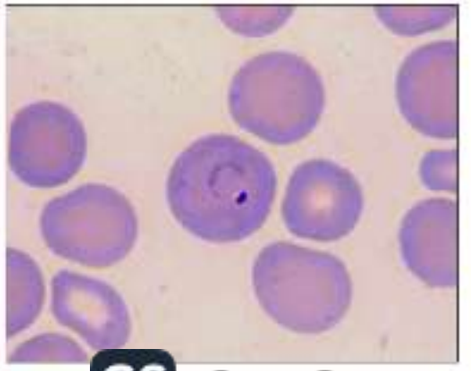
Trypanosoma

Sporozoa

Toxoplasma



Plasmodium



CamScanner

Lab
3

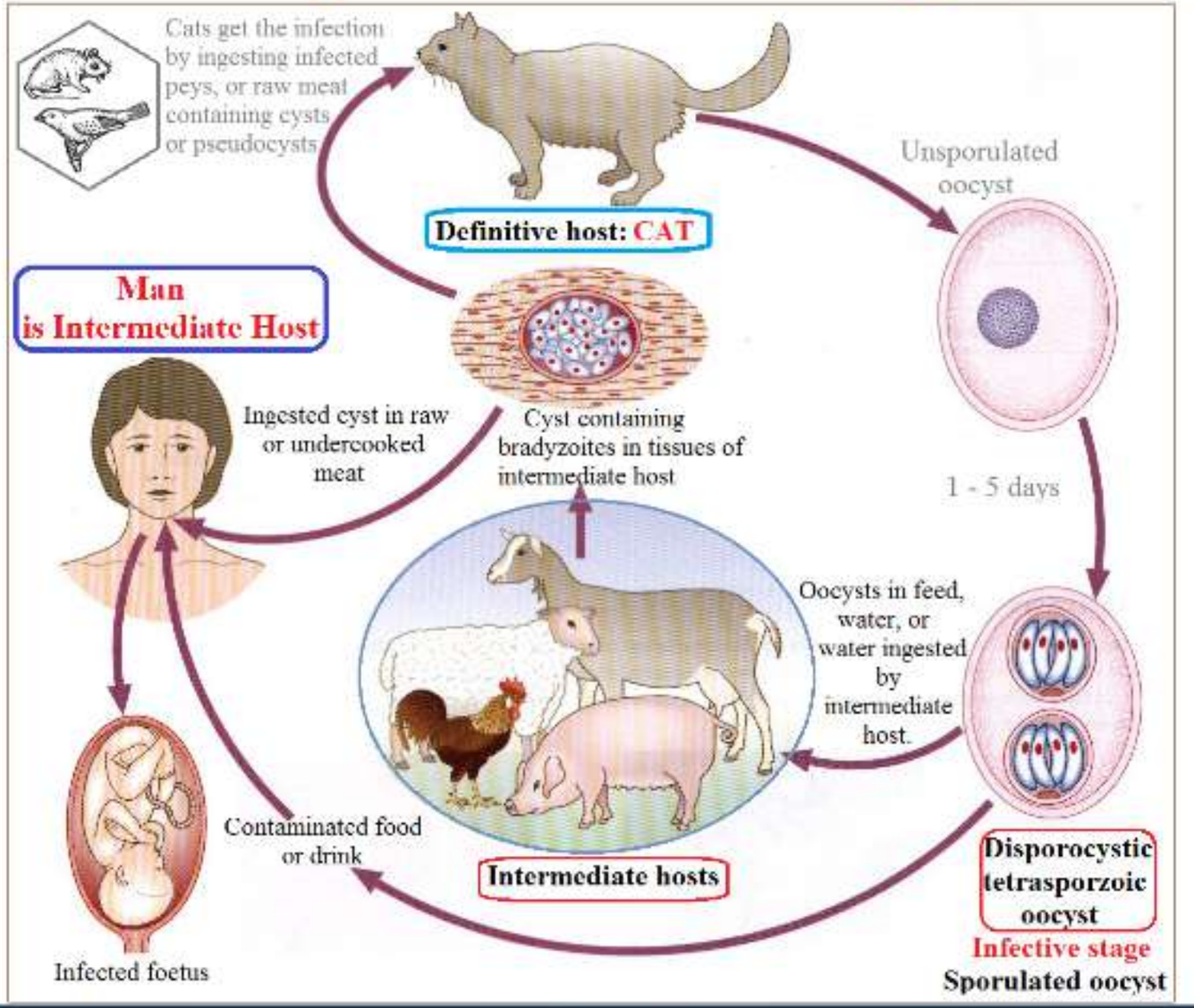
Toxoplasma gondii
(2 slides)

- Tachyzoites (or trophozoites)
- Cyst in brain

CamScanner

Lab 3

Life Cycle of *Toxoplasma*



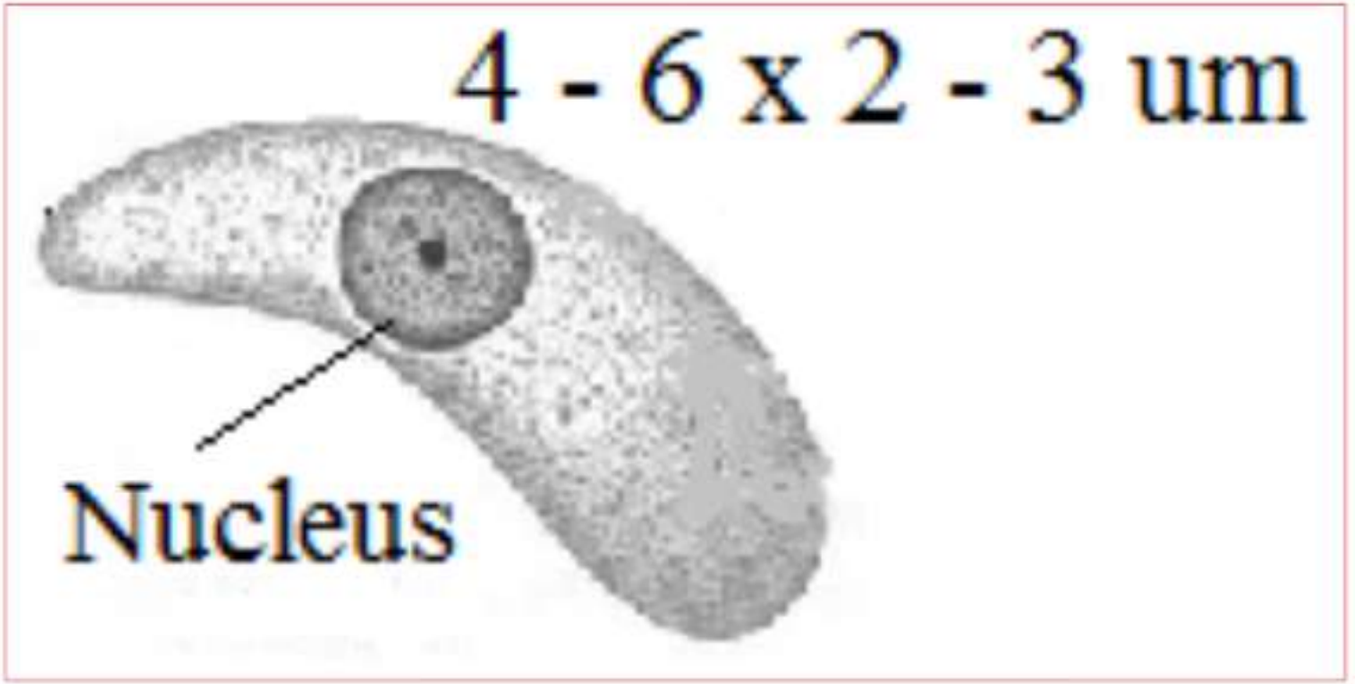
CamScanner

Lab 3

Toxoplasma gondii

Tachyzoite

- 4 x 2 µ
- Crescent shape
- Tapered anterior end, blunt (Rounded) posterior end.
- Central nucleus



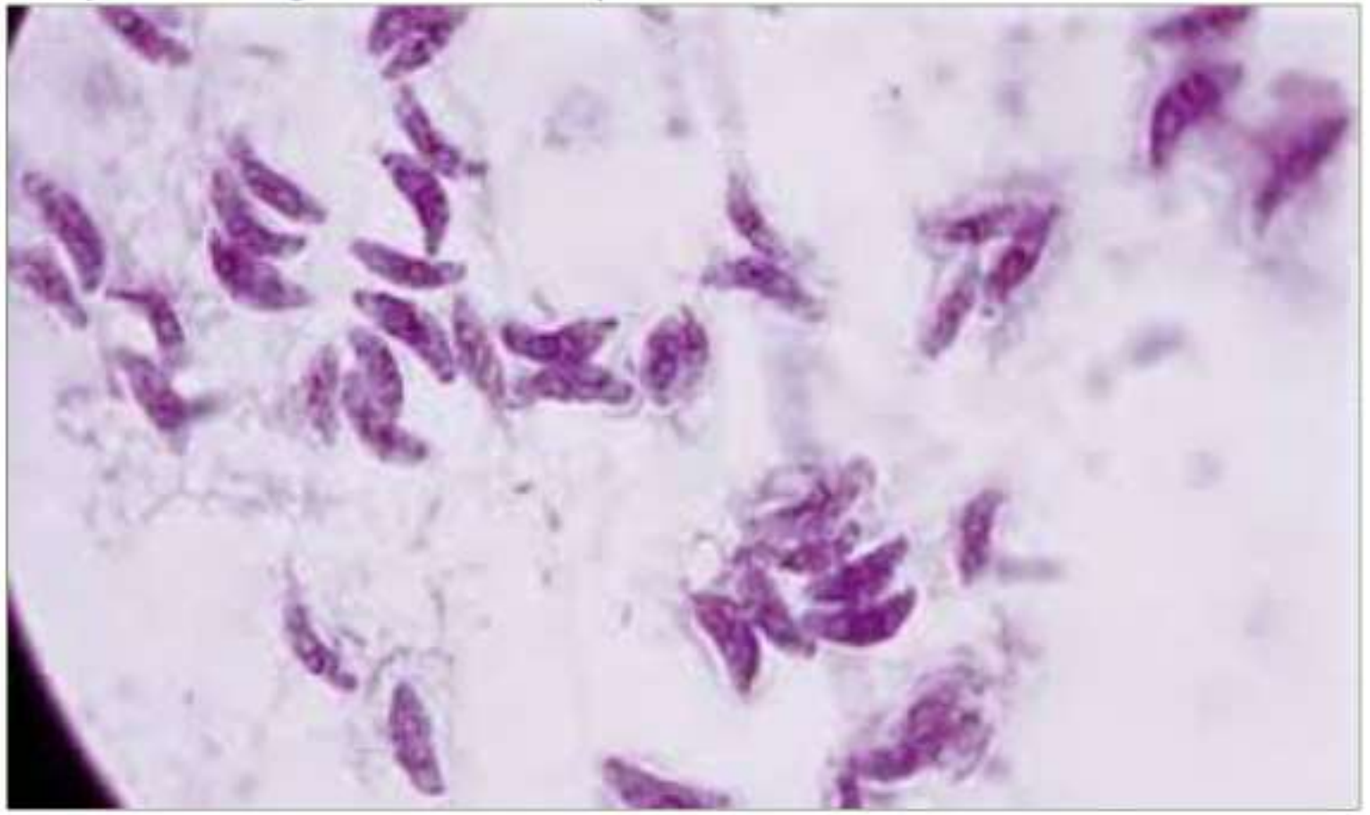
CamScanner

Lab 3

Toxoplasma gondii

Trophozoites (tachyzoites) in Smear

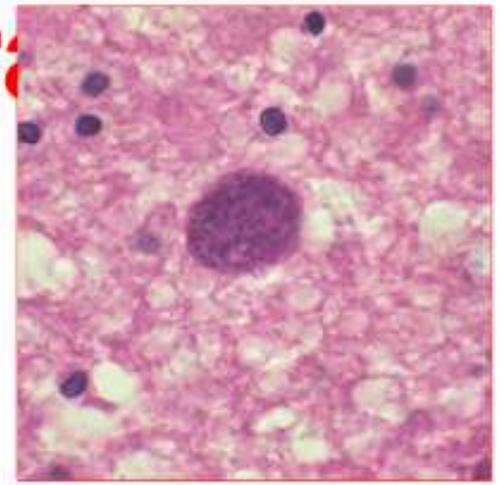
- 4 x 2 µ
- Crescent shape
- Tapered anterior end, a blunt (Rounded) posterior end.
- Central nucleus

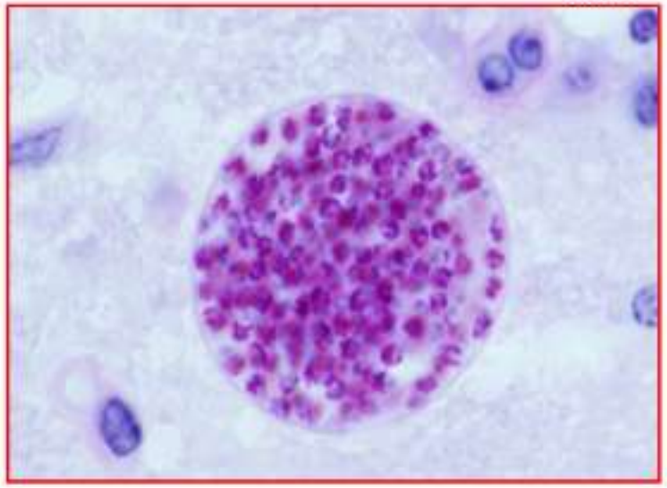


CamScanner

Lab 3

Toxoplasma gondii Cyst in Br





- §50-200 µm
- §Rounded
- §Well defined cyst wall
- §Thousands of bradyzoites

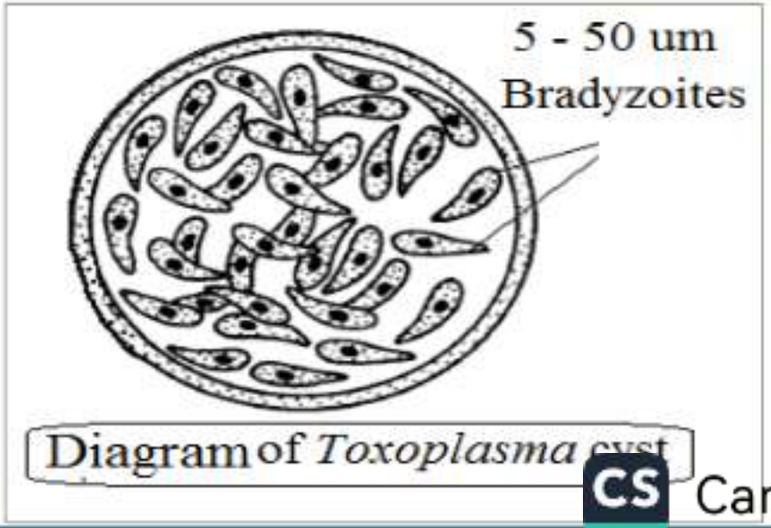


Diagram of *Toxoplasma* cyst

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- https://ain-shams.smartzoom.com/share/fd7effc9f6426a1bbc059849b56ddddf565ac308d0ecc40a064d57e2154ad320
- https://ain-shams.smartzoom.com/share/de67e212c601aceeb7494cea022e3e4e7393dc4080848f09a0467845fc5babf4
- https://ain-shams.smartzoom.com/share/836d300ba17ba26650125f2337e6fd3d0347c91153f01d666692561526c744af
- https://ain-shams.smartzoom.com/share/f84a25b549e9e3de5c53a3cad737e9a0326a58829a91de691b28599d74caa5bf

CamScanner

Helminths

Nematodes in lymphatic system

Wuchereria bancrofti

CamScanner

Wuchereria bancrofti

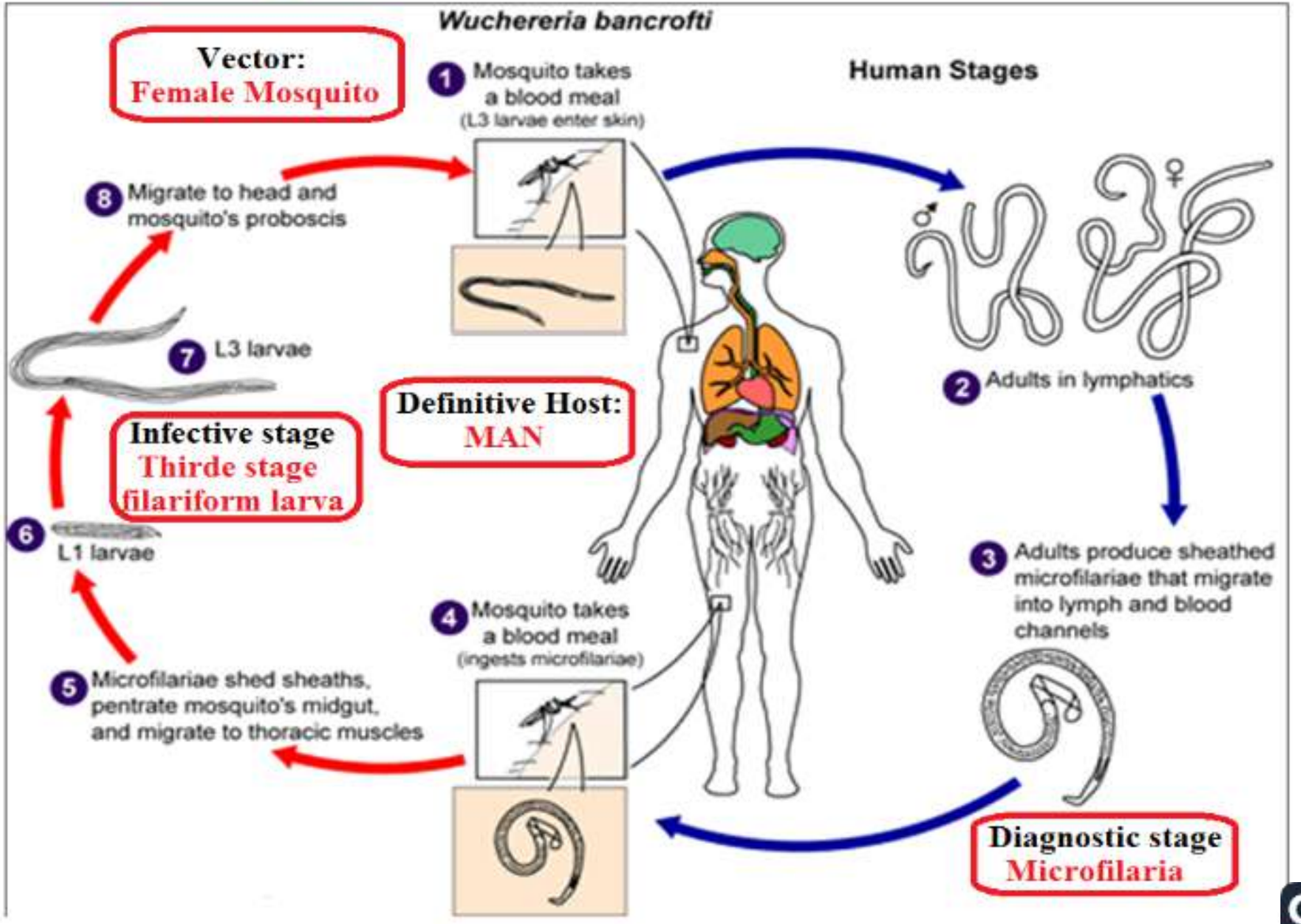
causing

Elephantiasis of Lower Limbs



CamScanner

Life Cycle of *Wuchereria bancrofti*



CamScanner

Microfilaria of *W. bancrofti*

Diagnostic stage



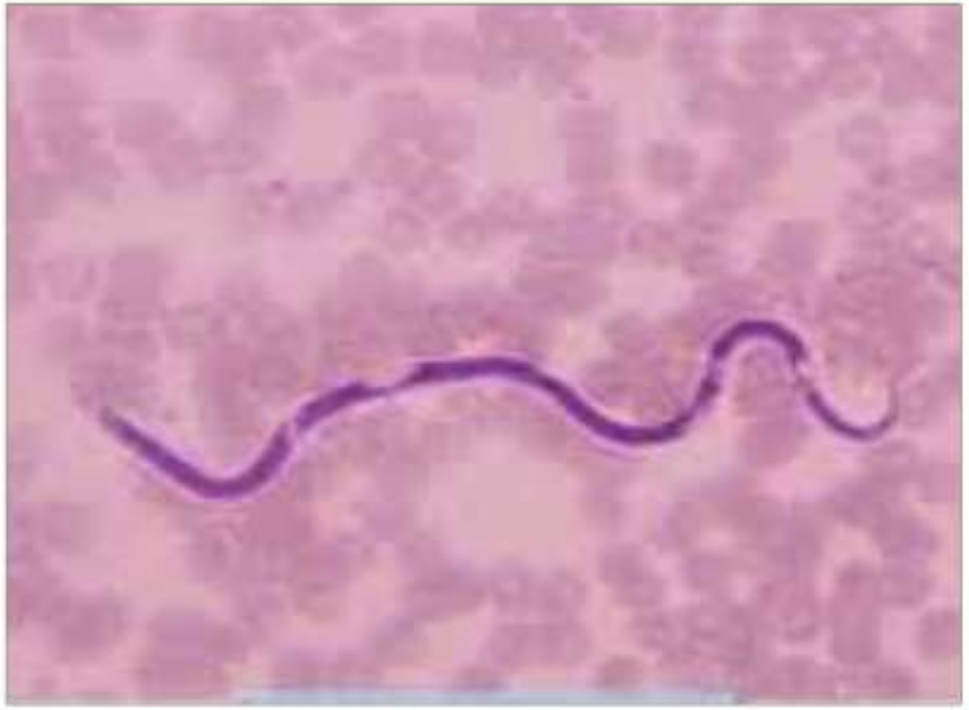
Size: 250 - 300 X 7 – 10 um.

Shape:

- Thread-like with graceful curves.
- The body is full of nuclei arranged in rows.
- The anterior end is rounded and devoid of nuclei.
- The posterior end is tapering and devoid of nuclei.
- It has a redundant sheath.

CamScanner

Microfilaria of *W. bancrofti* in blood film





CamScanner

LIVERPOOL SCHOOL OF TROPICAL MEDICINE

Lymphatic Filariasis Support Centre

Life cycle of Wuchereria bancrofti:

a causative agent of lymphatic filariasis



click to play

CamScanner

- https://ain-shams.smartzoom.com/share/b21d1bef5ec4c5588f9abee9a9ec23b73d41b429dd461f50afd54f5bc63cc495
- https://ain-shams.smartzoom.com/share/664b1f08a22649e7ebee32ff148f857ee845d40c8b7d9b7680bf4b9926692361

CamScanner

To wrap up

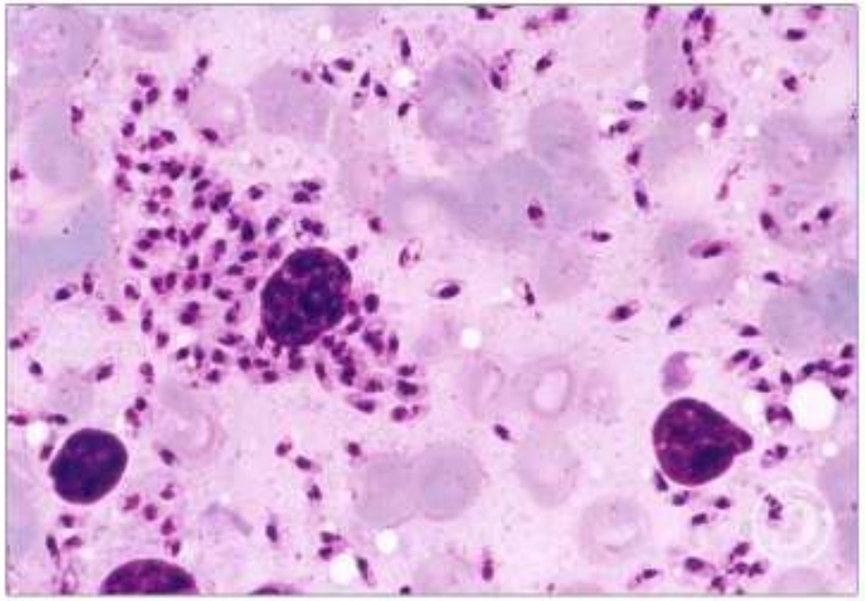
CamScanner

2 forms of *Leishmania spp.*

Promastigotes in culture



Amastigotes in spleen smear



CamScanner



Sand fly female



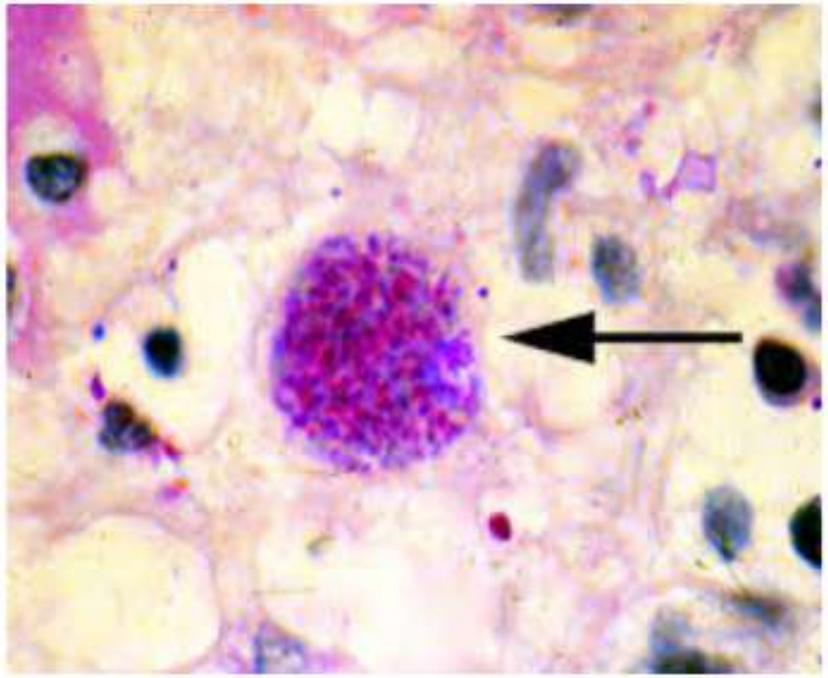
Sand fly male



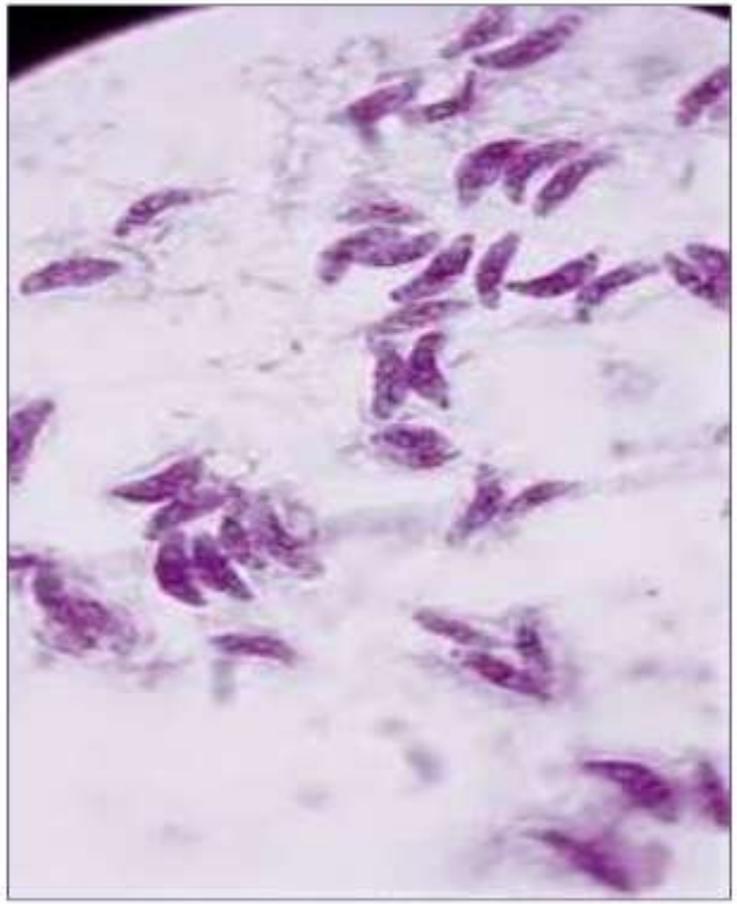
Culex Mosquito Female

CamScanner

Lab 3



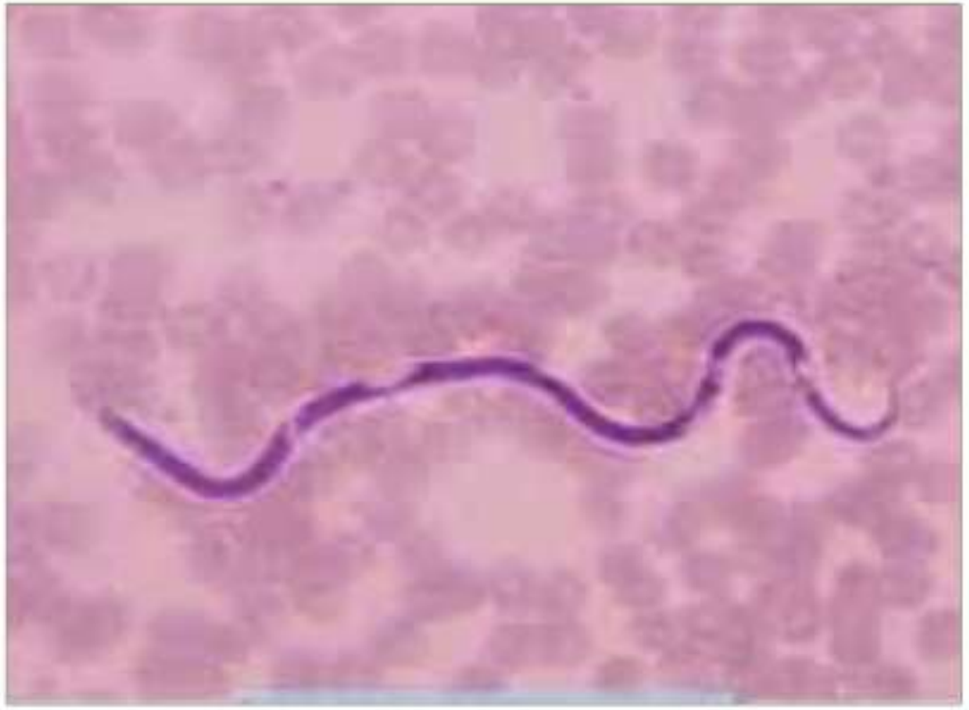
Toxoplasma cyst in brain



Toxoplasma trophozoites in smear

CamScanner

Microfilaria of *W. bancrofti* in blood film





CamScanner


TIME TO
THINK







CamScanner

Choose the correct answer:

Infective stage of *Leishamania* species in visceral leishmaniasis is:

a. Metacyclic form.

b. Epimastigote form.

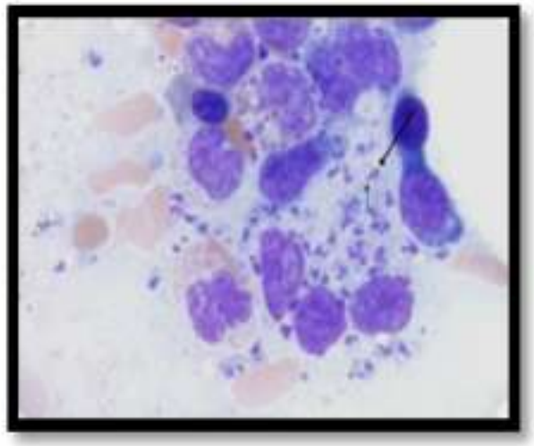
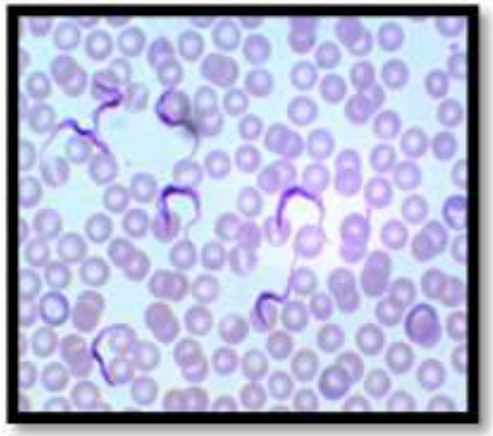
c. Promastigote form.

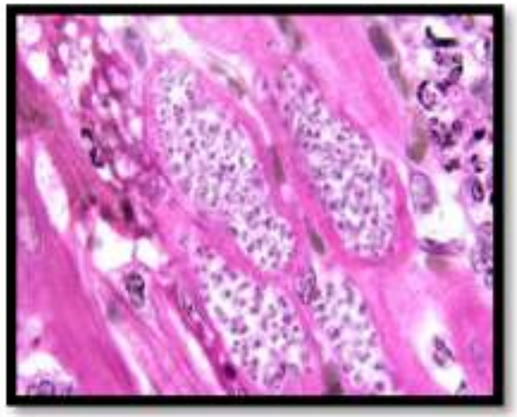
d. Amastigote form.

Answer: C

CamScanner

- **A 10 year old child presents with fever and hepatosplenomegaly since 4 weeks. Blood picture shows pancytopenia. Culture of bone marrow aspirates shows:**

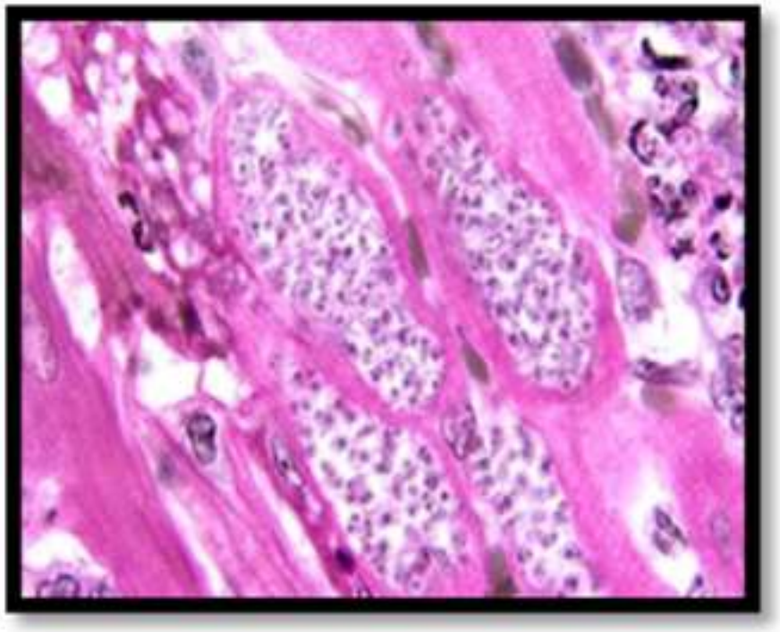


Answer: C

CamScanner

- Detection of this parasitic stage indicates:

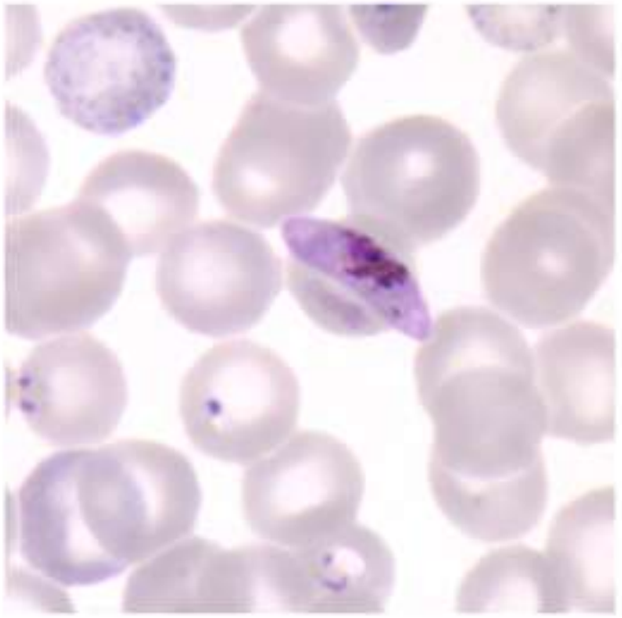
- a. African trypanosomiasis
- b. American trypanosomiasis
- c. Visceral leishmaniasis
- d. Toxoplasmosis
- e. Malaria



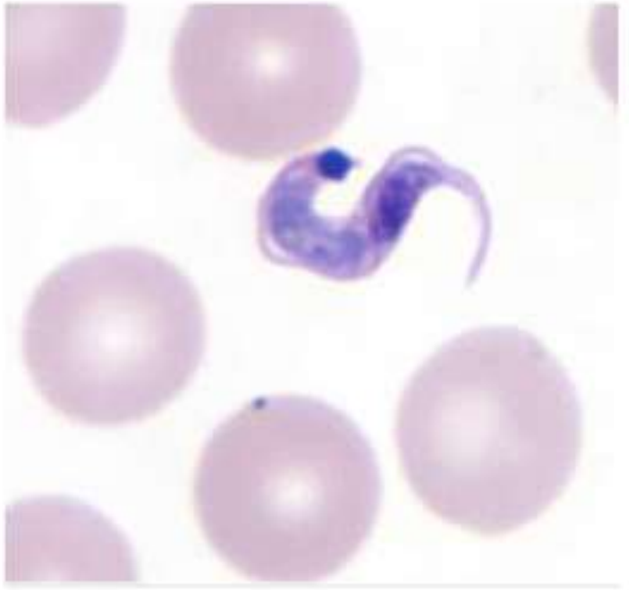
CamScanner

Match the shown clinical sign with the diagnostic stage of the causative parasite.





A



B



C

Answer: b

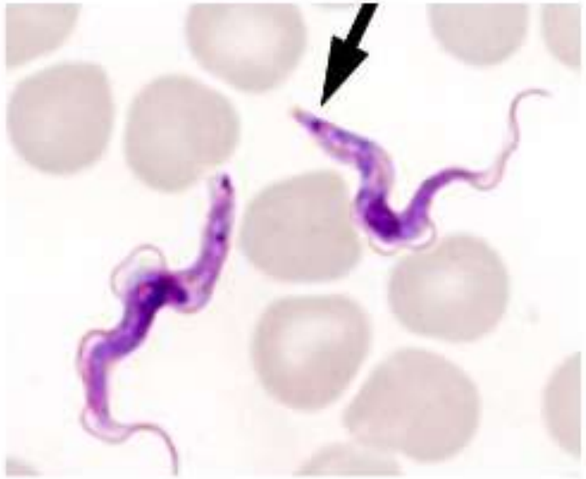
CamScanner

A patient presents with intermittent high fever , and hepatosplenomegaly. He showed enlarged posterior cervical lymph node as shown in the figure. Which of the following figures represents the parasitic stage that is found on examination of thin blood film?

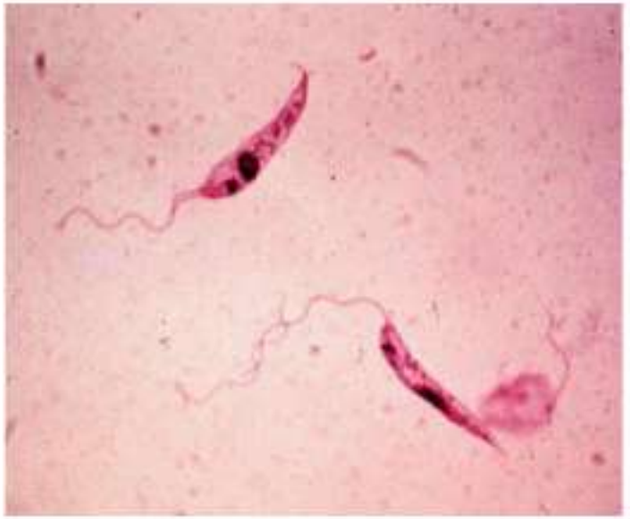




A



B



C

Answer: B

CamScanner

Lab 3

Choose the correct answer:

In chronic Chagas' disease, the main lesions are in:

A- Digestive and respiratory tracts

B- Heart and liver

C- Heart and digestive tract

D- Liver and spleen

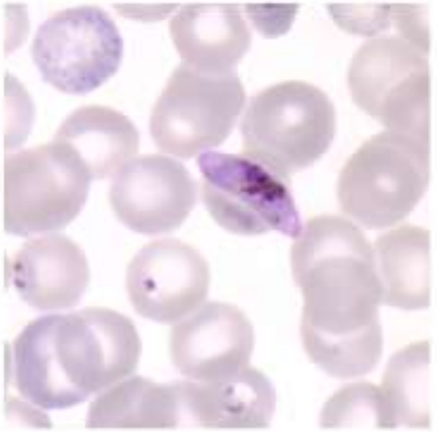
E- CNS

Answer: C

CamScanner

Match the shown clinical sign with the diagnostic stage of the causative parasite.





A

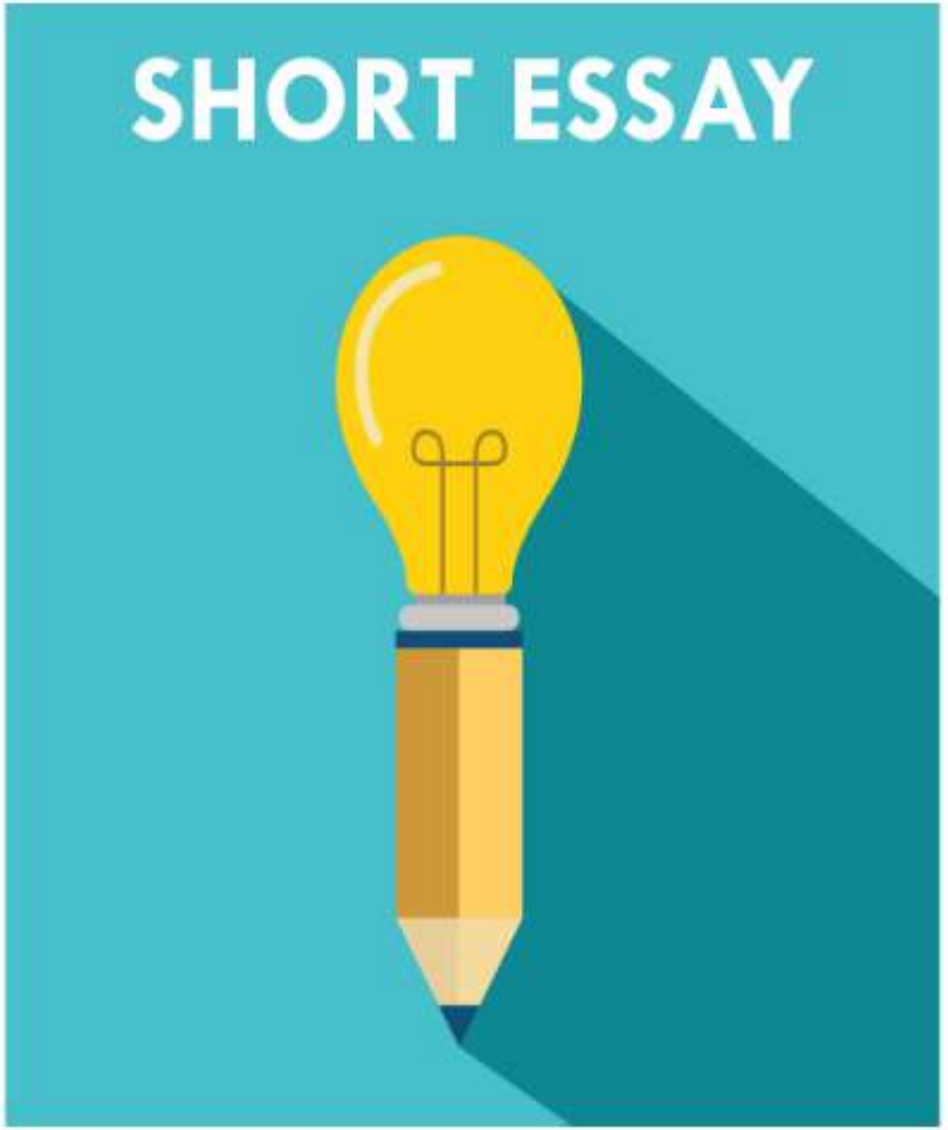


B



C

CamScanner



CamScanner

I. Interpret the statement:

The litter box of indoor cats should be emptied daily and not by a pregnant women.

In order not to give the chance to Toxoplasma oocysts to sporulate and become infective (few days are needed for sporulation).
And not by pregnant mother to avoid her from catching *Toxoplasma* infection in case she is seronegative.

CamScanner

II. Give an account on:

Sabine-Feldman methylene blue dye test (DT).

The DT is a sensitive and the most specific test for toxoplasmosis.
live organisms + patient's serum + complement + blue dye IgG T. gondii-

Specific antibody lyse the trophozoites so they do not take the blue stain of
methylene blue = +ve reaction
In contrary, in negative reaction the parasite stained blue.
Its main disadvantages are its high cost and the human hazard of using live
organisms.
N.B. It measures the early –formed IgG antibodies.

CamScanner

III. Mention the different infective stages and modes of *Toxoplasma gondii* Infection.

Cats are infected by:

- Oocysts in soil.
- Tissue cysts or pseudocysts in rodents.

Man and accidental hosts are infected by:

Oral route:

- Contaminated water or food by cat's stool (oocysts).
- Undercooked meat (tissue cysts).

Parentral route: (pseudocysts):

- Transplacental from mother to fetus.
- Organ transplant (rare).
- Blood transfusion (rare).

N.B. All forms of the parasite are infective.

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IV. Comment on the following statement:

Detection of IgG antibodies only can diagnose recent infection of toxoplasmosis.

A single detection of anti-Toxoplasma IgG can not diagnose a recent infection because it rises two weeks after infection and persists for life.
Its detection can diagnose a recent infection only if:
1- Detection of IgG rising titers or total antibody rise by a factor of 4 to 16 in serum taken 2 to 4 weeks after the initial serum collection.
2- IgG-avidity test.

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Best Wishes

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